CONTROLLING THE OBSERVATION CLASS OF A RUNTIME MONITOR: SELECTION, SELF-TUNING, AND PHYSICAL GROUNDING

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Abstract. The observation class of a temporal-implication monitor is a *controlled quantity* — optimally selected and held against drift by a convergent self-tuning loop that steers the monitor across the observation-class partition without manufacturing truth. Where prior runtime verification *fixes* the observation class, we *steer* it. The paper [9] shows that monitorability of a bounded-eventuality temporal implication is a property of the pair (formula, observation model), and that the observation space partitions into eight machine-verified window classes (the (A,B,C) cells). This paper takes the engineering stance and treats class assignment as an object to be selected, stabilised, and physically constrained. Our single headline contribution is the first end-to-end engineering of a monitor for this observation task: optimal monitor-type *selection* paired with a convergent *self-tuning* loop, both operating over a machine-checked generative signature space (the $390625 \to 50000 \to 8$ enumeration). This realises the active-tuning third source of finite confirmation that the classification companion paper [7] identifies but defers (its O5 direction); the selection, tuning, and generative-space results stand on their own, and the layers that follow support and bound them. First, over an exhaustively enumerated catalogue we identify the monitor type that minimises diagnostic ambiguity and opener sensitivity while offering the finest timing-error detection resolution among the minimisers (a *witnessed* selection result on $(\mathcal{A}, \mathcal{S})$; the stability radius is analytic). Second, we give a closed self-tuning loop that recovers timing phase by classical symbol-synchronisation machinery (early–late gate, Gardner and Mueller–Müller detectors); we prove almost-sure convergence of the self-tuning loop to the true phase within a basin of attraction under explicit stochastic-approximation hypotheses (a local guarantee; global convergence of the band-counting detector is conjectured and witnessed only on a finite simulation prefix), yielding a monotone migration of the observed class toward the static-like regime. Third, we constrain both selection and tuning by the physics of the observed process: a causal-coherence penalty forbids the loop from converging to an epistemically invalid operating point, and a window-length sweep exhibits the admissible corridor between premature falsification and causal inflation. We cast the whole as a formal specification of eight property classes in the distributed-algorithms style, safety-dominant by design with convergence as its sole liveness obligation discharged by proof. We complement these with the eight-class Schmitt-trigger worked gallery, a signature-level taxonomy of causal pathologies, and instrument-grounded case studies. The engineering thesis is that a temporal-implication monitor can be *steered* across the observation-class partition of the core paper, with provable guarantees and a hard physical boundary that keeps tuning from manufacturing truth.

1. Introduction

1.1. **The Control Problem.** A monitor for the bounded-eventuality implication "always, a trigger is followed by a confirmation within a bounded window" does not observe the world directly: it observes a tick-grid projection of the world through a sampling clock subject to latency, skew, drift, and jitter. The paper [9] establishes the consequence precisely. Monitorability is *observation-relative*: which formula faithfully represents the implication — the bounded safety formula Φ_k under a finite, known offset, the unbounded liveness formula $\mathbf{G}(w_p \to \mathbf{F}\, w_q)$ under an unbounded one — and which definitive verdicts its monitor can reach depend on the observation model, while the property class of each fixed formula does not; and the space of possible observations partitions into eight classes (the (A,B,C) cells), complete by construction and machine-verified. Which class a given trace lands in is not a fact about the formula; it is a fact about the pair (formula, observation model).

Headline question. This paper answers one question: given that the observation class of a bounded implication is joint in the pair *(formula, observation model)* [9], can that class be treated as a *controlled* quantity — selected, self-tuned, and physically bounded — with guarantees? — a question of *engineering*, downstream of the observation theory it builds on.

The core paper [9] settles, at the level of principle, what can be *observed*, what can be *read*, and what may be *attributed True*; the present paper is its engineering counterpart, and takes a single further step: it treats the observation class as a *quantity to be controlled*. If class assignment is a property of the observation model, and the observation model is under the monitor's control, then a deployed monitor is a control system whose set-point is a window class. Three design problems follow immediately, and this paper is organised around them.

Selection.: Among the monitor types that a small set of design axes generates, which one minimises diagnostic ambiguity — the number of window classes compatible with an observed signature — while remaining robust to timing perturbation? (Section 7.)

Self-tuning.: Once a type is chosen, how does the monitor keep its sampling phase aligned against drift at runtime, and can the resulting loop be guaranteed to converge? (Section 8.)

Physical grounding.: What forbids the tuning loop from converging to an operating point that reports confirmations the physics cannot support? (Sections 9 and 10.)

We flag one scope limit at the outset: the causal-incoherence penalty $\mathcal{C}(w)$ of the physical-grounding layer is a design objective whose calibration is [HEURISTIC] and is *not* numerically evaluated in the case studies of this paper, consistent with the tags on Contribution 4.

1.2. **Background: the Observation Class as a Controlled Quantity.** Runtime verification classically *fixes* the monitor's observation model — its sampling clock, threshold, and phase — and asks the single question *does the observed trace satisfy φ?* The observation model is assumed given, fixed, and correct. For a bounded-eventuality implication "a trigger is always followed by a confirmation within a bounded window", however, *which class the implication falls into is not a fact about the trace alone*: it depends on where the monitor samples and where it thresholds. This is the observation-relative monitorability of [9]: read through a mis-set observation model, one and the same physical implication migrates through an eight-class window partition — from a clean static-like reading (1,0,0) down into the miscalibration pathologies (the A=0 classes) — and across the four temporal families safety, cosafety, liveness, and coliveness [7].

The observation model is therefore a *degree of freedom*, and the engineering task is its *control*: choose the monitor that reads the implication most sharply, hold that reading against drift at runtime, and bound it by the physics of the observed process so that no amount of tuning ever manufactures a confirmation the signal does not support. No existing tool occupies this loop. Monitor synthesis [4, 19] builds a decider for a *fixed* specification; classical timing recovery [13, 23] aligns a receiver clock but carries no property-class semantics; runtime enforcement [29, 12, 6] constrains the monitored *system*, not the monitor's reading of itself. What is new here is the closed loop that treats the monitor's own observation model as the controlled variable, under a formal convergence obligation and

a causal-coherence bound with no analogue in receiver design. (The same control-under-conformance obligation is familiar industrially — e.g. the ETSI Reconfigurable Radio Systems framework [22], where a fielded device that re-parameterises itself over its lifetime must still be certified against its declared specification — but there reconfiguration is discrete and governed out-of-band, whereas ours is a continuous, in-band, closed loop.)

What the paper establishes, in one pass. The argument builds in six steps, each a section, and the reader can follow the whole arc before meeting a single proof. We first *pin down what is observed* — the tick grid, the crossing and touch primitives, and the eight window classes with their feature triples (A, B, C) (Section 3). We then *specify the monitor* as eight temporal obligations with explicit LTL schemas and four-family classes, over a rigid world model whose only licensed relaxations are the tuning operators and which is bounded below by an inviolable event-soundness floor: the tuning calculus provably *cannot fabricate a token* (Section 5, Theorem 5.8). We *map the signature space* — a generative automaton enumerates the $390625 \to 50000 \to 8$ chain from raw signatures through the admissible catalogue to the eight window classes (machine-checked), and lifting each clopen window pattern sorts it into the safety, cosafety, liveness, and coliveness classes (Section 6). We *select*: over the exhaustive k=3 catalogue a single monitor type minimises signature ambiguity and opener sensitivity together (Theorem 7.4, witnessed by enumeration). We *tune*: an early–late timing-recovery loop converges almost surely to the true phase within a basin of attraction (Theorem 8.8, local convergence proven, global conjectured) — which is exactly the migration of the observed class up the ladder toward (1,0,0) that Figure 9 plots (Corollary 8.10). We *ground it in physics*: an admissibility constraint forbids convergence to an epistemically invalid operating point (proven), tempered by a causal-coherence penalty (heuristic) (Section 9). Finally we *diagnose* — five causal pathologies become signature-level predicates, complete by construction over the catalogue, with a window-length corridor (Section 10). A forward-looking *enforcement* reading, which would steer the observed class to a chosen target, is reached for but kept conjectural — its transduction has no closed operational semantics yet — and nothing in the selection, tuning, or enumeration results depends on it.

Division of labour. Every scientific primitive we rely on is proved elsewhere and cited, never re-derived: the touch and crossing observation primitives with their capability calculus are [8]; the safety/cosafety/liveness/coliveness classification is [7]; the observation-relative monitorability theorem is [9]. The present paper contributes the engineering control layer — selection, self-tuning, physical bounding, and the machine-checked signature space that supports them — over that observation class. The itemised contributions, each with its epistemic status, follow next.

1.3. **Contributions.** *Primary contribution. The observation class of a temporal-implication monitor is a controlled quantity — optimally selected and held against drift by a convergent self-tuning loop that steers the monitor across the observation-class partition without manufacturing truth.* Where prior runtime verification *fixes* the observation class, this paper *steers* it. Concretely, the single headline result is the first end-to-end *engineering* of a monitor for the bounded-implication observation task: optimal monitor-type *selection* (Theorem 7.4, WITNESSED on $(\mathcal{A}, \mathcal{S})$ over the exhaustive k=3 catalogue) paired with a *self-tuning* convergence loop (Theorem 8.8, local a.s. convergence PROVEN, global CONJECTURED), both operating over a machine-checked generative signature space (the

390625 → 50000 → 8 enumeration). This realises the active-tuning third source of finite confirmation that the classification companion paper [7] identifies but explicitly defers (its O5 direction, Remark 8.16). The companion core paper [9] carries the complementary half of the diptych: it proves that monitorability is observation-relative; this paper builds the optimal, self-tuning monitor that operates under that relativity. The itemised contributions below develop and support this core.

(1) **A worked eight-class gallery** (Section 4). A visual, per-class development of the single-threshold edge-triggered encoding, with token rows, feature triples (A, B, C), the trace-subclass enumeration, a calibration tree, and a reachability matrix. [WITNESSED]
(2) **A formal specification of the self-tuning monitor** (Section 5), in the distributed-algorithms style: eight properties with explicit LTL schemas, four-family classes, and verdict degrees, together with a rigid world model whose licensed relaxations are exactly the tuning operators, bounded below by an inviolable soundness floor (Section 5.4). The suite is safety-dominant by design, its unique liveness obligation (convergence, a persistence property) being discharged by proof rather than runtime checking. [CLASSIFICATION PROVEN-BY-INHERITANCE; ALGORITHMS 1–2 NEW]
(3) **An optimal-selection result** (Section 7). Over the exhaustively enumerated k=3 catalogue, the monitor type (ρ=0, ω=0, π=1) simultaneously attains zero signature ambiguity and zero opener sensitivity, and among the minimisers has the finest timing-error detection resolution (half-tick). The ($\mathcal{A}$, $\mathcal{S}$) result is *witnessed* over the finite catalogue; the stability radius $\mathcal{R} = 1/2$ is analytic, not witnessed. [WITNESSED ON ($\mathcal{A}$, $\mathcal{S}$)]
(4) **A convergent self-tuning loop** (Section 8). A first-order phase-recovery loop driven by an early–late timing-error signal, shown to converge almost surely to the true phase offset within a basin of attraction via Robbins–Monro stochastic approximation (C1–C5), with global convergence conjectured, inducing a monotone migration of the observed class toward the static-like regime. [LOCAL A.S. CONVERGENCE PROVEN UNDER THE SIGN/REGRESSION CONDITION (C5) ON A BASIN; GLOBAL CONVERGENCE CONJECTURED, WITNESSED ON A FINITE PREFIX]
(5) **Physical grounding of selection and tuning** (Section 9). A physical-admissibility constraint that forbids convergence to an epistemically invalid operating point [PROVEN], together with a causal-incoherence penalty whose calibration is [HEURISTIC].
(6) **Pathology diagnostics** (Section 10). A trace-level taxonomy of five causal failure modes, each captured by a signature-level predicate whose coverage is complete by construction over the catalogue, plus a window-length sensitivity sweep exhibiting the admissible corridor. [WITNESSED] (window-length sweep; taxonomy coverage is complete by construction, not an empirical result)

The engineering thesis, in one line: *the observation class of a temporal-implication monitor can be selected to minimise ambiguity, held phase-aligned at runtime with provable convergence, and bounded by the physics of the observed process.* All three levers move the same quantity — the monitor's position in the eight-class partition of the core paper [9].

Scope note (what the engineering core does not rest on). The selection result (Theorem 7.4), the self-tuning convergence result (Theorem 8.8), and the machine-checked generative signature space (390625 → 50000 → 8) are established *independently* of the forward-looking

material of this paper. In particular they do not depend on the enforcement framework (Definition 8.14: its restricted class-quotient core yields a PROVEN-CONDITIONAL *controllability* fact, Proposition 8.21, while the *adversarial* enforcement reading stays CONJECTURED and its include/exclude reading HEURISTIC) or on the O5-closure positioning of Remark 8.16: those are interpretive and forward-facing, not premises of any selection, tuning, or enumeration claim. The engineering contribution stands on its own.

1.4. **Problem statement and research questions.** We can now state the engineering problem precisely. Fix the bounded-eventuality implication "always, a trigger is followed by a confirmation within a bounded window" and its observation on the tick grid. A deployed monitor projects the world through a sampling clock $\mathcal{C}(L, S, D, J)$ subject to latency, skew, drift, and jitter, and by the core paper [9] the observed window class is a property of the pair (formula, observation model). The task of this paper is to *engineer a monitor for this observation task*: to choose a monitor type that resolves each reachable observation into a single, physically supportable window class, and to keep it there at runtime under timing and observation uncertainty. Success is declared against three criteria, in decreasing strength of guarantee: (i) *minimal diagnostic ambiguity* — the chosen type maps every reachable signature to a single window class; (ii) *runtime phase alignment* — the type is held at its nominal operating point against drift; and (iii) *physical honesty* — no operating point reports a confirmation the observed process cannot support. Criteria (i) and (ii) are optimisation and convergence targets; (iii) is an inviolable side constraint.

The paper is organised around five research questions, each answered by a specific section or result; no question is posed whose answer is not already developed below.

RQ1.: Among the monitor types generated by a small set of design axes, which minimises diagnostic ambiguity and opener sensitivity, and how robust is that minimiser to timing perturbation? *(Answered by the optimal-selection result, Section 7.)*

RQ2.: Can the space of observation signatures be generated, exhaustively enumerated, and classified into property classes with explicit LTL schemas? *(Answered by the band-language generator and quantifier lift of Section 6 and the specification of Section 5.)*

RQ3.: Once a type is chosen, can a self-tuning loop be guaranteed to converge to the true operating point at runtime? *(Answered by Section 8 and the convergence theorem Theorem 8.8, locally; global convergence is conjectured.)*

RQ4.: What forbids self-tuning from converging to an operating point that manufactures confirmations the physics cannot support? *(Answered by the physical-grounding constraint of Section 9 and the pathology diagnostics of Section 10.)*

RQ5.: What can the monitor's tuning powers *enforce* over the observation-class partition, and with what epistemic status? *(Answered, in restricted form, by the class-quotient controllability proposition Proposition 8.21: on the finite class-quotient arena, and on a cooperative (edge-affording) signal, recurrence (*$\mathbf{G}\,\mathbf{F}\,\sigma^*$*) and persistence to the absorbing class (*$\mathbf{F}\,\mathbf{G}\,(1{,}0{,}0)$*) are controllable by the monitor via memoryless strategies,* PROVEN-CONDITIONAL *on the estimate staying in the convergence basin, on an explicit edge-affordance assumption, and on the causal-coherence conditions of the core paper.* Adversarial *enforcement — steering against an environment that may withhold edges — and the* unconditional *(basin-free, CC-free) readings remain conjectured (Definition 8.14 and Remark 8.17).)*

The three success criteria map onto these questions as follows: RQ1–RQ2 establish the selectable, classified design space in which criterion (i) is met; RQ3 discharges criterion (ii); and RQ4 enforces criterion (iii), with RQ5 lifting the whole to an ω-level enforcement reading whose resolution is left open (Sections 13 and 14).

2. Related Work

This paper sits inside a mature body of work on runtime verification and a classical body of work on timing recovery, and borrows the enforcement vocabulary of security automata. Its contribution is neither a new monitoring semantics nor a new synchroniser, but a *control layer* that selects, self-tunes, and physically bounds a monitor's position in the observation-class partition of the core paper [9]. The theory-side positioning of that partition — what can be observed, read, and attributed true — is given in the companion paper's own related-work discussion [9]; we do not repeat it here and concentrate on the engineering lineage.

Runtime verification and monitor synthesis. The apparatus for synthesising monitors from temporal specifications is well established: the three-valued LTL/TLTL monitor constructions of Bauer, Leucker, and Schallhart [4], the survey account of runtime verification [19], which also covers their four-valued refinement RV-LTL, rewriting-based monitor generation [16], and the specification-based monitoring of cyber-physical systems surveyed by Bartocci et al. [3], with the challenges beyond software collected by Sánchez et al. [28]. This literature answers "given a property, build a monitor that decides it." It does *not* treat the monitor's own observation model as a tunable object: the sampling clock, threshold, and phase are typically assumed fixed and correct. The present paper takes the observation model as the controlled quantity and asks which monitor type to *select* and how to *self-tune* it — monitor engineering on top of an existing monitor semantics, not a replacement for it.

Runtime enforcement, shielding, and security automata. Beyond passive detection, a monitor may *enforce*: Schneider's security automata characterise the safety policies enforceable by a monitor that halts on violation [29]; runtime enforcement has since been systematised, and the tool landscape taxonomised by Falcone et al. [12]; shield synthesis [6] produces a reactive component that corrects an output stream on the fly to satisfy a specification; the canonical Safety–Progress account of which properties can be *verified* and which *enforced* at runtime is due to Falcone, Fernandez and Mounier [11]. Assumption-relative monitorability, in which a declared assumption A restricts the admissible trace space, is treated by Henzinger and Saraç [17]; the declared observation theory under which this paper's enforcement objectives are stated is a physically-instantiated special case. Our enforcement framework (Section 8.3) is deliberately more modest and orthogonal: we do not reinvent enforcement, we *cost-classify monitor adaptation*. The tuning powers are transductions on band-words (Section 8.2), and whether they can enforce a target observation class is stated as a conjectured ω-reachability problem (Definition 8.14 and Remark 8.17), grounded in the convergence result rather than in a shield-synthesis construction. Where shielding corrects outputs, we correct the *observation kernel* that feeds the monitor, under the inviolable soundness floor of Section 5.4.

Timing recovery, clock/data recovery, and adaptive detection. The self-tuning loop of Section 8 is, structurally, a symbol-synchronisation loop. Its timing-error detectors are the classical ones: the Gardner detector [13] and the Mueller–Müller detector [23], the general theory of digital-receiver synchronisation [21], and the phase-locked-loop and clock/data-recovery design tradition [5, 25]. This literature recovers a sampling phase for a communication receiver; we repurpose it as a runtime-monitor mechanism, where the "symbol" is a window-class token and the recovered phase migrates the observed class toward the static-like regime. The novelty is not the detector but its placement inside a monitor with a formal convergence obligation (S4, Theorem 8.8) and a hard causal-coherence bound that has no analogue in receiver design.

The companion papers. This paper is the engineering member of a co-deposited pair. The core paper [9] proves the observation-relative monitorability theorem and the eight-class partition this paper controls; the classification of each monitored obligation as safety, cosafety, liveness, or coliveness is grounded in the intersection-algebra paper [7]; and the touch/crossing primitives that generate the tuning powers, together with the capability calculus (w, b, k), come from the dual-modalities paper [8]. We cite these three throughout and re-derive none of their results: the present paper adds only the control layer above them.

3. Background in Brief

This section fixes notation and recalls, without proof, the objects this paper controls. Every statement below is established in the cited papers and is cited, not re-derived; the reader is referred to [9, 8] for the constructions.

3.1. **Tick grid, crossing, and touch.** A monitor samples continuous channels on a discrete tick grid whose clock imperfections are the observation-model 4-tuple (L, S, D, J) — latency, skew, drift, jitter — of the core paper [9] (the case studies use assignments such as $(L{=}1, S{=}0, D{=}0, J{=}1)$). The antecedent and consequent channels $\tilde{p}, \tilde{q}$ are compared against a semantic threshold θ. Two primitive observation modalities are distinguished in [8]: a *crossing* (a channel transits θ: state-bearing, memory-carrying, cost $(w, b, k) = (1, 1, 0)$) and a *touch* (a local extremum against a probe level: memoryless, window-bearing, cost $(3, 0, 1)$). The crossing modality generates the confirmation/refutation token alphabet used throughout; the touch is used here only to disambiguate the silent window (Section 4). These two primitives form a *complete* observation basis: by the ternary-rung tuning-closure theorem of [8], every ternary, instant-level, one-step-lookback threshold modality is event-stream-equivalent under threshold tuning to crossing or to touch, so no third primitive is required and the crossing alphabet Σ_{ev} together with the touch probe may be adopted without loss of generality (WITNESSED by the cited closure theorem).

3.2. **The eight window classes and the (A, B, C) triple.** Relative to the opener tick x — the first tick at which the trigger w_p holds — the windowed observation is summarised by a Boolean feature triple (A, B, C): $A = 1$ iff a Q-validation ($Q^{\top}$) is observed in the window, $B = 1$ iff a Q-invalidation ($Q^{\perp}$), and $C = 1$ iff a P-retraction ($P^{\perp}$). The eight assignments of (A, B, C) define the eight window classes of [9], complete by construction; their machine verification over the 50000-signature catalogue at window length $k = 3$ is this paper's (Remark 6.5); we adopt them as given and draw their canonical realisations in Section 4. We

steer between the *static-like* class (1,0,0) (ideal confirmation), the *bounded-temporal* classes (A=1 with a spurious B or C), and the *miscalibration* classes (A=0). The block name *miscalibration* is used here in the sense of Corollary 8.10, i.e. under its ground-truth-implication assumption; the finer per-class reading of [9] labels these individually ((0,0,0) miscalibration, (0,1,0) monitor-refuted (out-of-window onset), (0,0,1) late retraction, (0,1,1) complex/race), the cell placements agreeing throughout. Three vocabulary levels structure the catalogue: the eight *window classes* (the (A, B, C) cells) inherited from the core [9], the *signatures* (the 50000 eight-tick instances at $k = 3$), and the trace-level realisations of Section 4.

3.3. **Observation kernel and tuning operators.** The projection from continuous channels to observed tokens factors through an *observation kernel* $\mathcal{K}_{\boldsymbol{\theta}}$ parameterised by a tunable vector $\boldsymbol{\theta}$ (sampling interval, pre-filter / debounce strength, phase estimate, decision threshold, gating margins), formalised in [9]. The kernel suppresses noise but never inserts evidence: it changes *which* measurements pass, not the ground-truth signal. A finite family of elementary tuning operators (threshold shift, phase alignment, window-length adjustment, debounce, margin extension, hysteresis, oversampling, Robbins–Monro auto-convergence, causal-coherence filtering) moves $\boldsymbol{\theta}$ — and hence the observed window class — without altering the underlying crossing/touch classification. These operators are the actuators of the control problem.

4. The Worked Gallery: Eight Schmitt-Trigger Classes

This section illustrates the analogue-to-discrete encoding adopted by the decision stage $\mathcal{D}_{\theta_3}$ of the observation kernel — one of the kernel stages ($\mathcal{F}_{\theta_1}$ pre-filter, $\mathcal{A}_{\theta_2}$ phase-align, $\mathcal{D}_{\theta_3}$ decision, $\mathcal{M}_{\theta_4}$ gating) of [9] — and visualises the resulting feature triples (A, B, C) on eight representative window classes.

Encoding model — explicit choice. Of the three encoding families—single-threshold (M1, level-triggered $\mathrm{M1}_{\mathsf{lvl}}$ or edge-triggered $\mathrm{M1}_{\mathsf{edge}}$), Schmitt-trigger with hysteresis (M2), and extremum-based (M3), whose edge-detector, Schmitt and extremum encoders are formalised in [8]— the figures below adopt the *single-threshold, edge-triggered* variant, denoted $\mathbf{M1}_{\mathsf{edge}}$:

- *One semantic threshold* θ per channel (the red dashed line in the figures, set to $+0.5$);
- *Edge-triggered events*: the kernel emits a confirmation token whenever the signal crosses θ *going up* in a sampling interval $(t-1, t]$, a refutation token whenever the signal crosses θ *going down*, and the no-observation token $\varnothing$ at every other tick. Formally for the consequent channel:

$$\sigma_{\mathrm{obs}}(t) \;=\; \begin{cases} Q^{\top} & \text{if } Q \uparrow \theta \text{ in } (t-1, t], \\ Q^{\perp} & \text{if } Q \downarrow \theta \text{ in } (t-1, t], \\ \varnothing & \text{otherwise;} \end{cases}$$

  analogously for P.
- *No dead-band as a primitive*: the kernel is blind to the absolute level of the signal and sensitive only to threshold transitions. The second (blue) threshold of M2 is retained in some figures as an optional *hysteresis floor* for engineering robustness against chattering noise, but plays no role in the formal observation semantics.

$\mathrm{M1}_{\mathsf{lvl}}$ is the level-triggered analogue: $\sigma_{\mathrm{obs}}(t) = Q^{\top}$ whenever $Q(t) > \theta$, etc. Under $\mathrm{M1}_{\mathsf{lvl}}$ every tick produces a binary token (above/below) and the silent-window class (0,0,0) is unreachable. M3 (extremum-based) is the limit where the threshold θ is replaced by the local running mean of the signal.

Constitutive ambiguity of edge-triggered observation. Under $\mathrm{M1}_{\mathsf{edge}}$, the kernel *cannot distinguish constant levels*: every signal trajectory whose graph remains strictly on one side of θ throughout the window produces the *same* observation (only the trigger token at `X`, followed by $\varnothing$ at every other tick of the window). In particular, the silent-window class (0,0,0) admits *two qualitatively distinct* physical realisations: *(A) Q stays above θ* (in its "True" level throughout the window) and *(B) Q stays below θ* (in its "False" level throughout the window). Both realisations yield the same token sequence; the sensor cannot tell them apart. This is not a defect of the encoding but the *constitutive condition* of edge-triggered sensing: only changes are observable, and constant states are by construction invisible. The Peled–Havelund *Quaestio* cell ($SFR \cap SFS$) captures this ambiguity *operationally* — Figure 3 therefore comprises two panels, one per realisation, accompanied by their (identical) token row.

This same family of ambiguities applies, in weaker forms, to every class: any pre-window trajectory of Q that does not cross θ within the window is observationally indistinguishable from any other such trajectory, regardless of its absolute level. The constitutive blindness of edge-triggered sensing is shared with several systems, among them event cameras (dynamic vision sensors) and integrate-and-fire neuron encoders; see [8] for the encoder taxonomy.

Common legend (applies to all eight figures). All eight figures Figures 1 to 8 share the same visual vocabulary, listed below. In-figure annotations are kept to a minimum: only the 4-band labels (`pre` | `bx` | `win` | `post`) appear inside the plots. Everything else is decoded here.

Visual vocabulary

$\tilde{p}(n)$ (antecedent / P-channel)	$\theta = +0.5$ (single semantic threshold)
$\tilde{q}(n)$ (consequent / Q-channel)	discrete sample of $\tilde{p}$ at integer tick
trigger event ($\tilde{p} \uparrow \theta$ in `X`)	discrete sample of $\tilde{q}$ at integer tick
rising-edge crossing of θ ($Q^{\top}$ or $P^{\top}$)	falling-edge crossing of θ ($Q^{\perp}$ or $P^{\perp}$)

`Pre` | `X` | `Body` | `Post`: 4-band partition (the `Body` band is $n \in \{1, \ldots, k\}$, the observation interval)

The horizontal axis is the discrete tick index $n \in \{-3, \ldots, 8\}$, partitioned into four bands by the trigger: `pre` (n<0), `bx` (n=0, the trigger tick), `win` ($n \in \{1, 2, 3\}$ with k=3; the in-band window $W = \{1, 2, 3\}$), `post` (n>3). In `pre` both signals oscillate strictly inside the dead-band (no Schmitt event fires). At n=0 the $\tilde{p}$-signal crosses θ_{high}: this *is* the trigger and defines `bx`. The token row below each plot records the Schmitt output $\sigma_{\mathrm{obs}}(n)$ at integer ticks $n \in \{0, \ldots, 8\}$ (with Q-priority at co-occurrences per the observation-alphabet definition [9]); pre-band ticks emit $\varnothing$ and are omitted from the row. The feature triple (A, B, C) is the existential aggregation over the in-band window $W = \{1, 2, 3\}$.

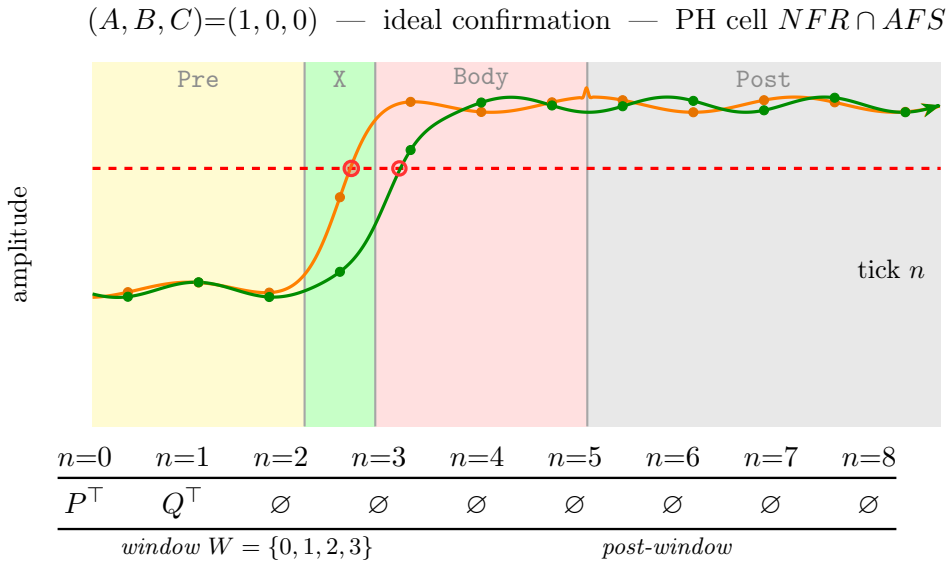


FIGURE 1. Class (1,0,0) — the canonical *ideal confirmation* under edge-triggered single-threshold encoding (M1_{edge}). *Pre.* Both signals oscillate below the threshold θ, on slightly different frequencies. Since neither crosses θ, no event fires — the kernel is present but *silent*. Note that the absolute level of the pre-band oscillation is not observable: any pre-band trajectory that stays below θ throughout produces the same kernel state, regardless of its exact shape. *Bandx (X).* The antecedent P rises and crosses θ going up during the bandx interval. By the trigger convention of the observation-alphabet definition [9], this rising edge of P is recorded as the token $P^\top$ at tick $n = 0$; the observation window opens. *Body.* P continues to rise asymptotically and does *not* fall back through θ during the window, so no $P^\perp$ event fires ($C = 0$). The consequent Q, hitherto oscillating below θ, rises smoothly and crosses θ going up at a tick interior to the window. The kernel records this rising edge as the single $Q^\top$ token in the token row; Q then continues above θ without re-crossing, so no further Q-channel events fire ($B = 0$). *Post.* Both signals remain above θ; with no further crossings, the kernel records only ∅. *Verdict.* Existential aggregation gives $(A, B, C) = (1, 0, 0)$, placing the trace in $NFR \cap AFS$ — the open dense cell of properties admitting a finite confirmation but no finite refutation [7]. *Ambiguity note.* The figure depicts one canonical realisation, but any pre-window trajectory of Q that ends below θ at $n = 0$ and crosses θ once within the window produces the *same* observation; the in-pre level is invisible to the sensor.

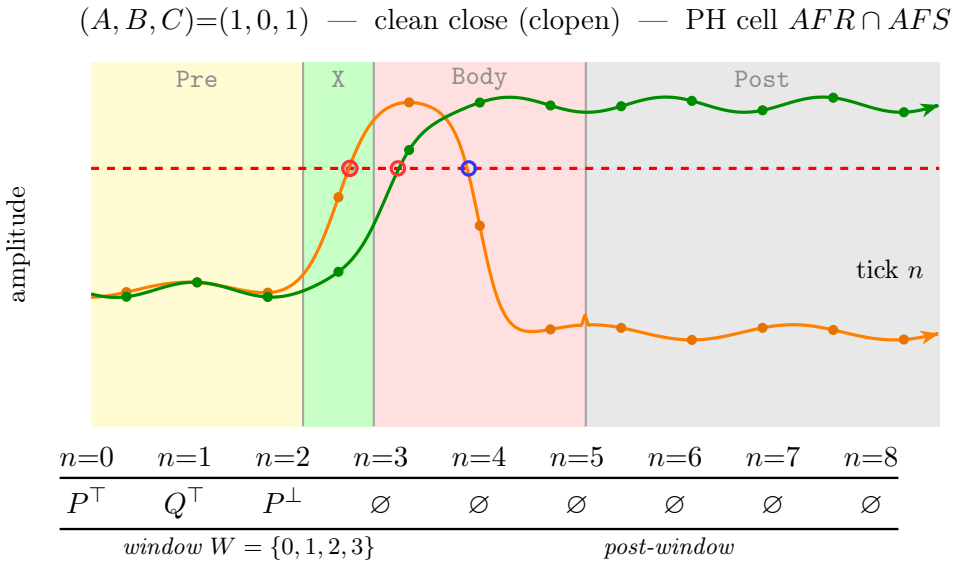


FIGURE 2. Class (1,0,1) — *clean close*, the unique *clopen* cell of the PH partition. *Pre.* As in (1,0,0), both signals oscillate below θ and the kernel emits nothing. *Bandx (X).* P rises through θ inside the bandx interval; the trigger token $P^{\top}$ is recorded at $n = 0$. *Body.* Two distinct edge events now unfold in succession, both visible in the window. First, Q rises through θ at an interior tick of the window: the kernel records a single $Q^{\top}$ token (whence $A = 1$, $B = 0$ because Q never re-crosses θ). Then, still within the window, P bends sharply downwards and crosses θ going down: the kernel records a single $P^{\perp}$ token, a finitely observed *retraction* of the antecedent (whence $C = 1$). *Post.* Both signals are now on the side of θ that suppresses further events; the kernel records only $\varnothing$. *Verdict.* Aggregation gives $(A, B, C) = (1, 0, 1)$, placing the trace in $AFR \cap AFS$ — the unique *clopen* cell, the only cell of the PH partition in which *both* verdict axes are finitely decided within the window. *Ambiguity note.* The figure depicts one canonical realisation. Several variations produce the *same* token row: (i) reordering of the events (Q rise and P fall) *within their ticks* is unobservable; (ii) the exact pre-window levels of P and Q are immaterial provided neither crosses θ during pre; (iii) at the moment P crosses θ downward, the kernel could in principle be observing either channel — the priority rule of the observation-alphabet definition [9] resolves ties in favour of the Q channel, which is why, in this design, $P^{\perp}$ is observed at $n = 2$ only after Q has already plateaued and is no longer producing edges.

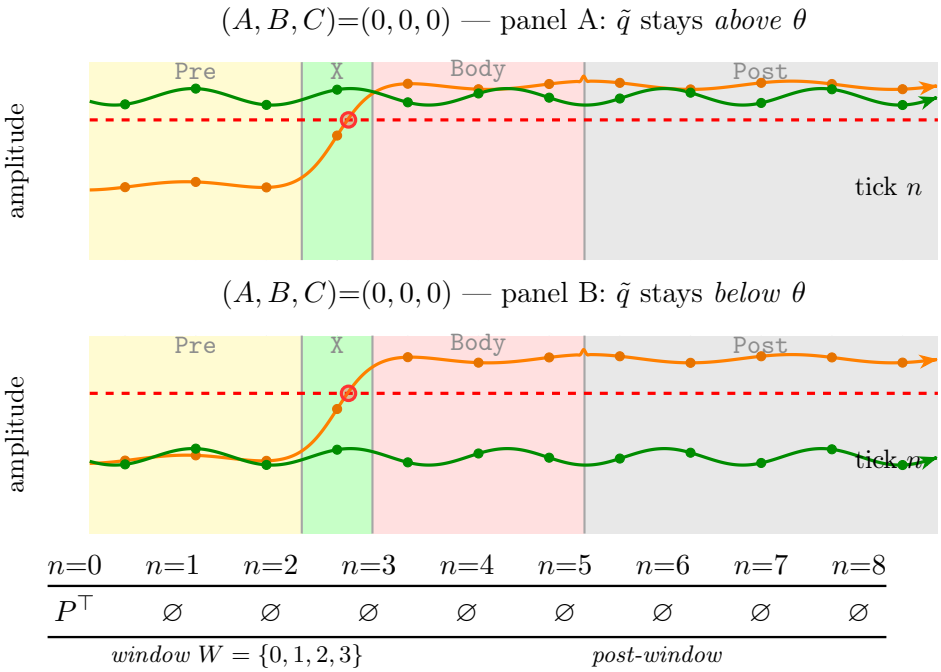


FIGURE 3. Class (0,0,0) — the *silent window* and the *constitutive ambiguity* of edge-triggered observation. The two panels above depict two qualitatively different physical realisations. *Panel A* shows a consequent that is *already True* when the trigger fires and remains True for the whole observation: $\tilde{q}$ stays strictly above θ, oscillating quietly in its "high" regime. *Panel B* shows the symmetric situation: a consequent that is *already False* when the trigger fires and remains False throughout: $\tilde{q}$ stays strictly below θ. The antecedent P is identical in both panels: it oscillates below θ in pre, crosses θ at bandx (the trigger), and saturates above θ thereafter without ever falling back. Neither panel produces any Q-channel edge crossing of θ within the window. Consequently the kernel emits the *same* token row for both panels: a single $P^{\top}$ at the trigger, followed by $\varnothing$ at every subsequent tick. The aggregated feature triple is $(A, B, C) = (0, 0, 0)$ in both cases. The Peled– Havelund partition places both realisations in the *Quaestio* cell $SFR \cap SFS$: a cell of double "Sometimes" uncertainty in which neither verdict axis is finitely decided. This is not a defect of the encoding but the *constitutive condition* of edge-triggered sensing — only changes are observable, and constant states are by construction invisible. (0,0,0) is therefore the figure that makes *observational vacuity* concrete: classical LTL collapses both panels to "vacuously true" (no antecedent retraction), but the two underlying realities are operationally distinct, and the kernel cannot tell them apart. The Quaestio cell formalises this gap. The hysteresis relaxation Θ_2 (Section 5.4) addresses the ambiguity by introducing a second threshold (the hysteresis floor) that lifts the encoding to M2 [8] and partially distinguishes the two realisations.

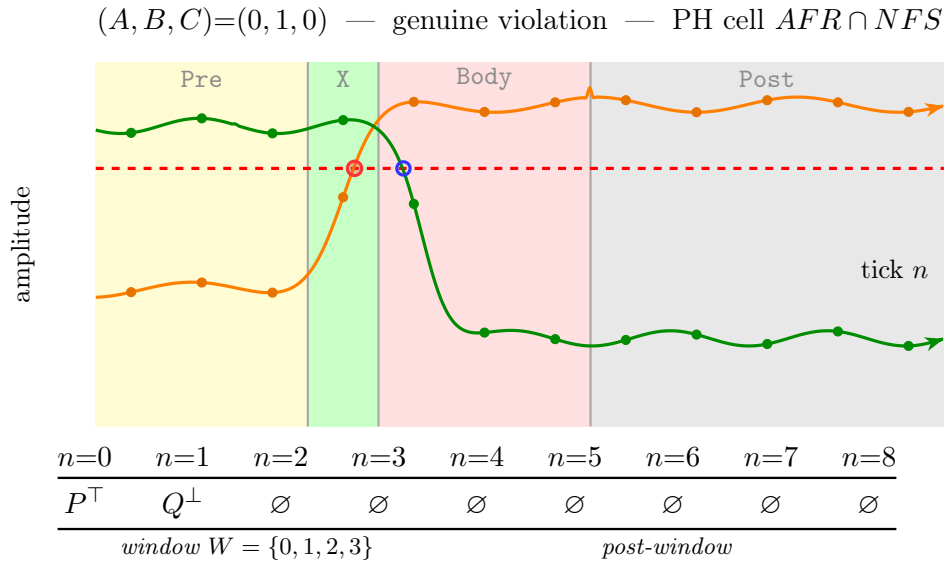


FIGURE 4. Class (0,1,0) — *genuine violation*, exhibited through the dual encoding template: (1,0,0) shows "Q becomes True"; here (0,1,0) shows "Q becomes False". *Pre.* The antecedent P oscillates below θ as in the other classes. The consequent Q, however, oscillates *above* θ: the system is in a state where the consequent currently holds, but the antecedent has not yet fired. No crossings, no events. *Bandx (X).* P rises through θ going up; the trigger token $P^{\top}$ is recorded at $n = 0$. *Body.* Immediately after the trigger, the consequent Q begins to descend. At an interior tick of the window, Q crosses θ going down. The kernel records this falling edge as a single $Q^{\perp}$ token — the violation observation (whence $B = 1$, $A = 0$ because no rising edge of Q occurred in the window). P remains above θ throughout (no $P^{\perp}$; $C = 0$). *Post.* Q remains below θ and P above; no further crossings; the kernel records only $\varnothing$. *Verdict.* Aggregation gives $(A, B, C) = (0, 1, 0)$, placing the trace in $AFR \cap NFS$ — the closed cell of properties that admit a finite refutation but no structural prospect of a finite confirmation [7]. *Ambiguity note.* The exact pre-window history of Q above θ is unobservable: any trajectory that approaches the window-start tick from above θ produces the same observation. Furthermore, a single falling crossing inside the window produces the same $Q^{\perp}$ token regardless of where exactly it occurs — the token row records *which* tick the crossing was observed at, but not the precise sub-tick time of the crossing.

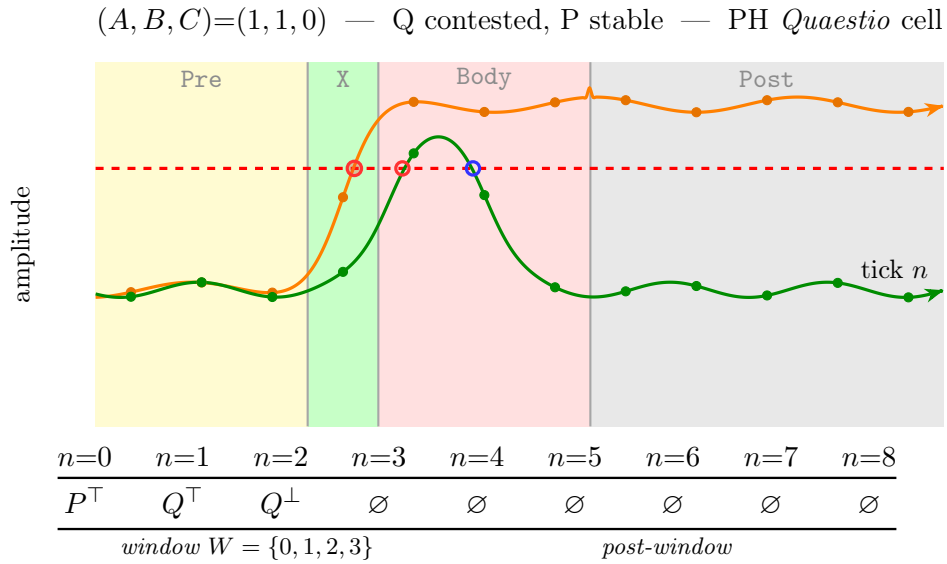


FIGURE 5. Class (1,1,0) — the consequent is briefly observed to "become True" and then "become False" again within the same window; the antecedent remains active throughout. *Pre.* P and Q both oscillate below θ; no events. *Bandx.* P rises through θ: trigger token $P^\top$ recorded. *Body.* Two Q-channel events unfold in sequence: Q crosses θ going up (token $Q^\top$, signalling tentative confirmation), and then later in the same window Q crosses θ going down (token $Q^\perp$, signalling that the confirmation has been retracted). P remains above θ throughout: no $P^\perp$ event fires. *Post.* P stays high, Q stays low; no further events. *Verdict.* $(A, B, C) = (1, 1, 0)$, placing the trace in the *Quaestio* cell of the Peled–Havelund partition. The classical implication has been both confirmed and refuted within the same window, leaving the monitor with contested evidence. *Ambiguity note.* A dual variant exists in which Q starts *above* θ and the order of events is reversed: $Q^\perp$ first (the initially-True consequent falls), then $Q^\top$ later (it recovers). The two scenarios are physically distinct but both land in the same Quaestio cell; their token rows differ in event order.

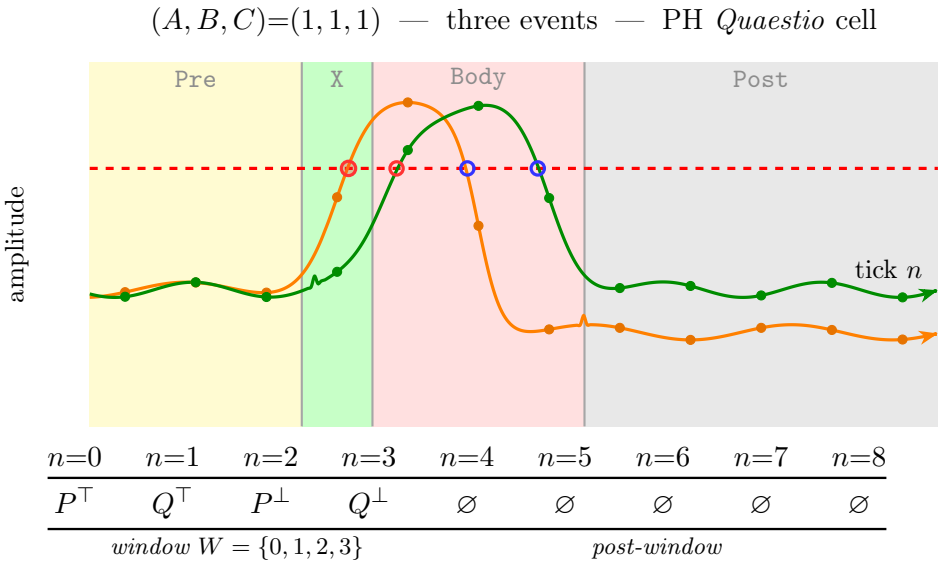


FIGURE 6. Class (1,1,1) — all three observable events occur within the same window: a Q-confirmation, an antecedent retraction, and a Q-refutation. This is the most event-rich of the eight classes. *Pre.* Both signals oscillate below θ. *Bandx.* P rises through θ: trigger $P^{\top}$. *Body.* Three events unfold in succession. Early in the window, Q rises through θ (token $Q^{\top}$, tentative confirmation). Some ticks later, P falls through θ (token $P^{\perp}$, antecedent retraction). Shortly afterwards Q also falls through θ (token $Q^{\perp}$, refutation). *Post.* Both signals are now low; no further events. *Verdict.* $(A, B, C) = (1, 1, 1)$, also a Quaestio cell of PH. This is the combinatorially richest configuration — nothing about the implication has stabilised within the window. *Ambiguity note.* The three events admit six possible orderings in principle; if Q starts above θ, the Q-channel order is forced to be $Q^{\perp} \to \ldots \to Q^{\top}$. All orderings land in the same Quaestio cell but produce distinct token rows.

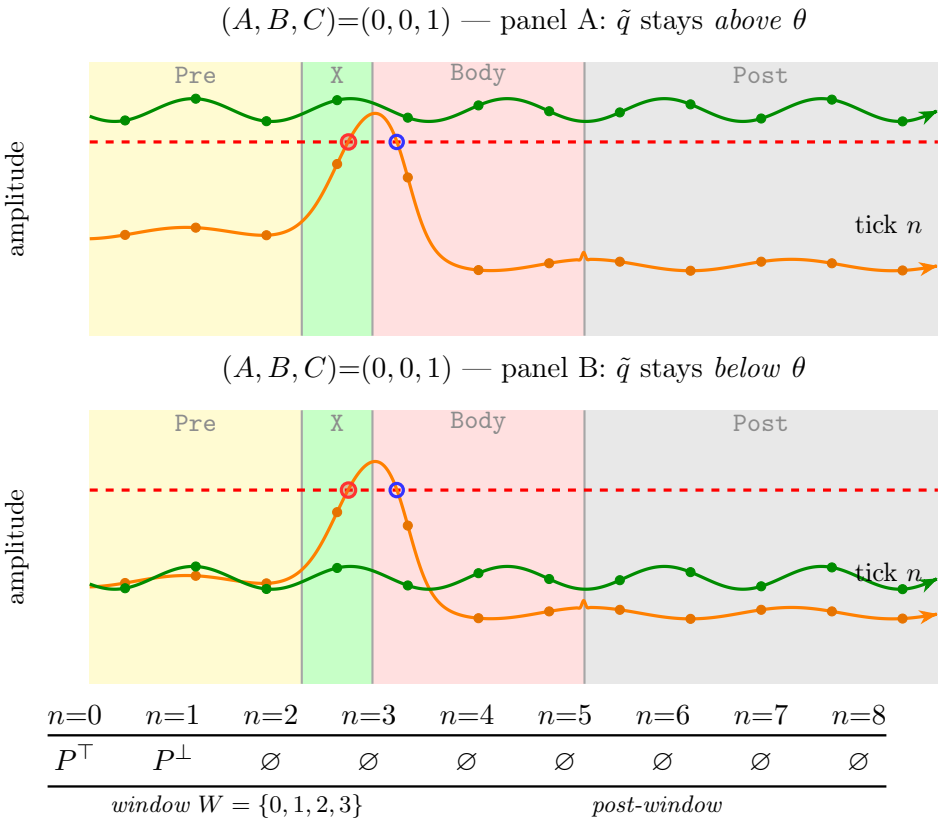


FIGURE 7. Class (0,0,1) — the antecedent retracts inside the window before the consequent does anything observable; the consequent emits no edge event. Like (0,0,0) this class exhibits the *constitutive ambiguity* of edge-triggered sensing: the kernel cannot tell whether Q was True or False throughout the window, because no Q-edge occurs. Two realisations (panel A with Q above θ, panel B with Q below) produce the *same* token row. *Pre.* P below θ; Q in one of its two possible constant levels. *Bandx.* P rises through θ: trigger $P^{\top}$. *Body.* P subsequently falls back through θ (token $P^{\perp}$). Q never crosses θ; no Q-event fires. *Verdict.* $(A, B, C) = (0, 0, 1)$, placing the trace in $SFR \cap NFS$ of the PH partition — finite refutation by retraction, but no structural prospect of a finite confirmation, plus the silent-window ambiguity on Q.

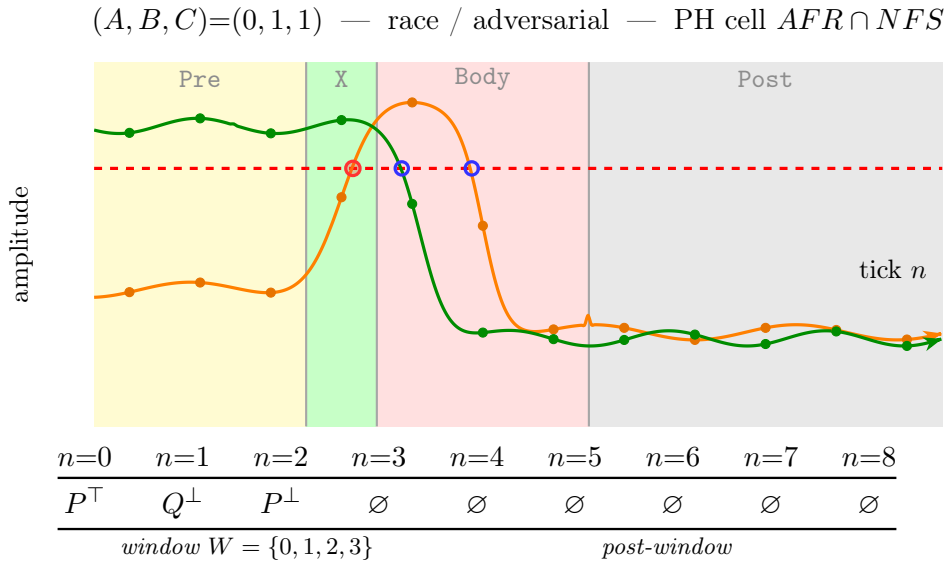


FIGURE 8. Class (0,1,1) — both the consequent and the antecedent collapse within the window: two falling events, no rising events. Combinatorially the richest of the A=0 classes. *Pre.* P oscillates below θ; Q oscillates above θ (the consequent currently holds). *Bandx.* P rises through θ: trigger $P^{\top}$. *Body.* Two falling events follow. First, Q crosses θ going down: token $Q^{\perp}$ (the consequent fails). Then, later in the window, P also crosses θ going down: token $P^{\perp}$ (the antecedent retracts as well). *Post.* Both signals below θ; no further events. *Verdict.* $(A, B, C) = (0, 1, 1)$, placing the trace in $AFR \cap NFS$ — the same PH cell as (0,1,0), but with the additional observation of antecedent retraction. This is the operational signature of a system in "compound failure mode". *Ambiguity note.* The order of the two falling events (Q-fall first vs. P-fall first) is observable from the token row; the figure depicts the Q-first canonical variant. Both orderings land in the same PH cell.

5. Formal Specification and its Property Classes

We now specify the self-tuning threshold monitor in the style of distributed algorithms: properties are stated first, and the algorithms of the later sections are read as their realisations rather than as the primary objects. This section specifies the *monitor's* eight tuning obligations — its world model together with the licensed relaxations — and does *not* re-formalise the implication property of the core paper [9]: the objects differ, a trace property there versus a monitor-obligation here. The monitor is the triple $\mathcal{M} = (\mathcal{K}_{\boldsymbol{\theta}}, \mathbb{V}, \mathcal{T})$ of observation kernel, verdict domain, and tuning-operation catalogue, observing the world through a channel $\mathcal{C}(L, S, D, J)$ under a declared observation theory $\mathcal{O}$ — the kernel, the channel, and the observation ladder are all those of the core paper [9]. Its per-cycle state is $\sigma_n = (\theta_n, k_n, \text{margins}, \hat{\phi}_n)$: the active threshold, the active window length, the gating margins, and the current phase estimate.

The specification is written in the rely/guarantee discipline. The *rely* collects the assumptions the monitor makes about its environment: bounded drift per epoch; the physical premises CC1–CC4 of the core paper [9]; a declared observation rung that is fixed for the duration of an epoch; and fairness of the monitoring cycles. Each *guarantee* below carries its LTL schema and its class in the four-family intersection algebra [7], together with the set d of finitely reachable definitive verdicts — the very classification machinery that the monitor itself implements. The class is a property of the *trace* (the ω-language the obligation denotes), whereas d is a property of the *monitor* that decides it; these two axes, which the trace–monitor cross-axis independence remark of [7] keeps apart, are not conflated below. We write $d = 1$ when a single-sided definitive verdict (a finite violation certificate) is reachable and $d = 0$ when no finite prefix decides the property.

A word on the status of these annotations, in the interest of honesty about what is and is not settled here. The eight obligations below are, first and foremost, the *constructive scaffold* of the design: each is selected because the monitor must discharge it, each is realised by one of the algorithms of the later sections, and each is shown to sit under a safety-dominant cell that keeps it runtime-enforceable. In that expository role the definitions already do their work — they select, organise, and realise the design, and they expose what the monitor owes its environment. The four-family class and the verdict degree d attached to each item are offered in the *same* constructive spirit: as a *provisional* reading rather than a definitive classification of each formula's exact position. The reading is exact wherever the named predicate resolves to the bounded, prefix-decidable condition the monitor actually tests — as it does where both the shape and the predicate are simple (for instance S4's $\mathbf{F}\,\mathbf{G}$ legit is a genuine persistence read-off in $\mathsf{L} \cap \mathsf{K}$ once legit is taken as a state predicate) — and it is to be *refined* wherever a predicate such as $\text{closed}_{\mathcal{O}}$, $\text{silent}_{\mathcal{O}}$, $\text{cert}_{\mathcal{O}}$, feasible, or legit hides more temporal structure than a single class name can carry. A fully rigorous classification would unpack each such predicate — bounded-window closure, silence over the window, certification, feasibility of a class, legitimacy of the loop state — and confirm that it is indeed prefix-decidable before pronouncing a formula's exact position; we leave that unpacking to later refinement. We therefore read the annotations as a constructive, provisional map of the specification, not as its final classification: useful now for selecting, organising, and realising the design, and honest about the work a precise placement of each formula still requires.

The monitor's observables can be organised as a point/line/curve ontology — observation points (temporal-depth-0 atoms), threshold lines (invariantly held point-properties),

and regularity-constrained curves, with crossings and touches as their incidences — expressed entirely in LTL; its point stratum reconciles with the *point semantics* of the core paper [9]. Since no selection, convergence, or pathology result below depends on this stratification, we omit its development here and refer the reader to the core paper for the point-semantics bridge, proceeding directly to the specification.

5.1. **The specification.** Each entry lists the property name, its LTL schema, its four-family class, its verdict degree d, and the assurance route by which the obligation is discharged. We abbreviate the O-dilated closure and silence predicates of the core paper as $\mathrm{closed}_{\mathcal{O}}$, $\mathrm{silent}_{\mathcal{O}}$, and $\mathrm{cert}_{\mathcal{O}}$, and write $\mathbf{W}$ for the weak-until operator.

S1 — Verdict soundness (safety, $\mathsf{S} \cap \mathsf{K}$, $d = 1$):

$$\mathbf{G}\big(\,\mathrm{emit}(\bot, x) \;\rightarrow\; \mathrm{closed}_{\mathcal{O}}(x) \wedge \mathrm{silent}_{\mathcal{O}}(x)\,\big).$$

A definitive $\bot$ is emitted only at the $\mathcal{O}$-dilated closure of a silent window; a wrong $\bot$ is a finite bad prefix of the specification, and is therefore runtime-checkable.

S2 — Bounded verdict completeness (safety by boundedness, $\mathsf{S} \cap \mathsf{K}$, $d = 1$):

$$\mathbf{G}\big(\,\mathrm{cert}_{\mathcal{O}}(x) \;\rightarrow\; \mathbf{F}_{\leq k_{\mathcal{O}}+\varepsilon}\,\mathrm{emit}(\bot, x)\,\big).$$

The response is deliberately *bounded*, chosen so that the obligation stays inside the enforceable cell; the unbounded variant would be a response/liveness property and would leave that cell.

S3 — Non-circularity of tuning (safety over the two-layer trace, $\mathsf{S} \cap \mathsf{K}$, $d = 1$):

$$\mathbf{G}\big(\,\mathrm{latch}(\theta', t) \;\rightarrow\; \mathrm{crossing_cont}(s_q, \theta', t)\,\big).$$

Every latched crossing following a tuning action corresponds to a θ'-crossing of the *same* continuous signal. It is enforced by a pre-application guard (the feasibility clamp) and verified offline (notebook `nb04_schmitt_gallery_reachability`): the continuous layer is not visible to $\mathcal{M}$ at runtime.

S4 — Convergence (LIVENESS — persistence in the Manna–Pnueli hierarchy [20], $\mathsf{L} \cap \mathsf{K}$, $d = 0$):

$$\mathbf{F}\;\mathbf{G}\;\mathrm{legit}.$$

From any initial kernel state, and under a stationary channel, the loop eventually reaches and holds the legitimate state (phase error below tolerance, faithful reading regime). The property is zero-sided: no finite prefix refutes or confirms it — which is exactly why it is discharged *offline* by the Robbins–Monro theorem (Theorem 8.8, local, under the symmetry and sign/regression hypotheses (C3, C5)) and WITNESSED on a finite simulation prefix (global convergence conjectured; see Remark 8.9), never by runtime checking.

S5 — Closure (safety, $\mathsf{S} \cap \mathsf{K}$, $d = 1$):

$$\mathbf{G}\big(\,\mathrm{legit} \wedge \neg\mathrm{drift} \;\rightarrow\; \mathbf{X}\;\mathrm{legit}\,\big).$$

Once legitimate and absent drift, the loop stays legitimate; checkable per cycle.

S6 — Calibration validity (safety invariant, $\mathsf{S} \cap \mathsf{K}$, $d = 1$):

$$\mathbf{G}\big(\,k_n \;\geq\; \mathrm{bound}(\hat{\Delta}_n)\,\big).$$

The active window always respects the calibration bound for the current channel estimate.

S7a — Feasibility honesty (safety, $\mathsf{S} \cap \mathsf{K}$, $d = 1$):

$$\mathbf{G}\big(\,\mathrm{claim}(C_j) \to \mathrm{feasible}(C_j)\,\big).$$

No window class is claimed outside the feasible orbit of the current tuning. The class is a direct skeleton read-off: the invariant shape $\mathbf{G}(p \to q)$ lies in $\mathsf{S} \cap \mathsf{K}$.

S7b — Probe-gated non-escalation (safety, $\mathsf{S} \cap \mathsf{K}$, $d = 1$):

$$\mathbf{G}\big(\,\mathrm{detected}((0{,}0{,}0)) \to (\neg \mathrm{escalate})\ \mathbf{W}\ \mathrm{probe}\,\big).$$

On the ambiguous silent-window class (0,0,0), the monitor does not escalate until a touch-probe resolves it. The class is a direct skeleton read-off: the guarded weak-until ordering $\mathbf{G}(p \to (\neg r)\ \mathbf{W}\ s)$ lies in $\mathsf{S} \cap \mathsf{K}$.

S8 — Epistemic transparency (safety, $\mathsf{S} \cap \mathsf{K}$, $d = 1$):

$$\mathbf{G}\big(\,\mathrm{emit_def} \ \to\ \mathrm{labeled}(\theta_n, \hat{\Delta}_n, \mathcal{O})\,\big).$$

Every definitive verdict carries its three coordinates; verdicts are never emitted bare. This operationalises the core paper's observation-relative monitorability theorem as a protocol obligation.

Remark 5.1 (Safety-dominance by design). Every runtime-enforceable obligation above is drawn from the $d = 1$ cell. By the verdict-reachability read-off of the core paper [9], only properties with a finite violation certificate can be refuted by a runtime check; the specification is therefore built *inside* the theory the monitor implements, rather than layered on top of it.

Remark 5.2 (Liveness is proved, not monitored). S4 is the unique zero-sided property of the suite, and precisely the one discharged by theorem plus simulation. The pair (S4, S5) is a self-stabilisation specification in the classical sense [10, 2]: convergence together with closure, with the faithful reading playing the role of the legitimate state.

Remark 5.3 (Rely/guarantee and re-arming). When the rely is broken — drift resumes — S5 lapses and S4 re-arms; the suite is designed to recur. The only environmental liveness assumption is fairness of the monitoring cycles.

5.2. **Realisation map.** Table 1 maps each specified property to the algorithm that realises it, the proof obligation that realisation incurs, and the current epistemic status of that obligation.

5.3. **Two supporting algorithms.**

Algorithm 1 (Calibration clamp, realises S6). Each cycle:

(1) update the channel estimate $\hat{\Delta}_n$ from the timing-error statistics;
(2) set $k_{n+1} := \max\big(k_n,\ \mathrm{bound}(\hat{\Delta}_n)\big)$;
(3) if k grew, re-anchor open windows conservatively (no retroactive shrinking).

Proof (one line). bound$(\cdot)$ is monotone in $\hat{\Delta}$ and step (2) never decreases k below it; hence the invariant S6, $k_n \geq \mathrm{bound}(\hat{\Delta}_n)$, holds inductively. [PROVEN]

Property	Realising algorithm	Proof obligation	Status
S1, S2	$\mathcal{O}$-parametric window automaton	soundness / completeness	PROVEN on the synchrony rung [9]; rung-parametric variant CONJECTURED (to follow by consolidating the per-rung ladder propositions of [9])
S3	pre-application feasibility clamp (linear-time synthesis)	guard correctness	PROVEN [8]
S4	self-tuning loop (Section 8)	a.s. convergence	PROVEN (local, basin) under the sign/regression condition (C5); global [CONJECTURED]
S5	margin adaptation + step-size decay	local contraction	WITNESSED (+ small-signal analysis, Section 8.7)
S6	Algorithm 1	invariant preservation	PROVEN (trivial)
S7a	constructive realisability / feasibility check	feasibility soundness	single-signal synthesis PROVEN [8]; two-signal realisability WITNESSED (Proposition A.1)
S7b	touch-probe subroutine	probe-before-escalate / dichotomy correctness	PROVEN (Proposition A.3)
S8	verdict-labelling protocol (Algorithm 2)	by construction	design obligation [HEURISTIC calibration of label granularity]

TABLE 1. Realisation map: each property class of Section 5.1 and the algorithm, proof obligation, and epistemic status that discharge it. The suite is safety-dominant; the sole liveness obligation (S4) is proved offline, never monitored.

Algorithm 2 (Verdict labelling, realises S8). On emission of a definitive verdict, attach the tuple $(\theta_n, \hat{\Delta}_n, \mathcal{O}\text{-rung id}, \text{feasibility flag from the S7a check})$; consumers *must* treat unlabeled verdicts as inconclusive. This is a design obligation discharged by construction.

5.4. **The world-model axioms and the relaxation lattice.** The rigid base monitor is characterised by nine structural axioms about its world model, grouped as threshold (Θ), grid (Γ), and eventing (Σ) axioms; the tuning operations they license are catalogued in Table 10. Their purpose is economy: *every* elementary tuning operation of that catalogue is the licensed relaxation of exactly one axiom, and each relaxation *adds* a power to the tuning algorithm. The axioms thus organise the tuning space as a *relaxation lattice* whose bottom is the rigid monitor and whose one forbidden direction is the single inviolable axiom. Axiom maintenance is itself trivially a safety obligation — $\mathbf{G}$ (axiom-holds), class $\mathsf{S} \cap \mathsf{K}$ — which is why the base monitor is a safety device to begin with. The (A,B,C) classification exists

to be *controlled*: the observation class is the observable that the self-tuning loop reads and steers, driving the monitor toward a *faithful* capture of the causal structure $P^{\uparrow} \to Q^{\uparrow}$ fixed by the core paper [9] — in short, tuning the monitor to produce the right traces.

Θ_1 — **Threshold constancy:** $\theta(t) = \theta$: the threshold is a horizontal line. *Relaxation:* a time-varying threshold $\theta(t)$. *Power:* moving-threshold sensitivity / per-channel threshold shift (T1) [8]. *Parameter:* θ_p, θ_q. *Effect:* recovers sub-threshold crossings — the (0,0,0)→(1,0,0) edge of the readability orbit.

Θ_2 — **Threshold continuity:** $\theta(t)$ has no jumps. *Relaxation:* a piecewise-constant threshold with jumps. *Power:* the hysteresis pair $(\theta_{\text{up}}, \theta_{\text{down}})$ — the Schmitt encoder of [8], whose external realisation keeps the crossing capability $\kappa = (1, 1, 0)$ and whose internal realisation reaches $(3, 1, 1)$. *Parameter:* the hysteresis margins M^-, M^+. *Effect:* a dead-band against glitches; partially separates the two silent-window realisations.

Θ_3 — **Threshold uniqueness:** one decision line per channel. *Relaxation:* auxiliary lines. *Power:* the touch probe at $\theta - \delta$ deciding the (0,0,0) dichotomy (Proposition A.3), and in general the augmented monitor $(\theta_u, \theta_d, \pi)$ [8]. *Parameter:* δ, the probe line. *Effect:* resolves δ-escapable from δ-trapped silent windows — a distinction invisible to any single line.

Γ_1 — **Grid uniformity:** ticks are equally spaced (Δt constant). *Relaxation:* refined / variable sampling. *Power:* oversampling and sub-bin edge recovery. *Parameter:* Δt. *Effect:* separates same-bin events, defeating the priority-max collapse.

Γ_2 — **Phase rigidity:** the grid phase is fixed. *Relaxation:* an adjustable phase. *Power:* the self-tuning loop itself — the Stage-C PLL update (Section 8) is the licensed violation of phase rigidity, governed by S4/S5. *Parameter:* $\hat{\phi}$. *Effect:* the regime migration $(A{=}0) \to (A{=}1 \wedge (B \vee C)) \to (1,0,0)$.

Γ_3 — **Horizon fixity:** the window bound k is fixed at design time. *Relaxation:* an adaptive k. *Power:* the calibration clamp (Algorithm 1, S6) and the window-widening operations. *Parameter:* k. *Effect:* avoids premature falsification while respecting the calibration bound.

Σ_1 — **Event completeness:** every spec-conformant behaviour of the signal (a genuine crossing of the effective threshold) generates its event. *Relaxation:* selective eventing. *Power:* debounce / minimum-pulse suppression ($\tau_{\min}$). *Parameter:* $\tau_{\min}$. *Effect:* suppresses glitch-induced retractions (the $C\colon 1 \to 0$ migrations).

Σ_2 — **Event soundness:** every emitted event corresponds to a real behaviour of the continuous signal. *No relaxation exists.* This is the unique inviolable axiom: its "relaxation" would be the minting of events, exactly what non-circularity (S3) forbids. The lattice has a floor.

Σ_3 — **Latched memory:** the crossing modality carries one bit of persistent state. *Relaxation:* a memoryless reading. *Power:* the touch modality (the modality bit π of the augmented monitor); the crossing/touch duality is a memory-axiom relaxation [8]. *Parameter:* π. *Effect:* reads extrema instead of transitions (whence the count asymmetry $N_{\mathsf{X}} \leq N_{\mathsf{T}} + \mathbf{1}[\text{the window ends inside an excursion}]$, the plateau-form bound of [8]; the boundary tick is necessary because a single monotone crossing has $N_{\mathsf{X}} = 1$ while its interior carries no completed touch).

Remark 5.4 (The Σ_3 relaxation is a proven latch construction, WITNESSED). The crossing/touch memory-axiom relaxation of Σ_3 is not a slogan: it is realised by two explicit constructions that [8] proves correct. In one direction, a touch monitor augmented with

a one-bit alternation latch simulates the crossing monitor exactly at the level of the event sequence (type, order, and counts, timing aside); in the other, a gated crossing monitor simulates the touch monitor. The Σ_3 bit *is* that latch, so relaxing latched memory to a memoryless touch reading, and restoring it by re-latching, are exactly those two proven simulations rather than an informal analogy. The grounding is WITNESSED.

Remark 5.5 (Relaxation calculus). Each elementary tuning operation is the relaxation of exactly one axiom, and the universal-readability orbit (Proposition A.1) is generated by composing at most two relaxations (three for the two exceptional pairs): the orbit is the join-closure of the lattice above the floor.

Remark 5.6 (The lattice top is the general capability join, WITNESSED). The join-closure of Remark 5.5 is not merely an informal lattice: its top element is the capability triple $\kappa = (3, 2, 1)$ of the general augmented monitor of [8], which that paper proves to be the componentwise join over its full hub-reduction table of modality triples — *not* merely the join of the two crossing/touch primitives, whose own join $(1, 1, 0) \vee (3, 0, 1) = (3, 1, 1)$ lacks the second state bit $b = 2$. That extra bit is supplied by the DIRRES ternary engaged state (the multiplicity/alternation axis beyond the two primitives), so $(3, 2, 1)$ is the join over the whole table, one rung above the primitive pair. Read through the capability calculus (w, b, k), each power is a bounded coordinate increment $(\Delta w, \Delta b, \Delta k)$ towards that top, and the per-knob hub-reduction table of [8] records exactly which component each frozen knob drops from the join. Grounding ME's relaxation lattice in that proven capability join is WITNESSED.

Remark 5.7 (The inviolable floor). Σ_2 is the unique axiom with no licensed relaxation; a device relaxing it would not be a more powerful monitor but a forger of evidence. Every other axiom's relaxation is truth-preserving precisely because Σ_2 stays.

Theorem 5.8 (The retuning calculus cannot fabricate a token: Σ_2 is closed under re-parameterisation, PROVEN)**.** No-synthesis premise (declared modelling assumption). *Every sample is a real evaluation $\tau(t_n)$ at a tick time t_n; no tuning operation interpolates or synthesises samples. In particular T_7 (Table 10) is an* acquisition-interval *change — it re-acquires the physical signal τ at the new Δt —* not *a digital upsampling of a fixed buffer. (A sinc/spline reconstruction of a captured buffer can produce a synthetic strict peak by ringing overshoot, with no real interior extremum of τ, fabricating a touch token; the acquisition reading forbids this. Crossing tokens survive interpolation by the intermediate value theorem, so the premise is load-bearing only for touch tokens.)*

Let τ be a continuous signal sampled at the monitor ticks, and let $\mathsf{Sound}(\tau)$ *be the set of position-typed tokens* (n, type)*,* type $\in \{Q^{\uparrow}, Q^{\downarrow}\}$*, for which the real signal exhibits, at the firing neighbourhood of n, a genuine threshold transit (a crossing,* [8]*) or a genuine strict extremum (a touch,* [8]*) certifying* type*. Then, under the no-synthesis premise, no composition of the re-parameterisation operations $T_1, \ldots, T_8$ of Table 10 with the band-word operation* EXCLUDE *of Definition 8.11 can emit a validation/invalidation token — a crossing or a touch — that is not backed by a real crossing or a real strict extremum of τ: every such composition maps any observed word to a word whose token set is contained in* $\mathsf{Sound}(\tau)$*. Equivalently — the load-bearing statement — the retuning cannot* manufacture *an event the signal does not contain; the event-soundness floor Σ_2 (Remark 5.7) is closed under the re-parameterisation calculus (T_1–T_8,* EXCLUDE*).*

Proof. By the crossing and touch semantics of [8] each token emitted by a kernel configuration is a *deterministic function of the real signal* τ evaluated against the current parameter vector $\boldsymbol{\theta} = (\theta, \hat{\varphi}, k, \tau_{\min}, M^{\pm}, \Delta_{\text{hyst}}, \Delta t)$. A crossing token fires at position n only when two real samples straddle the wall, $\tau_{n-1} < \theta \leq \tau_n$ (or the downward dual); a touch token fires only at a real strict sampled extremum, $\tau_{n-1} < \tau_n \geq \tau_{n+1}$ with $\tau_n \geq \theta$ (or the dual). Because τ is continuous, a straddling pair witnesses — by the intermediate value theorem — a genuine crossing of the continuous signal in $[(n{-}1)\Delta t,\, n\Delta t]$, and a strict sampled extremum interior to $[(n{-}1)\Delta t,\, (n{+}1)\Delta t]$ witnesses a genuine local extremum there (the extreme value theorem certifies its *existence*; strictness is the touch convention of [8], and a plateau maximum is still a real feature backing the token). Here the no-synthesis premise is load-bearing for touches: because each τ_n is a real acquisition, a strict sampled extremum reflects a real interior extremum of τ, not a reconstruction artefact (sinc/spline ringing); for crossings the premise is not needed — the intermediate value theorem alone certifies the transit. (The intersample interval is the excluded interval of this discrete-to-continuous reading; the intermediate value theorem reintegrates it as *existence*, not timing — the residual timing slack is a Σ_1/completeness matter, not a Σ_2 one.) Hence every token a configuration emits lies in $\mathsf{Sound}(\tau)$ for the parameters in force.

We show the invariant "current word $\subseteq \mathsf{Sound}(\tau)$" is preserved, by induction on the length of the composition. *Base:* the initial reading is emitted by one configuration, so its tokens lie in $\mathsf{Sound}(\tau)$ by the paragraph above. *Step, re-parameterisation T_j ($j \in \{1, \ldots, 8\}$):* each T_j alters only a coordinate of $\boldsymbol{\theta}$ and leaves τ itself untouched (Table 10). It therefore changes only *which* real transits/extrema of τ satisfy the firing predicate — which fall in-window $(k, \hat{\varphi}, \Delta t)$, clear the wall $(\theta, M^{\pm})$, or survive suppression $(\tau_{\min}, \Delta_{\text{hyst}})$ — and never introduces a transit or extremum that τ does not have; the re-read word is again $\subseteq \mathsf{Sound}(\tau)$. *Step,* EXCLUDE($e$)*:* this deletes a token from the current word (Definition 8.11); a subset of a subset of $\mathsf{Sound}(\tau)$ is again a subset. In neither step is a token *inserted* that is not backed by a real transit/extremum, so the composed word lies in $\mathsf{Sound}(\tau)$, as claimed. The single δ-trapped case of Proposition A.3(ii) — no threshold shift of magnitude $\leq \delta$ can manufacture the absent Q-crossing — is exactly the instance of this closure for the threshold power T_1 on a silent-window signal. □

Remark 5.9 (Scope of Theorem 5.8). The closure covers the re-parameterisation catalogue T_1–T_8 and the band-word EXCLUDE. It deliberately does *not* depend on the operational semantics of the full transduction T: the open rationality question Remark 8.13 is untouched, and its resolution is not required for the floor to be preserved. INCLUDE is the one band-word power *outside* the theorem — it re-attributes a real token's band under a declared premise rather than re-reading τ — and is treated conditionally in Remark 5.10. Structurally the result is a *closure*: the substantive step is the discrete-to-continuous base case (the intermediate/extreme value theorem grounding a token in a real transit or extremum), and the induction over T_1–T_8 and EXCLUDE merely propagates it. We state it as a theorem for the base case's role in the soundness argument, not for depth in the closure induction, which is elementary.

Remark 5.10 (Two soundness levels: syntactic Σ_2 versus meaningful soundness, and the CC3 power-boundary). Theorem 5.8 preserves *syntactic* soundness: an emitted token is a real transit or extremum. The guarantee proved here is *deliberately weak*: it is the anti-forgery floor — no token without a real transit, and in particular no firing on a flat signal — *not* meaningful confirmation. Whether an in-window token is a *meaningful* confirmation, above

the instrumental resolution, is a *separate*, stronger layer — the CC3 condition that the core paper [9] develops *on top of* this floor. It is not required for, and does not enter, the closure proved here, which is self-contained; that meaningful layer is where the substantive content of soundness lives. This holds *unconditionally*, even at arbitrarily fine threshold precision (a large window/gain budget): a tiny but real oscillation crossed by a finely placed θ is a *real* transit, not a fabricated one, so no amount of re-parameterisation breaks Σ_2. *Meaningful* soundness is a stronger, separate demand: confirming from an oscillation *below the instrumental resolution* manufactures a meaningful truth even though the token is syntactically real. That is exactly what condition CC3 — threshold detectability, θ above the instrumental resolution — of the core paper [9] forbids; a noise-induced transit below resolution violates CC3. Hence extreme threshold precision (unbounded window/gain) does *not* break Σ_2 but *does* break CC3: the power-boundary for meaningful soundness is CC3, not Σ_2. Accordingly the Σ_2-closure is PROVEN for crossing tokens premise-free (by the intermediate value theorem) and, for touch tokens, conditional on the no-synthesis premise (Theorem 5.8). That premise is not an external accident: it *withholds a signal-write power* — the power to modify or synthesise the monitored signal τ itself (buffer interpolation/upsampling in its minimal form, active injection on the line in its strongest), the Layer-3 signal-modification power of the strict tuning-layer hierarchy of [8], as against the read-only (Layer-1) re-parameterisation this closure covers. A monitor *granted* that power can *make* a transit real and fire a token backed by the signal it now controls — syntactically sound relative to the written τ, fabricated relative to the world; "unconditional" is unlocked exactly by it. So the closure is *read-only-conditional*: it holds for every budget that only *reads* τ (re-parameterisation, Definition A.7), and the signal-write power is its boundary — the *syntactic* twin of CC3. Meaningful soundness is CONDITIONAL on CC3 — imported from [9] and not re-proven here, hence CONJECTURED relative to CC3 as an engineering claim about the retuning calculus. Two powers, two boundaries: the signal-write power bounds *syntactic* Σ_2; CC3 bounds *meaningful* soundness. INCLUDE inherits the same split: it preserves Σ_2 (the re-attributed token is real, Definition 8.11), but its meaningful soundness is likewise governed by CC3, not by the syntactic floor.

Remark 5.11 (Powers versus threats). The same structural axioms can be broken from the *channel* side: skew, drift, and jitter are adversarial violations of Γ_1/Γ_2 by the world, not by the monitor. Monitor-side relaxations are *powers*; channel-side violations are *threats*. The declared theory ladder of the core paper [9] is precisely the accounting that keeps the two ledgers separate, and the calibration bound (S6) is where they meet.

6. The Signature Space and its Property Classes

Boundary. This paper develops, as the general engineering layer, the observation-*model* itself — the band-delimited signature space and its generative automaton, the deciding automaton, the optimal monitor-type selection, and the tuning calculus that *transforms* one observation class into another. The core paper [9] *instantiates* this space for the implication reading and supplies its causal *validity*: which observed orderings the forward reading is well-posed on. Construction and transformation are the work of this paper; causal validity is settled in the core paper.

Having fixed the specification and its realisation, we now develop the combinatorial object those obligations range over: the band-delimited signature space, its generative automaton, the clopen prefix-filtration of its window patterns, and the quantifier lift that assigns each monitored property its four-family class. The chain is elementary and self-contained, and every count is machine-checked by exhaustive enumeration in this paper's script `eightick_derive.py` (Remark 6.5).

Trace properties versus monitor properties. Throughout this section we keep two axes strictly apart, following the trace–monitor cross-axis independence remark of the intersection-algebra paper [7]. The clopen prefix-filtration marks Φ_ℓ and the four-family class of a monitored obligation are *trace properties*: sets of ω-words — equivalently, predicates on a trace and its finite prefixes — living in the class families $\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}$ of that paper's *Class Families* definition, with the finite-prefix reading FIN-INV $\Leftrightarrow \mathsf{S}$, FIN-CONF $\Leftrightarrow \mathsf{C}$, NO-FIN-INV $\Leftrightarrow \mathsf{L}$, NO-FIN-CONF $\Leftrightarrow \mathsf{K}$ of its *Polarity Schemas*. The band automaton, the prefix trie, and their determinacy, verdict degree, and state count are by contrast *monitor properties*: features of the machine that decides a trace property — a separate axis that the same paper keeps independent of the trace class (its trace-axis/monitor-axis independence remark). A trace property is *in* $\mathsf{S} \cap \mathsf{C}$; a monitor *decides* that property — the two are never identified, and in particular the minimal state count $|Q|$ (the Nerode index) is a fact about the monitor's memory, not about the family the trace belongs to.

6.1. **The band-delimited signature language.** We fix the formal object over which selection, self-tuning, and causal coherence all act: the language of *band-delimited event words*. The development is elementary automata theory; it is recorded here because every later obligation is a property of words of this language.

Alphabet. Write $\Sigma = \Sigma_{\mathrm{ev}} \cup \Sigma_{\mathrm{band}}$, with

$$\Sigma_{\mathrm{ev}} \;=\; \{\, P^{\uparrow},\; P^{\downarrow},\; Q^{\uparrow},\; Q^{\downarrow} \,\}, \qquad \Sigma_{\mathrm{band}} \;=\; \{\, \langle \mathrm{pre}\rangle,\; \langle x\rangle,\; \langle \mathrm{win}\rangle,\; \langle \mathrm{post}\rangle,\; \langle \mathrm{end}\rangle \,\}.$$

The four *boundary (crossing) events* are, in the directional rendering of the reference class of the core paper [9], the antecedent validate/retract and consequent validate/invalidate crossings: $P^{\uparrow}$ (antecedent validates), $P^{\downarrow}$ (antecedent retracts), $Q^{\uparrow}$ (consequent validates), $Q^{\downarrow}$ (consequent invalidates). We use the directional tokens throughout precisely because they resolve the symbol clash with the in-window feature triple (A, B, C) of Section 3: the letter A names a Q-validation *feature*, whereas $P^{\uparrow}, \ldots, Q^{\downarrow}$ name *events*, so the two never collide. The band delimiters mark the start of each band and a single terminal; because bands are contiguous, the family of start/finish delimiters collapses to five starts plus one terminal.

Band-words. A *band-word* is a string

$$w \;=\; \langle \mathrm{pre}\rangle\, u_{\mathrm{pre}}\, \langle x\rangle\, u_x\, \langle \mathrm{win}\rangle\, u_{\mathrm{win}}\, \langle \mathrm{post}\rangle\, u_{\mathrm{post}}\, \langle \mathrm{end}\rangle, \qquad u_\bullet \in \Sigma_{\mathrm{ev}}^*,$$

where each *band content* $u_\bullet$ is a finite event sequence. The word is *well-formed with bound* N when $|u_\bullet| \le N$ for every band $\bullet \in \{\mathrm{pre}, x, \mathrm{win}, \mathrm{post}\}$. Write L^N_{band} for the set of well-formed band-words with per-band bound N.

Proposition 6.1 (Regularity of the band language, PROVEN, elementary)**.** *For each fixed N the language L^N_{band} is regular, recognised by a deterministic finite automaton with one control state per band plus a bounded counter.*

Proof. By construction. Take states $Q = \{q_{\text{pre}}, q_x, q_{\text{win}}, q_{\text{post}}\} \times \{0, 1, \ldots, N\} \cup \{q_0, q_{\text{acc}}\}$: one control state per band tagged with a counter $c \in \{0, \ldots, N\}$, an initial state q_0, and an accepting state q_{acc}. The transitions read a delimiter to switch band and reset the counter to 0, read an event $e \in \Sigma_{\text{ev}}$ to increment the counter (and reject when it would exceed N), and read $\langle\text{end}\rangle$ from q_{post} to accept:

$$q_0 \xrightarrow{\langle\text{pre}\rangle} (q_{\text{pre}}, 0), \quad (q_b, c) \xrightarrow{e} (q_b, c{+}1) \ \ (c < N), \quad (q_{\text{pre}}, c) \xrightarrow{\langle x\rangle} (q_x, 0),$$

and analogously $\langle\text{win}\rangle$, $\langle\text{post}\rangle$ advance the band, with $(q_{\text{post}}, c) \xrightarrow{\langle\text{end}\rangle} q_{\text{acc}}$. The automaton has $4(N{+}1) + 2$ states — finite — so L^N_{band} is regular. □

Reference class and rich fan. Inside L^N_{band} the *reference class* R is the sublanguage

$$R \;=\; \{\, w \;:\; u_{\text{pre}} = u_{\text{post}} = \varepsilon,\ u_x = P^{\uparrow},\ u_{\text{win}} \text{ an ordering of } \{P^{\downarrow}, Q^{\uparrow}, Q^{\downarrow}\} \,\} :$$

empty pre/post bands, the opener band carrying the single antecedent validation $P^{\uparrow}$, and the window carrying an ordering of the three remaining events. R is finite, hence regular, and is the directional-token image of the antecedent-validated reference class of the core paper [9]. The *rich fan* is the full L^N_{band}: bounded event content is now permitted in the Pre and Post bands as well, so the reference class is one regular sublanguage of a strictly larger regular language. The subsequent property classes, selection, and self-tuning are all obligations over L^N_{band}.

Reduced (confirming) reference signatures. Between the full reference class R and the rich fan sits the *confirming family*: the signatures whose in-band window carries *at least* the consequent validation $Q^{\uparrow}$. They arise from R by *dropping* window events — the antecedent retraction $P^{\downarrow}$, the consequent invalidation $Q^{\downarrow}$, or both — with each dropped event accounted for either out-of-band (observed in a Pre or Post tick) or simply *undetected* (falling outside the monitored window). Keeping $Q^{\uparrow}$ and dropping one event leaves the two-event references $\{Q^{\uparrow}, Q^{\downarrow}\}$ and $\{P^{\downarrow}, Q^{\uparrow}\}$; dropping both leaves the single-event reference $\{Q^{\uparrow}\}$. In the (A, B, C) classification this is the reduction lattice

$$(1,1,1) \;\longrightarrow\; (1,1,0),\ (1,0,1) \;\longrightarrow\; (1,0,0),$$

over the four $A{=}1$ cells, with the confirmation $A{=}1$ ($Q^{\uparrow}$ present) invariant and the six-signature reference R as its top cell $(1,1,1)$. These four cells — the confirming family — comprise 24400 of the 50000 admissible eight-tick signatures; their complement (the $A{=}0$ cells) are the non-confirming observations (violation when $Q^{\downarrow}$ is present, silence otherwise): valid observations that lie outside the reference family. The causal validity of each reference form is settled in the core paper [9].

6.2. **The generative automaton for the eight-tick signature space.** The band language of Section 6.1 is the *enforcement* view: four content bands (Pre, opener, window, Post) delimited for the selection and self-tuning obligations. The core paper's *classification* view is finer — the eight-tick partition `pre`/`x-1`/`x`/`x+1`/`m2`/`cl`/`cl+1`/`post` of [9] — and its purpose is not to bound event content but to *generate exactly* the catalogue of observation signatures that the classification enumerates. We record that generator here as a finite automaton (equivalently, a right-linear grammar), because every count the paper relies on, and every property class of Section 6.4, is a property of *its* language.

Observation alphabet and token reconciliation. The classification runs over the five-token observation alphabet of the core paper [9],

$$\Sigma_{\text{obs}} \;=\; \{\, \varnothing,\; P^{\top},\; P^{\perp},\; Q^{\perp},\; Q^{\top} \,\}, \qquad \varnothing \prec P^{\top} \prec P^{\perp} \prec Q^{\perp} \prec Q^{\top},$$

each tick emitting exactly one symbol and the priority order being the band-token order of the core paper (its BAND-MAX rule). Reducing the competing per-tick events to a single fired symbol is a timing- and labelling-convention choice in the sense of the convention algebra of [8]. This is the same crossing alphabet as the directional events Σ_{ev} of Section 6.1 under the fixed renaming

$$P^{\uparrow} \equiv P^{\top}, \quad P^{\downarrow} \equiv P^{\perp}, \quad Q^{\uparrow} \equiv Q^{\top}, \quad Q^{\downarrow} \equiv Q^{\perp},$$

with $\varnothing$ (the silent tick) having no directional counterpart. We use the $\top/\perp$ tokens in this subsection to stay verbatim with the core paper's signature domain, reserving the directional tokens for the enforcement language. No symbol is redefined: the two token sets name the same four crossings, and the reconciliation above is a bijection on events; neither this map nor the generator below re-defines the in-window feature triple (A, B, C) of Section 3, which continues to name *features* rather than *events*.

The signature generator. Fix the window bound k (the reference instance is $k = 3$). Resolving the band partition to one *tick* per position gives the eight-tick grid `pre` · `x-1` · `x` · `x+1` · `m2` · `cl` · `cl+1` · `post`. An *eight-tick signature* is a word

$$s \;=\; (r,\, a,\, o,\, b,\, m,\, c,\, d,\, e), \qquad r, a, o, b, m, c, d, e \in \Sigma_{\text{obs}},$$

one token per tick from the five-token observation alphabet $\Sigma_{\text{obs}} = \{\varnothing, P^{\top}, P^{\perp}, Q^{\top}, Q^{\perp}\}$. The two *out-of-band* ticks `pre`, `x-1` precede the opener, the opener is `x`, the *in-band window* is $\{\texttt{x+1}, \texttt{m2}, \texttt{cl}\}$, and `cl+1`, `post` are the two post ticks. The *signature generator* $\mathcal{G}$ emits one Σ_{obs}-token per tick in the fixed left-to-right order; its language $L(\mathcal{G})$ is the set of all eight-tick signatures. (This is the eight-tick classification model of [9] — every band a single tick, with the pre/post bands free to carry observation content; the window resolution and the causal validity of the signatures are those of the core paper.)

Proposition 6.2 (Generator cardinality, PROVEN)**.** *The raw eight-tick signature space has*

$$|L(\mathcal{G})| \;=\; 5^{8} \;=\; 390625,$$

and the map sending each generated word to the tick-labelling of a triggered observation is a bijection onto the raw signature domain.

Proof. The eight ticks range independently over Σ_{obs} with $|\Sigma_{\text{obs}}| = 5$; since $\mathcal{G}$ is acyclic with exactly one word per tick-tuple, $|L(\mathcal{G})| = 5^8 = 390625$. The tick-tuple is precisely the coordinate record of a raw eight-tick signature, so the correspondence is a bijection. □

The admissibility automaton. Not every generated word is a realisable triggered observation. Two opener rules prune the raw space, acting on the tick coordinates:

$$\text{adm}(s) \;=\; (o = P^{\top}) \;\wedge\; (r \neq P^{\top}) \;\wedge\; (a \neq P^{\top}),$$

i.e. the opener tick `x` carries the antecedent validation $P^{\top}$ (rule V1, the opener definition of [9]), and *no earlier* tick (`pre`, `x-1`) carries a $P^{\top}$ (rule V2, *opener minimality*: the opener is the first antecedent validation). A *later* $P^{\top}$ — a re-trigger inside or after the window — is *admitted*: it is the multiplicity content that the monitor-selection result of this paper ranges

over (a single-trigger restriction would collapse the selection scores to a tie). The predicate is a finite Boolean function of the tick coordinates, recognised by a finite automaton $\mathcal{A}_{\text{adm}}$ run in product with $\mathcal{G}$.

Proposition 6.3 (Admissible cardinality, PROVEN)**.** $|L(\mathcal{G}) \cap L(\mathcal{A}_{\text{adm}})| = 4 \cdot 4 \cdot 5^5 = 50000$.

Proof. The opener tick is pinned to $P^\top$ (one choice); the two pre-opener ticks `pre`, `x-1` each range over $\Sigma_{\text{obs}} \setminus \{P^\top\}$ (4 tokens); and the five remaining ticks (`x+1`, `m2`, `cl`, `cl+1`, `post`) each range over all of Σ_{obs} (5 tokens). Hence $1 \cdot 4^2 \cdot 5^5 = 50000$. Verified by exhaustive enumeration. □

The classification morphism. Forgetting *where* in the in-band window each event occurred sends an admissible signature to its presence triple; write $\kappa : s \mapsto (A, B, C)$, with $A = \exists\, Q^\top$, $B = \exists\, Q^\perp$, $C = \exists\, P^\perp$ over the in-band window $\{\texttt{x+1}, \texttt{m2}, \texttt{cl}\}$.

Proposition 6.4 (Window-class quotient, PROVEN)**.** *κ maps the* 50000 *admissible signatures onto the eight window classes (the* (A,B,C) *cells), with per-class counts*

$$((1,0,0), \ldots, (0,1,1)) = (7600,\ 7200,\ 7200,\ 2400,\ 3200,\ 7600,\ 7600,\ 7200) \quad (\textit{sum}\ 50000),$$

and it is well-defined: every admissible signature lands in a single class. The eight-class partition is therefore invariant *of the band/tick resolution — it is the same* (A, B, C) *classification the eight-tick model of* [9] *induces, so the eight window classes (and every downstream result stated over them) are unchanged.*

Proof. (A, B, C) is a determinate Boolean function of the three in-band tick coordinates, so κ is a total single-valued function into the eight cells; the per-class counts are the exhaustive enumeration (summing to 50000). □

Remark 6.5 (The eight-tick cardinality chain, PROVEN). Reading the three propositions in sequence,

$$L(\mathcal{G})\ (390625) \xrightarrow{\mathcal{A}_{\text{adm}}} (50000) \xrightarrow{\kappa} 8 \text{ classes},$$

the raw 5^8 tick-word space, its opener-minimal admissible sublanguage $4^2 \cdot 5^5$, and the eight (A, B, C) window classes. Each stage is machine-checked by exhaustive enumeration (`eightick_derive.py`); the two counts are PROVEN, not merely asserted. The eight-class partition coincides with the (A, B, C) classification of the eight-tick model of the core paper [9]: the two papers share the same window classes and the same *class-level* reachability results stated over them; the signature-level catalogue that monitor selection ranges over is the eight-tick one.

Two band structures, one projection. The eight-tick partition used here is the fine *classification* model of the core paper (its `pre`/`x-1`/`x`/`x+1`/`m2`/`cl`/`cl+1`/`post` ticks, [9]); the four-content-band delimiter language L^N_{band} of Section 6.1 is the coarser *enforcement* view (Pre, opener, window, Post). They are not rival definitions. The projection $\pi_{8\to4}$ merges $\{\texttt{pre}, \texttt{x-1}\}$ into `Pre`, keeps `x` as the opener, merges the three window ticks $\{\texttt{x+1}, \texttt{m2}, \texttt{cl}\}$ into a single `win` band, and merges $\{\texttt{cl+1}, \texttt{post}\}$ into `Post` by the priority-max rule, preserving the crossing/touch features (A, B, C) and hence the eight-class enumeration; so the generator $\mathcal{G}$ refines L^N_{band} along $\pi_{8\to4}$. The enforcement language bounds per-band event content for the selection obligations, while $\mathcal{G}$ resolves the window into its timing ticks for the classification counts. Nothing below depends on choosing one over the other.

6.3. **Clopen prefix-filtration: trie equals deterministic monitor.** The property classes of the specification (Section 5) are stated over the infinite trace, but their per-window skeleton is a finitary object we can construct outright. This subsection does so; the next reads off the four-family class of the lifted window pattern. The development is elementary.

Setup. On the *bounded* observation window — the opener band x together with the window ticks — fix the opener $A = P^{\uparrow}$ at window-local position 0. Under the antecedent-validated regime this opener is a clopen property that is always $\top$, so it may be factored out and the window carries the other three events $\{P^{\downarrow}, Q^{\uparrow}, Q^{\downarrow}\}$. We work with *group* (a) of the reference class — consequent affirms first, $Q^{\uparrow}$ before $Q^{\downarrow}$ — whose window sequences are the three orderings

$$\sigma_1 = P^{\downarrow}\, Q^{\uparrow}\, Q^{\downarrow}, \qquad \sigma_2 = Q^{\uparrow}\, P^{\downarrow}\, Q^{\downarrow}, \qquad \sigma_3 = Q^{\uparrow}\, Q^{\downarrow}\, P^{\downarrow}.$$

Proposition 6.6 (Every window event is clopen, PROVEN, elementary)**.** *Over the bounded window, "event e has occurred by tick i" is decided by a finite prefix; hence it is both finitely confirmable and finitely refutable — clopen, i.e. a* trace *property in* $\mathsf{S} \cap \mathsf{C}$*, the fully finitely-decided class of* [7]*. In local Linear Temporal Logic (LTL) over the window clock, "the i-th window event is e" is* $\mathbf{X}^i\, e$*, and a full signature is the clopen conjunction* $\varphi_\sigma = A \wedge \mathbf{X}\, e_1 \wedge \mathbf{X}^2\, e_2 \wedge \mathbf{X}^3\, e_3$*.*

Proof. Occurrence by tick i is a Boolean function of the first $i{+}1$ letters, so it stabilises on any prefix of length $i{+}1$; both verdicts are reached in bounded time, which is exactly clopen $(\mathsf{S} \cap \mathsf{C})$. A signature conjoins finitely many such clopen atoms and is therefore clopen. □

This is the bounded-window specialisation of the finite-horizon *clopen collapse* of [7], whose bounded projections all reduce to the single non-trivial cell $\mathsf{S} \cap \mathsf{C}$; the pointer is a WITNESSED cross-reference and adds no proof obligation of its own.

The filtration. For depth $\ell \in \{1, 2, 3\}$ and a signature set S define the *prefix filtration*

$$\Phi_\ell^S \;=\; \bigvee_{\sigma \in S} \mathrm{pre}_\ell(\sigma), \qquad \mathrm{pre}_\ell(\sigma) \;=\; A \wedge \bigwedge_{i \le \ell} \mathbf{X}^i\, e_i(\sigma),$$

each Φ_ℓ^S clopen because bounded and prefix-decided. For group (a), $S = \{\sigma_1, \sigma_2, \sigma_3\}$, the levels are

$$\Phi_1 = A \wedge \mathbf{X}\big(P^{\downarrow} \vee Q^{\uparrow}\big),$$

$$\Phi_2 = A \wedge \big[(\mathbf{X}\, P^{\downarrow} \wedge \mathbf{X}^2\, Q^{\uparrow}) \vee (\mathbf{X}\, Q^{\uparrow} \wedge \mathbf{X}^2\, P^{\downarrow}) \vee (\mathbf{X}\, Q^{\uparrow} \wedge \mathbf{X}^2\, Q^{\downarrow})\big],$$

$$\Phi_3 = A \wedge \big[(\mathbf{X}\, P^{\downarrow} \wedge \mathbf{X}^2\, Q^{\uparrow} \wedge \mathbf{X}^3\, Q^{\downarrow}) \vee (\mathbf{X}\, Q^{\uparrow} \wedge \mathbf{X}^2\, P^{\downarrow} \wedge \mathbf{X}^3\, Q^{\downarrow}) \vee (\mathbf{X}\, Q^{\uparrow} \wedge \mathbf{X}^2\, Q^{\downarrow} \wedge \mathbf{X}^3\, P^{\downarrow})\big].$$

The inner conjunctions are positional — event at tick i and the next at $i{+}1$ — so the $\mathbf{X}$-nesting encodes event order.

Proposition 6.7 (Filtration structure, PROVEN, elementary)**.** *The filtration satisfies:* (i) Φ_ℓ^S *is clopen at every level;* (ii) *the refining chain* $\Phi_3 \Rightarrow \Phi_2 \Rightarrow \Phi_1$ *holds, and the class* $R_a = \bigcap_\ell \|\Phi_\ell\|$ *is reached at the finite depth* $\ell = |\mathit{window}| = 3$*;* (iii) *the disjuncts of* Φ_ℓ^S *are exactly the depth-*ℓ *nodes of the prefix trie of* S*, so the disjunction width is the number of live prefixes — here* $2, 3, 3$*.*

Proof. (i) by Proposition 6.6 each disjunct is a bounded clopen conjunction and a finite disjunction of clopen sets is clopen. *(ii)* $\mathrm{pre}_{\ell+1}(\sigma) \Rightarrow \mathrm{pre}_{\ell}(\sigma)$ by dropping the last conjunct, and taking disjunctions preserves the implication; the intersection is attained once ℓ equals the window length because no further letters remain to constrain. *(iii)* two signatures share a depth-ℓ disjunct iff they agree on their length-ℓ prefix, which is the defining identification of trie nodes; the surviving disjuncts count the distinct live prefixes, 2 at depth 1 ($\{P^{\downarrow}\}, \{Q^{\uparrow}\}$) and 3 at depths $2, 3$. □

It is the deterministic monitor. Read the filtration operationally: at tick ℓ evaluate Φ_ℓ; the trace stays alive iff it holds, and it is accepted iff Φ_3 holds at window end. Both $\bot$ (the trace has fallen out of every live prefix) and $\top$ (the trace has matched a full signature) are reachable on a finite prefix, so the filtration *is* the deterministic monitor that decides this clopen trace property: the property decided lies in $\mathsf{S} \cap \mathsf{C}$, while determinacy and the verdict degree $d = 1$ (both verdicts definitive, reached on a finite prefix) are properties of the monitor, not of the family — the two axes that the trace–monitor independence remark of [7] keeps apart. The construction generalises to any signature set S — any subset of the 24 orderings, or of the band-extended words of Section 6.1: every finite-window trace class equals the filtration of clopen disjunctions given by its trie. Group (b) ($Q^{\downarrow}$ before $Q^{\uparrow}$) is the symmetric filtration, and the full reference class of the core paper [9] is the union $\Phi_\ell^{(a)} \vee \Phi_\ell^{(b)}$.

Remark 6.8 (Trace property versus deciding monitor, PROVEN). The filtration makes the two axes concrete. The marks Φ_ℓ and the window class R_a are *trace properties* — clopen sets of words, in $\mathsf{S} \cap \mathsf{C}$ by Proposition 6.6 and Proposition 6.7(i), matching the finitely-decided class of [7]. The trie of Proposition 6.7(iii) is the *monitor* that decides them; its determinacy, its verdict degree, and its minimal state count ($|Q|$, the Nerode index of S) are monitor properties, on the monitor axis that the trace–monitor independence remark of [7] declares independent of the trace axes. "Monitorable" is accordingly a predicate on the deciding machine, never a synonym for the safety class S: a property may fail to be finitely monitorable (it may sit in $\mathsf{L} \cap \mathsf{K}$) and still be a perfectly well-defined trace property.

6.4. **The four families of the lifted window pattern.** The filtration shows that on the window *everything is clopen* ($\mathsf{S} \cap \mathsf{C}$): the monitor decides both verdicts on a finite prefix. The four-family marks of the specification — and coliveness K in particular — therefore do *not* arise at the window level. They arise only under the *quantifier lift* to the infinite trace. That the universal $\mathbf{G}$ turns a clopen window skeleton into coliveness is the trace-side, causal-validity result of the core paper [9]; we restate it below in the form we use, to fix the verdict horizon the monitor's clopen levels range over.

Proposition 6.9 (Quantifier lift of a clopen window pattern)**.** *Let φ be a nontrivial clopen per-window pattern — both $\top$ and $\bot$ reachable on bounded prefixes, as in Propositions 6.6 and 6.7. Then, in the four-family intersection algebra of* [7]*:*

(a) $\mathbf{G}\,\varphi$ *is a safety property that is never finitely confirmable:* $\mathbf{G}\,\varphi \in \mathsf{S} \cap \mathsf{K}$ (coliveness). [PROVEN]
(b) $\mathbf{F}\,\varphi \in \mathsf{C} \cap \mathsf{L}$ (liveness). [PROVEN]
(c) *an unbounded eventuality, e.g.* $\mathbf{G}\,\mathbf{F}\,\varphi$*, lies in* $\mathsf{L} \cap \mathsf{K}$. [PROVEN]

Proof. (a) Any violation of $\mathbf{G}\ \varphi$ is a window at which φ fails; since φ is clopen that failure is decided by a finite prefix, so $\mathbf{G}\ \varphi$ has a finite bad prefix and is a safety property with $d = 1$ refutation. No finite prefix confirms $\mathbf{G}\ \varphi$, because after any prefix strictly more windows remain to be checked; the property is therefore never finitely confirmable, which is coliveness K. Hence $\mathbf{G}\ \varphi \in \mathsf{S} \cap \mathsf{K}$. *(b)* Once a satisfying window occurs — finitely decided, as φ is clopen — $\mathbf{F}\ \varphi$ is confirmed by that finite good prefix, so it is cosafety C with $d = 1$ confirmation; and no finite prefix refutes it, since a satisfying window may still arrive later, so it is liveness L. Hence $\mathsf{C} \cap \mathsf{L}$. *(c)* $\mathbf{G}\,\mathbf{F}\ \varphi$ admits neither a finite good prefix (no finite observation establishes "infinitely often") nor a finite bad prefix (a satisfying window may always recur), so it is neither cosafety nor safety: pure $\mathsf{L} \cap \mathsf{K}$. □

Parts (a)–(c) specialise the canonical-witness memberships of [7] — $\Phi_{Gp} \in \mathsf{S} \cap \mathsf{K}$, $\Phi_{Fp} \in \mathsf{C} \cap \mathsf{L}$, $\Phi_{GFp} \in \mathsf{L} \cap \mathsf{K}$ — to a clopen per-window pattern φ in place of a raw atom p; the pointer is a WITNESSED cross-reference.

The trace-side derivation that coliveness is what the universal $\mathbf{G}$ adds to the clopen window skeleton ($\mathsf{S} \cap \mathsf{K} = \mathbf{G}(\text{clopen})$) is established in the core paper's causal-validity section [9]; here we *use* it to characterise the monitor's verdict horizon, the proposition above being the specialisation of that trace-side result to a clopen per-window pattern. Each mark $\mathsf{S}\cap\mathsf{K}$, $\mathsf{C}\cap\mathsf{L}$, $\mathsf{L}\cap\mathsf{K}$ invoked here is a property of the *ω-trace* — a class family in the sense of [7] — decided, where it can be, by a monitor whose verdict degree is the separate monitor-axis quantity of Remark 6.8; the lift changes the trace class, not the identity of the deciding machine. The enforcement objectives of the later sections are then the liveness/coliveness lifts $\mathbf{G}\,\mathbf{F}\ \sigma^*$ / $\mathbf{F}\,\mathbf{G}\ \sigma^*$ of patterns acting on this skeleton, whose four-family class is the trace-side result cited above and whose deciding monitor is the object this paper builds.

6.5. **The four-distinct-event family as a regular sub-language.** The generator $\mathcal{G}$ of Section 6.2 produces the full 390625-signature space, in which many words carry *fewer* than four distinct crossings: a window may repeat the silent token $\varnothing$, omit $Q^\top$ (the A=0 classes), or present only one consequent event (the presence classes, A=1 with a crossing absent). Inside that space sits one distinguished family in which all four crossings occur, each exactly once.

The family. Call a triggered trace *four-distinct* when the opener carries $P^\top$ and the window carries the three remaining crossings $P^\perp, Q^\top, Q^\perp$ — each exactly once, no silence, no repetition. This is the regular-language rendering of the antecedent-validated reference class of [9]: the $\top/\perp$ image of that class, presented here as the reference language R of Section 6.1 and the object on which the prefix-filtration of Section 6.3 was built. The reference class itself is the core paper's; what this paper contributes is its recognising automaton and deterministic monitor. At the window-sequence level a four-distinct trace is an ordering $e_1e_2e_3$ of $\{P^\perp, Q^\top, Q^\perp\}$ prefixed by the opener, so the family is the finite set

$$F \;=\; \{\, P^\top e_1e_2e_3 \;:\; \{e_1,e_2,e_3\} = \{P^\perp, Q^\top, Q^\perp\} \,\}, \qquad |F| = 3! = 6.$$

Proposition 6.10 (The reference family is a regular sub-language, PROVEN)**.** *F is a regular sub-language of the generator, recognised by the acyclic permutation automaton $\mathcal{A}_F$ that reads $P^\top$ from the start state and then consumes each of $P^\perp, Q^\top, Q^\perp$ exactly once, its states being the subsets of $\{P^\perp, Q^\top, Q^\perp\}$ already seen (a Boolean lattice of $2^3 = 8$ states, of which the $3! = 6$ maximal-chain paths are the accepted words). Being finite, F is regular; being a set of well-formed opener-plus-window words, it is contained in the signature space of $\mathcal{G}$.*

Proof. $\mathcal{A}_F$ has one state per subset $U \subseteq \{P^\perp, Q^\top, Q^\perp\}$, an edge $U \xrightarrow{e} U \cup \{e\}$ for each $e \notin U$, the opener edge into $U = \varnothing$, and the single accepting state $U = \{P^\perp, Q^\top, Q^\perp\}$. Accepted words are exactly the maximal chains of the subset lattice, i.e. the orderings of the three events, so $L(\mathcal{A}_F) = F$ with $|F| = 3! = 6$. A finite language is regular, and every word of F is an opener-plus-window word of $\mathcal{G}$. □

Its deterministic monitor is the clopen filtration. The prefix-filtration of Section 6.3 (Proposition 6.7) is exactly the deterministic monitor of this sub-language. Splitting F by which consequent event fires first gives the two groups of Section 6.3: group (a) ($Q^\top$ before $Q^\perp$) is the three orderings $\sigma_1, \sigma_2, \sigma_3$ whose trie has live-prefix widths $2, 3, 3$, and group (b) ($Q^\perp$ before $Q^\top$) is the symmetric filtration; their union $\Phi_\ell^{(a)} \vee \Phi_\ell^{(b)}$ recognises all six orderings of F. Thus $\mathcal{A}_F$ read as a monitor is the trie of Proposition 6.7: it accepts on Φ_3 at window end and rejects the moment a trace falls out of every live prefix, both verdicts on a finite prefix (the decided trace property lies in $\mathsf{S} \cap \mathsf{C}$; the deciding monitor has verdict degree $d = 1$).

Remark 6.11 (Crossing-closed reference core). F is the *crossing-closed* core of the generated space: the unique family exercising all four crossings once, hence the maximal-information reference against which the presence classes (A=1, some crossings absent) and the $Q^\top$-absent classes (A=0; individually miscalibration, monitor-refuted, late retraction, and complex/race in [9]) are read as degradations. The larger language $L(\mathcal{G})$ is what the generator adds on top of this core: silence, omission, and repetition are precisely the moves that carry a trace out of F and into the other window classes counted in Remark 6.5.

6.6. **LTL property classes over the generated traces.** The generator, its admissibility filter, and the reference family are all finitary. We close the loop by observing that the clopen skeleton (Section 6.3) and the quantifier lift (Section 6.4) apply verbatim to the traces $\mathcal{G}$ generates, so the $\mathsf{S}/\mathsf{C}/\mathsf{L}/\mathsf{K}$ classification of the monitored property is fixed by generation alone.

Proposition 6.12 (Every generated window class is clopen, PROVEN)**.** *For each generated signature $s \in L(\mathcal{G})$ the per-window predicate "the observed window has feature vector $\kappa(s)$" is decided by the finite window prefix, hence clopen ($\mathsf{S} \cap \mathsf{C}$) with verdict degree $d = 1$; and the reference family F is monitored by the clopen filtration of Proposition 6.7.*

Proof. By Proposition 6.6 each window event "e has occurred by tick i" is a Boolean function of a bounded prefix, so the conjunction defining $\kappa(s)$ over the $k{+}1$ window ticks is a bounded clopen predicate; both verdicts are reached by window end. The reference-family clause is Proposition 6.7 applied to F (Proposition 6.10). □

This is the generated-trace specialisation of the *bounded-operator migration* of [7], which sends every property with all G, F, U, R bounded into $\mathsf{S} \cap \mathsf{C}$; the pointer is a WITNESSED cross-reference.

Lifting these clopen window predicates over the infinite stream of windows by Proposition 6.9 then yields, with φ any generated window pattern:

$$\mathbf{G}\ \varphi \in \mathsf{S} \cap \mathsf{K}, \qquad \mathbf{F}\ \varphi \in \mathsf{C} \cap \mathsf{L}, \qquad \mathbf{G}\,\mathbf{F}\ \varphi \in \mathsf{L} \cap \mathsf{K},$$

so $\mathbf{G}(\text{clopen}) = \mathsf{S} \cap \mathsf{K}$ holds on exactly the traces the generator produces (all tags PROVEN, reused from Proposition 6.9). The chain is therefore complete: the generator fixes the

finite catalogue ($390625 \to 50000 \to 8$), the filtration gives each window class its clopen skeleton, and the quantifier lift assigns the monitored property its four-family class — all on the standard eight-tick traces, with no primitive introduced beyond those of the core papers [9, 7].

7. Optimal Monitor Selection

A worked catalogue of the eight window classes with annotated figures is collected in Section 4.

The kernel and its tuning theory established *what can be read* in principle; this section realises it at runtime. In the terms of Section 5, it realises the ambiguity-minimisation side of S7b and instantiates the selection half of the specification. Two design problems remain: selecting, among the monitor types the axes below generate, the one that minimises diagnostic ambiguity; and keeping the selected monitor phase-aligned against drift by a closed self-tuning loop. We retain the statements and guarantees here; the full engineering treatment — canonical timing-error detector comparisons, the signal-theoretic parameter mapping, the loop architecture, and the small-signal stability analysis — is provided in the sections that follow.

7.1. Monitor-Type Axes.

We parameterise the monitor design space along three binary axes:

Definition 7.1 (Monitor-Type Axes)**.** A monitor type is a triple $(\rho, \omega, \pi) \in \{0,1\}^3$ where:

- ρ **(Repeat policy)::** $\rho = 0$ for *repeat-collapsing* (the monitor registers only the *first* transition of each proposition within a window); $\rho = 1$ for *multiplicity-sensitive* (every sample is recorded individually).
- ω **(Opener policy)::** $\omega = 0$ for *edge-triggered* (the observation window opens on the rising edge of w_p); $\omega = 1$ for *level-triggered* (the window remains open as long as w_p holds).
- π **(Probe)::** $\pi = 0$ for *no half-tick probe*; $\pi = 1$ for *half-tick probe enabled* (an additional sample at $t + \frac{1}{2}$ is taken between every two integer ticks within the window, effectively doubling the sampling rate).

The eight monitor types span from the cheapest (ρ=0, ω=0, π=0: edge-triggered, collapsing, no probe) to the most expensive (ρ=1, ω=1, π=1: level-triggered, multiplicity-sensitive, with half-tick probes).

Remark 7.2 (Monitor-type axes as capability resources). The three axes read naturally in the capability calculus (w, b, k) of [8]: the probe π buys *window* (it doubles the samples available per tick), the level-trigger ω buys *persistent state* (the open-window condition must be held), and multiplicity-sensitivity ρ buys *memory* (per-event recording instead of a single latched bit). The optimal type of Theorem 7.4, (ρ=0, ω=0, π=1), thus reads as "buy window, not state" — the same resource profile that distinguishes the memoryless, window-bearing touch primitive ($\kappa = (3, 0, 1)$) from the state-bearing crossing primitive ($\kappa = (1, 1, 0)$). In the increment reading of Remark 5.6, these three purchases read as coordinate steps in the capability calculus of [8]: the probe π a $+\Delta w$ step, the level-trigger ω a $+\Delta b$ step, and multiplicity-sensitivity ρ a further $+\Delta b$ step. This mapping is this paper's; the knobs of that paper's hub-reduction table are its two walls, its alternation schedule (also written π there) and its plateau convention, not these axes.

7.2. **Optimality and Selection.** Three scores rank the monitor types over the exhaustively enumerated k=3 catalogue: the *ambiguity* $\mathcal{A}(w)$ — the mean number of window classes compatible with a signature reachable by type w; the *timing robustness* $\mathcal{R}(w)$ — the largest phase perturbation (in ticks) under which the assigned class is stable; and the *opener sensitivity* $\mathcal{S}(w)$ — the fraction of signatures whose class changes when the opener tick shifts by one. The formal definitions are given in Section 7.3; the criterion and the selection they induce are as follows.

Definition 7.3 (Monitor Optimality)**.** A monitor type $w^\star$ is *optimal* for implication verification if it minimises a composite cost that balances ambiguity, sensitivity, and robustness:

$$w^\star \;=\; \operatorname*{argmin}_{w} \big[\, \mathcal{A}(w) \;+\; \lambda\, \mathcal{S}(w) \,\big], \qquad \lambda > 0, \tag{7.1}$$

subject to the tie-breaking rule that, among all minimisers, the type with the finest timing-error *detection resolution* is preferred (the half-tick probe of Section 8.1.2). Detection resolution is *not* the stability radius $\mathcal{R}$ of Definition 7.6: the probe *raises* detection resolution while *lowering* $\mathcal{R}$ (Remark 7.7), so $\mathcal{R}$ does not enter the maximised objective. The weight λ trades off classification stability (low $\mathcal{S}$) against ambiguity reduction (low $\mathcal{A}$). In practice any $\lambda \in (0, 1]$ selects the same winner because the catalogue already achieves $\mathcal{A} = 1$ and $\mathcal{S} = 0$ simultaneously (Theorem 7.4).

Theorem 7.4 (Optimal Monitor Type)**.** *Among the eight monitor types* (ρ, ω, π)*, the type*

$$w^\star \;=\; (\rho{=}0,\ \omega{=}0,\ \pi{=}1) \tag{7.2}$$

(edge-triggered, repeat-collapsing, with half-tick probe) is optimal in the pair $(\mathcal{A}, \mathcal{S})$ *in the sense of Definition 7.3; the composite objective* $\mathcal{A} + \lambda\mathcal{S}$ *decides optimality only in that pair, and the specific monitor type* $w^\star$ *is fixed among the* $(\mathcal{A}, \mathcal{S})$*-minimisers by the detection-resolution tie-break of Remark 7.7. It attains the minimal composite cost, achieving zero signature ambiguity* $\mathcal{A}(w^\star) = 1$ *and zero opener sensitivity* $\mathcal{S}(w^\star) = 0$*, and among the minimisers it has the finest timing-error detection resolution (*$\frac{1}{2}$ *tick, from the half-tick probe). The optimality content is the pair* $(\mathcal{A}, \mathcal{S})$*, established by exhaustive enumeration over the* k=3 *eight-tick catalogue and hence* witnessed*. The per-type* $(\mathcal{A}, \mathcal{S})$ *figures are computed by the selection script* `eightick_selection.py` *over the* 50000*-signature catalogue: all eight types attain* $\mathcal{A} = 1$*; the four edge-triggered (*ω=0*) types attain* $\mathcal{S} = 0$ *and the four level-triggered (*ω=1*) types* $\mathcal{S} = 0.554$*, so* $w^\star = (\rho{=}0, \omega{=}0, \pi{=}1)$ *is a joint* $(\mathcal{A}, \mathcal{S})$*-minimiser at the floor* $(1, 0)$*, fixed among the minimisers by the detection-resolution tie-break. The stability radius of Definition 7.6 satisfies* $\mathcal{R}(w^\star) = \frac{1}{2}$*, which is* smaller *than the* $\mathcal{R} = 1$ *of an integer-tick type: the probe trades stability radius for detection resolution (Remark 7.7). Accordingly* $\mathcal{R}$ *is asserted analytically, is* not *part of the maximised objective, and is not independently witnessed by simulation; any residual tie among* $(\mathcal{A}, \mathcal{S}, \text{resolution})$*-equal types is broken by implementation cost (history buffer, sampling budget), a design preference outside the formal criterion. Finally, we state the narrowness of the criterion plainly: zero signature ambiguity* $\mathcal{A} = 1$ *is a property the* whole *catalogue shares — all eight types attain it, and four reach the* $(\mathcal{A}, \mathcal{S}) = (1, 0)$ *floor — so the theorem's real selection lever is edge-versus-level triggering (*ω*), refined by the probe (*π*); the minimum-ambiguity criterion discriminates only at the* ω/π *axes, not as a broad optimum over the type space.*

Proof. We evaluate each axis independently against the observation classes of the eight-class observation gallery [9], using the band partition anchored at the opener tick x (the opener-tick band definition [9]).

Repeat policy (ρ). At $k = 3$, all 50000 signatures map unambiguously to a single window class. Enabling multiplicity ($\rho = 1$) splits certain signatures into sub-variants that remain within the same class, so $\mathcal{A}$ is unchanged. However, the additional state doubles the history buffer with no information gain; hence $\rho = 0$ is preferred.

Opener policy (ω). Edge-triggering ($\omega = 0$) opens the window exactly once per rising edge of w_p, fixing the opener tick x unambiguously. Level-triggering ($\omega = 1$) may produce overlapping windows when w_p remains high, creating redundant signatures that do not reduce $\mathcal{A}$ but increase implementation cost. Crucially, overlapping openers would make x non-unique and break the early–late symmetry of the eight-tick partition around x. Hence $\omega = 0$.

Half-tick probe (π). The probe adds one sample between adjacent integer ticks. This is precisely the two-samples-per-symbol condition that enables a Gardner-style timing-error detector (cf. Section 8.1). With the probe, the finest *detectable* timing perturbation improves from 1 tick to $\frac{1}{2}$ tick: the additional sample straddles the `x`/`x+1` boundary and supplies sub-tick evidence for the early–late counts n_0, n_1 that feed the timing-error signal e_n (Equation (8.3)). This is a gain in *detection resolution*, and it is exactly why $w^\star$ is preferred among the $(\mathcal{A}, \mathcal{S})$-minimisers. It is *not* a gain in the stability radius $\mathcal{R}$ of Definition 7.6: making the classification sensitive to $\frac{1}{2}$-tick shifts means a signature's class can change already at $|\delta| = \frac{1}{2}$, so the probe *lowers* $\mathcal{R}$ from 1 to $\frac{1}{2}$ (Remark 7.7). Detection resolution and stability radius are distinct, oppositely-signed quantities here; the probe buys the former at the cost of the latter.

Opener sensitivity ($\mathcal{S}$). Because each signature's class label is determined by the Boolean values observed at discrete bands relative to x, and the eight-tick partition assigns distinct algebraic roles to `x` and `x+1` (the eight-tick temporal partition [9]), shifting x by ± 1 either produces an identical partition (if the trace has the same Boolean values at adjacent ticks) or maps to a different but still-unique class. Exhaustive enumeration over all 50000 signatures confirms $\mathcal{S}(w^\star) = 0$: no class migration occurs under ± 1-tick opener perturbation.

Combining: $(\rho{=}0, \omega{=}0, \pi{=}1)$ achieves $\mathcal{A}(w^\star) = 1$ (zero ambiguity, inherited from the 50000-signature catalogue) and $\mathcal{S}(w^\star) = 0$ (zero opener sensitivity). The composite cost $\mathcal{A} + \lambda\,\mathcal{S} = 1 + 0 = 1$ is minimal for any $\lambda > 0$. Among the minimisers, $w^\star$ is selected by the tie-break because its half-tick probe gives the finest detection resolution; its stability radius $\mathcal{R}(w^\star) = \frac{1}{2}$ is, by contrast, *lower* than the $\mathcal{R} = 1$ of an integer-tick type and is therefore not what the selection maximises. The $(\mathcal{A}, \mathcal{S})$ figures are witnessed by exhaustive enumeration; $\mathcal{R}(w^\star)$ is an analytic by-product, not a per-type measurement witnessed in any simulation of this paper. □

7.3. **Formal Selection Criteria.**

Definition 7.5 (Signature Ambiguity)**.** Let Σ_w denote the set of eight-tick signatures reachable by monitor type w. The *ambiguity* of w is

$$\mathcal{A}(w) \;=\; \frac{1}{|\Sigma_w|} \sum_{\sigma \in \Sigma_w} \big|\{\, C_j : \sigma \text{ compatible with } C_j \,\}\big|, \tag{7.3}$$

where the inner set counts how many window classes C_j ($j \in 1..8$) could have generated σ. A monitor with $\mathcal{A}(w) = 1$ has *zero* signature ambiguity: every observed signature maps to exactly one window class.

Definition 7.6 (Timing Robustness)**.** Let $\delta \in [-1, 1]$ denote a timing perturbation applied uniformly to the monitor's sampling instants. The *robustness* of type w is the largest $|\delta|$ under which no signature migrates to a different window class:

$$\mathcal{R}(w) \;=\; \max\{\, |\delta| : \forall\, \sigma \in \Sigma_w,\ \mathsf{cl}(\sigma) = \mathsf{cl}(\sigma^{\delta}) \,\}. \tag{7.4}$$

Remark 7.7 (Detection resolution versus stability radius). The stability radius $\mathcal{R}(w)$ of Definition 7.6 is a *benefit when large*: it is the largest sampling perturbation the classification tolerates before *any* signature changes class. Timing-error *detection resolution* — the smallest timing shift the TED can *sense* — is a benefit when *fine*. These two quantities move in *opposite* directions under the half-tick probe: the probe improves detection resolution from 1 tick to $\frac{1}{2}$ tick, but for exactly that reason the classification becomes responsive to $\frac{1}{2}$-tick shifts, so the stability radius *drops* to $\mathcal{R}(w^{\star}) = \frac{1}{2}$, below the $\mathcal{R} = 1$ of an integer-tick type. Treating the finer detection resolution as a larger $\mathcal{R}$ would invert Definition 7.6. Only detection resolution enters the tie-break of Definition 7.3; $\mathcal{R}$ is reported for completeness, is asserted analytically, and is not measured per monitor type in any simulation of this paper.

Definition 7.8 (Opener Sensitivity)**.** Let x be the opener tick (the opener-tick band definition [9]). For each signature $\sigma \in \Sigma_w$, let $\sigma^{x\pm1}$ denote the signature obtained by shifting the opener by ±1 tick (i.e. re-partitioning the trace with opener $x' = x \pm 1$). The *opener sensitivity* of monitor type w is the fraction of signatures whose window-class assignment changes under this perturbation:

$$\mathcal{S}(w) \;=\; \frac{1}{|\Sigma_w|} \sum_{\sigma \in \Sigma_w} \mathbb{1}\big[\, \mathsf{cl}(\sigma^{x-1}) \neq \mathsf{cl}(\sigma) \;\vee\; \mathsf{cl}(\sigma^{x+1}) \neq \mathsf{cl}(\sigma) \,\big], \tag{7.5}$$

where $\mathsf{cl}(\cdot)$ abbreviates $\mathsf{class}(\cdot)$. A monitor with $\mathcal{S}(w) = 0$ has *zero opener sensitivity*: no signature migrates to a different class when the opener shifts by one tick. Low sensitivity is desirable because it implies that the timing-error detector (Section 8.1.2) has a wider lock-in range around the nominal opener position.

Remark 7.9 (The Opener Tick x as the Sampling Instant). The optimality result depends on the eight-tick partition's explicit separation of the opener tick x (`x`) from the first response tick (`x+1`). Without the explicit `x`/`x+1` separation, the two ticks merge and the single-tick early–late count asymmetry $n_0 \neq n_1$ is undefinable. The introduction of `x` therefore serves a dual purpose: it anchors the optimality proof above and directly supplies the argument of the timing-error detector used in Self-tuning (Section 8.1.2). Placing the single-symbol-per-tick reduction inside the convention-algebra taxonomy of [8] is a declared timing-convention choice (WITNESSED); the opener-versus-first-response split then *corresponds to* the DIRRES$_{\mathsf{entry}}$ vs DIRRES$_{\mathsf{exit}}$ refinement, with `x` and `x+1` in the roles of the entry and exit edge of a run. We read this specific tick-to-edge identification as a structural analogy (HEURISTIC) rather than a literal instantiation — enough to show the `x`/`x+1` separation is principled rather than ad-hoc.

Remark 7.10 (Cost of the Half-Tick Probe). The half-tick probe doubles the number of samples per window (from k to $2k-1$). For the reference bound $k = 3$ this adds 2 samples — a negligible overhead for any monitor operating at microsecond or millisecond tick granularity.

8. Runtime Self-Tuning by Timing Recovery

8.1. **Self-Tuning by Timing Recovery.** The optimal monitor $w^\star$ selected in Section 7 still requires an accurate *phase alignment* between the monitor's sampling clock and the emitters' clocks. This section realises S4–S5 (convergence and closure) of Section 5 and, through the timing-error statistics, feeds the calibration clamp S6. In this section we design a closed-loop self-tuning algorithm that estimates and corrects the residual phase offset, drawing on the classical theory of symbol synchronisation in digital communications.

Connection to the observation kernel. In the language of the observation-kernel section [9], Self-tuning is the runtime adaptation of the parameter vector $\boldsymbol{\theta}$ of the observation kernel $\mathcal{K}_{\boldsymbol{\theta}}$. Each update $\boldsymbol{\theta}_n \to \boldsymbol{\theta}_{n+1}$ changes which layer-2 (measured) events become visible as layer-3 (observed) events, thereby reshaping the eight-tick signature that the monitor assigns to the current monitoring cycle. The design goal is convergence of $\mathcal{K}_{\boldsymbol{\theta}}$ to a *stationary operating point* at which the monitor's observations consistently occupy the `x+1`–`m2` subbands (Corollary 8.10), minimising timing ambiguity and maximising diagnostic resolution.

Remark 8.1 (Self-tuning invariance). Self-tuning modifies *only* the kernel parameter vector $\boldsymbol{\theta}$; it does not alter the crossing/touch classification (the crossing/touch observation primitives [9]), the pruning rules, or the eight-tick partition itself. Consequently, the Robbins–Monro convergence guarantee (Theorem 8.8) depends on the symmetry of the early/late counts and the gain schedule—both of which are properties of the *update rule*, not the classification. The revised detection-theoretic objective (the detection-theoretic objective [9]) is compatible with the existing proof because minimising eight-tick Hamming distance is equivalent to driving the timing-error signal e_n toward zero.

Self-tuning therefore affects only the observation kernel parameters and leaves the combinatorial and algebraic structure of the monitor classes unchanged. Its convergence analysis concerns parameter stability—specifically, the Lyapunov-type condition on the gain sequence $\sum \gamma_n = \infty$, $\sum \gamma_n^2 < \infty$—not semantic optimisation of Φ_k.

8.1.1. *Tunable parameters.*

Definition 8.2 (Tunable State)**.** The monitor maintains a tunable state vector $(\hat{\varphi},\, M^-,\, M^+,\, \theta,\, \psi)$:

- $\hat{\varphi} \in \mathbb{R}$: estimated phase offset (units: ticks) between the monitor's local clock and the emitter reference.
- $M^-, M^+ \in \mathbb{R}_{\geq 0}$: early and late timing margins. The monitor opens the observation window at $t_{\text{trigger}} - M^-$ and closes it at $t_{\text{trigger}} + k + M^+$.
- $\theta \in [0, 1]$: decision threshold for the timing-error detector (TED). A TED output $|e_n| > \theta$ is declared a significant timing error.
- $\psi \in \{0, \frac{1}{2}\}$: probe phase offset; $\psi = 0$ disables the probe, $\psi = \frac{1}{2}$ places it at the half-tick point (as selected by $w^\star$).

8.1.2. *Timing-error signal.* The half-tick probe of $w^\star$ yields two samples per symbol period: one at the opener tick x (`x`) and one at $x+\frac{1}{2}$ (straddling the `x`/`x+1` boundary). The eight-tick partition [9] anchors the early–late decomposition directly to the opener:

Definition 8.3 (Early and Late Confirmation Counts)**.** Let W successive trigger events each produce an eight-tick observation. The *early count* n_0 and *late count* n_1 are:

$$n_0 \;=\; \big|\{\text{ confirming samples in } \texttt{pre}, \texttt{x-1}, \text{ or } \texttt{x}\}\big|, \tag{8.1}$$

$$n_1 \;=\; \big|\{\text{ confirming samples in } \texttt{x+1}, \texttt{m2}, \text{ or } \texttt{cl}\}\big|, \tag{8.2}$$

accumulated over the sliding window of W events. A sample *confirms* if it observes $Q = \top$ (consequent present).

Because `x` and `x+1` occupy adjacent single-tick bands, the boundary between n_0 and n_1 coincides exactly with the opener tick x, giving the timing-error detector single-tick resolution. When the monitor's clock leads the emitter, confirmations cluster in `pre`/`x-1`/`x` ($n_0 > n_1$); when it lags, they cluster in `x+1`/`m2`/`cl` ($n_0 < n_1$).

Definition 8.4 (Early–Late Timing Error)**.** The timing-error signal is the normalised imbalance between early and late confirmation counts:

$$e_n \;=\; \frac{n_0 - n_1}{n_0 + n_1 + \epsilon}, \tag{8.3}$$

where $\epsilon > 0$ is a small constant preventing division by zero when no confirmations are available. At optimal alignment ($\hat{\varphi} = \varphi^\star$), confirmations are symmetric about x and $e_n \approx 0$.

The signal e_n is the discrete-time analogue of an *early–late* timing-error detector, which differences samples taken early and late of the nominal strobe [13]. The key insight is that the eight-tick partition provides a natural *decision boundary* at the opener tick x: everything in `x` or earlier counts as "early", everything in `x+1` or later counts as "late". It is the counterpart of the three-sample early–late (Alexander) phase detector of clock/data recovery, which decides from binary samples whether the clock leads or lags the data [25].

Remark 8.5 (Degeneracy on silent windows and the touch remedy). On a (0,0,0) trace the early–late error is degenerate: no sample confirms, so $n_0 = n_1 = 0$ and $e_n = 0/\epsilon$ carries no gradient — the loop receives the same null signal from a slightly-miscalibrated threshold as from a hopeless one. The touch channel of Proposition A.3 restores a usable error precisely here: when the probe fires at peak value $s_q(m)$, the *peak deficit* $\theta_q - s_q(m) \in (0, \delta]$ is a signed calibration error that can drive the Robbins–Monro update of Section 8.1.5 in place of e_n, at the capability cost $(3, 0, 1)$ of one memoryless window channel [8]. Two robustness observations of [8], which that paper presents as informal, heuristic sketches, qualify this remedy and its dead-band cure. First, the touch margin ρ_B is *quadratic* at extrema, against the crossing margin ρ_A linear in $|Q - \theta|$, so a touch read taken at a peak is intrinsically the more fragile of the two near the extremum: the peak-deficit signal used above is cheap in capability but shallow in noise margin. Second, the hysteresis dead-band of Θ_2 is exactly the *robustness-restoring* tuning that the same paper describes (the Schmitt-trigger construction). Both are imported from [8] as heuristic observations, not re-derivations here.

The early–late detector is the discrete cousin of the classical Gardner and Mueller–Müller detectors [23] of symbol synchronisation; the correspondence, the mapping of (L, S, D, J) onto standard signal-processing quantities, and the block architecture of the loop are developed below (Sections 8.4 and 8.6).

8.1.3. *PLL/DLL feedback loop.*

Definition 8.6 (Phase Update Rule)**.** The monitor updates its phase estimate after each trigger event using a first-order PLL/DLL loop:

$$\hat{\varphi}_{n+1} \;=\; \hat{\varphi}_n \;-\; \mu_n\, e_n, \tag{8.4}$$

where $\mu_n > 0$ is the step size at iteration n. An optional integrating (second-order) branch adds frequency tracking:

$$\hat{\varphi}_{n+1} \;=\; \hat{\varphi}_n - \mu_n^{(P)}\, e_n - \mu_n^{(I)} \sum_{i=1}^{n} e_i, \tag{8.5}$$

where $\mu_n^{(P)}$ and $\mu_n^{(I)}$ are the proportional and integral gains.

The update Equation (8.4) is a standard form for a numerically-controlled oscillator (NCO) in a digital PLL or a delay-locked loop (DLL). When the emitter drift D is non-zero, the second-order form Equation (8.5) is necessary to track the linearly increasing offset.

8.1.4. *Margin adaptation.* In addition to phase correction, the monitor adapts its early and late margins M^-, M^+ based on the asymmetry of confirmations around the opener tick x.

Definition 8.7 (Margin Update Rules)**.** After each trigger event, the margins are updated as:

$$M_{n+1}^- \;=\; M_n^- \;+\; \alpha_n \,\max\big(n_0 - n_1,\; 0\big), \tag{8.6}$$

$$M_{n+1}^+ \;=\; M_n^+ \;+\; \alpha_n \,\max\big(n_1 - n_0,\; 0\big), \tag{8.7}$$

where $\alpha_n > 0$ is a decaying margin step size satisfying the same Robbins–Monro conditions as μ_n (Items (C1) and (C2)).

The intuition is direct: if confirmations cluster before the opener ($n_0 > n_1$), the early margin M^- is too small and must grow to capture evidence arriving ahead of x. Conversely, if $n_1 > n_0$, the late margin M^+ must grow. Once the phase estimator converges ($e_n \approx 0$), both margins stabilise because $|n_0 - n_1| \to 0$.

Adaptivity interpretation. The five tunable parameters $(\hat{\varphi}, M^-, M^+, \theta, \psi)$ define the monitor's adaptive state. *Broadening* the observation window (increasing k or widening M^-, M^+) enlarges the set of reachable window classes, making more of the eight window classes reachable but increasing diagnostic ambiguity. *Restricting* the window (decreasing k, tightening margins, raising θ) narrows the reachable classes towards (1,0,0), reducing ambiguity at the cost of higher false-negative risk. Selection (Section 7) selects the monitor *type* that minimises ambiguity; Self-tuning adapts the monitor's *state* so that its observations converge to the static-like regime (Corollary 8.10). Neither stage introduces new patterns or alters the catalogue; they only control which subset of the eight window classes is reachable.

8.1.5. *Convergence.*

Theorem 8.8 (Local Convergence of the Phase Estimator)**.** *Let $\varphi^\star$ denote the true phase offset and write $h(\theta) = \mathbb{E}[e_n \mid \hat{\varphi}_n = \theta]$ for the mean-field of the timing-error signal. Assume:*

(C1) $\sum_{n=1}^{\infty} \mu_n = \infty$;

(C2) $\sum_{n=1}^{\infty} \mu_n^2 < \infty$;

(C3) $h(\varphi^\star) = 0$ *(equivalently $\mathbb{E}[e_n \mid \hat{\varphi}_n = \varphi^\star] = 0$), and $\varphi^\star$ is the only root of h in the basin B_ρ below;*

(C4) $\sup_n \mathrm{Var}(e_n \mid \hat{\varphi}_n) < \infty$;

(C5) (mean-field sign / regression condition.) *There is a radius $\rho > 0$ defining a basin $B_\rho = \{\theta : |\theta - \varphi^\star| \le \rho\}$ such that, for every $\varepsilon \in (0, \rho)$, there is a constant $c_\varepsilon > 0$ with*

$$\inf_{\varepsilon \le |\theta - \varphi^\star| \le \rho} (\theta - \varphi^\star)\, h(\theta) \;\ge\; c_\varepsilon > 0,$$

together with the growth bound $|h(\theta)| \le c\,(1 + |\theta|)$.

Then, from any initialisation $\hat{\varphi}_0 \in B_\rho$, the estimate converges to the true phase offset, $\hat{\varphi}_n \xrightarrow{\text{a.s.}} \varphi^\star$ as $n \to \infty$.

Proof. Conditions (C1)–(C5) adapt the hypotheses of the Robbins–Monro stochastic-approximation scheme [26] — which proves convergence in probability under uniformly bounded responses — to a bounded conditional variance with a growth bound, localised to the basin B_ρ. Writing the error signal as $e_n = h(\hat{\varphi}_n) + \xi_n$ with ξ_n zero-mean and of bounded variance ((C4)), condition (C3) places the unique mean-field root at $\varphi^\star$, and (C1)–(C2) are the gain-schedule conditions. Condition (C5) is the mean-field sign (regression) condition that Robbins–Monro requires and that the original four hypotheses omit: it makes $V(\theta) = (\theta - \varphi^\star)^2$ a strict stochastic Lyapunov function on B_ρ. Indeed, with the update Equation (8.4),

$$\mathbb{E}[V(\hat{\varphi}_{n+1}) - V(\hat{\varphi}_n) \mid \hat{\varphi}_n = \theta] \le -2\mu_n\,(\theta - \varphi^\star)\,h(\theta) + \mu_n^2\,\mathbb{E}[e_n^2 \mid \theta],$$

whose leading term is strictly negative away from $\varphi^\star$ by (C5) and whose second term is summable by (C2) and the growth bound. The Robbins–Siegmund almost-supermartingale lemma then yields $\hat{\varphi}_n \to \varphi^\star$ almost surely for every trajectory confined to B_ρ. The small-signal analysis of Section 8.7 exhibits the local contraction rate $1 - \mu_n K_d$, with $K_d > 0$ the S-curve slope at $\varphi = 0$; this is precisely the linearisation of (C5) at the root.

Condition (C3) is the TED's design symmetry: at optimal alignment the n_0/n_1 counts (Definition 8.3) about the opener tick x are *assumed* to be balanced, so that the `pre`/`x-1`/`x` and `x+1`/`m2`/`cl` halves of the eight-tick partition receive equal expected confirmation mass when $\hat{\varphi} = \varphi^\star$. This is a modeling hypothesis — made explicit here rather than derived — which the simulation of Section 8.1.6 supports empirically. A standard choice $\mu_n = \mu_0/n^\alpha$ with $\alpha \in (\frac{1}{2}, 1]$ satisfies both (C1) and (C2). □

Remark 8.9 (Global convergence: witnessed prefix, conjectured tail)*.* Theorem 8.8 is *local*: it certifies almost-sure convergence only from initialisations inside the basin B_ρ on which the mean-field sign condition (C5) holds. Global monotonicity of the band-counting mean-field h across the whole operating interval is *not* established here, and does not hold for a fixed-span eight-tick window: once $|\theta - \varphi^\star|$ exceeds the window support, confirmation mass leaves

the window, the n_0/n_1 counts saturate, and $h(\theta) \to 0$ — the same gradient collapse recorded for the silent-window degeneracy in Remark 8.5. This gradient collapse is no longer merely predicted: the de-circularised experiment of Section 8.1.6 (`convergence_experiment.py`) *measures* the band-counting mean-field $h(\theta) = \mathbb{E}[e_n \mid \hat{\varphi} = \theta]$ directly and finds exactly the conjectured shape — an odd, restoring S-curve that saturates at ± 1 over $|\theta| \in [1, 3]$ and then *collapses* back toward 0 for $|\theta| \gtrsim 4$ [WITNESSED]. The loop's basin is therefore finite and *measured*: convergence from an initial offset $|\hat{\varphi}_0| \lesssim 4$ ticks, and stall (initial gradient ≈ 0 in the collapse zone) for $|\hat{\varphi}_0| \geq 5$ [WITNESSED]. Global almost-sure convergence from an *arbitrary* initial offset is therefore [CONJECTURED], not proved — and is now *bounded* by the measured finite basin ($|\hat{\varphi}_0| \lesssim 4$) and by the measured first-order drift breakpoint ($D \approx 0.02$; Section 8.1.6). The finite-prefix experiment of Section 8.1.6 now [WITNESSED] exhibits convergence of the *real* band-counting detector from $\hat{\varphi}_0 = 2.5$ over $n \leq N = 200$ iterations (no longer the abstract identity-map recursion); the tail $n > N$ is explicitly uncovered, since no finite observation can discharge an $n \to \infty$ claim. This is consistent with S4 being a zero-sided obligation discharged offline, never by runtime checking (Section 5).

Corollary 8.10 (Regime Migration)**.** *Under the conditions of Theorem 8.8, and from an initialisation within the basin B_ρ, after a transient phase the monitor's observations migrate from the miscalibration regime (A=0) to the bounded-temporal regime (A=1 $\wedge$ ($B \vee C$)) and eventually to the static-like regime* (1,0,0)*, provided the emitters satisfy the ground-truth implication.*

Proof. As $|\hat{\varphi}_n - \varphi^\star| \to 0$, the effective observation offset decreases below the thresholds that separate the miscalibration, bounded-temporal, and static-like regimes (the block names of Section 3, over the static-like and bounded temporal regimes of the core paper [9]). The almost-sure convergence of the phase error to zero implies that each regime boundary is eventually crossed; the migration therefore holds in the almost-sure/eventual sense (the Robbins–Monro trajectory need not decrease monotonically). □

8.1.6. *Simulation evidence.* To illustrate the convergence behaviour of the phase-update recursion Equation (8.4) and the class migration of Corollary 8.10, we run a lightweight stochastic simulation with parameters: $\hat{\varphi}_0 = 2.5$ ticks (initial miscalibration placing the monitor in the (0,1,0)/(0,1,1) regime), $\varphi^\star = 0$, $\mu_n = 0.5/n^{0.6}$, noise $\sigma = 0.4$, $N = 200$ iterations.

What is and is not witnessed. Two data-generating processes bear on the convergence claim, and it is worth separating them. The original generator behind this section was the *abstract* stochastic-approximation recursion $\hat{\varphi}_{n+1} = \hat{\varphi}_n - \mu_n e_n$ under a *linear* restoring mean-field $h(\theta) = \theta$ with additive zero-mean noise ($e_n = \hat{\varphi}_n + \xi_n$, ξ_n zero-mean of scale σ, on a stationary root $\varphi^\star = 0$). Read alone, such a trajectory would [WITNESS] only convergence of the abstract recursion, with the restoring field *assumed* rather than derived from the n_0/n_1 counts — the circularity we previously flagged. *That circularity is now removed*, and Figure 9 now plots the *real* band-counting detector, not that recursion. The de-circularised experiment `convergence_experiment.py` never sets $e_n = \hat{\varphi}_n + \xi_n$; it generates confirmation events at positions fixed by the *true* offset $\varphi^\star$ (with optional emitter drift D), runs the actual detector — partitioning them at the opener tick, counting n_0 (early) against n_1 (late) exactly as Definition 8.3 and forming $e_n = (n_0 - n_1)/(n_0 + n_1 + \varepsilon)$ exactly as Definition 8.4 — and drives $\hat{\varphi}$ with the paper's own update Equation (8.4). The mean-field is therefore an

TABLE 2. Convergence checkpoints and regime classification.

n	$\hat{\varphi}_n$	e_n	μ_n	**Regime**
0	2.500	0.000	0.5000	M5–M8
10	0.351	0.667	0.1256	M2–M4
25	0.076	0.083	0.0725	M1
50	0.003	0.250	0.0478	M1
100	-0.018	0.083	0.0315	M1
150	-0.033	-0.500	0.0247	M1
200	-0.008	0.083	0.0208	M1

emergent property of the band geometry, not an input. Under this de-circularised generator we [WITNESS] the following, all over 500 independent runs and stable across seeds.

(i) Non-circular finite-prefix convergence. From the paper's own initialisation $\hat{\varphi}_0 = 2.5$ ($\varphi^\star = 0$), the real band-counting loop converges in 100% of 500 runs over $n \le N = 200$, with mean final error ≈ 0.012 ticks (max 0.054). The [WITNESSED] finite-prefix convergence of Theorem 8.8 now holds for the *real* detector, not merely the abstract recursion.

(ii) The conjectured field is measured. Direct Monte-Carlo measurement of $h(\theta) = \mathbb{E}[e_n \mid \hat{\varphi} = \theta]$ yields an odd, restoring S-curve: near-linear through the origin, saturating at ± 1 over $|\theta| \in [1, 3]$, and *collapsing* back toward 0 for $|\theta| \gtrsim 4$ — exactly the saturate-then-collapse mechanism Remark 8.9 predicted. The conjectured mechanism is now a measured fact [WITNESSED].

(iii) Finite basin. The loop converges for $|\hat{\varphi}_0| \lesssim 4$ ticks and stalls for $|\hat{\varphi}_0| \ge 5$: a sharp basin wall between $\hat{\varphi}_0 = 4$ and 5, where the initial gradient lies in the measured collapse zone (window half-support = 3 ticks) [WITNESSED].

(iv) Drift stressor and the second-order loop. Under a non-zero emitter drift $\varphi^\star_n = D\, n$, the first-order loop Equation (8.4) converges up to $D = 0.01$ (steady-state lag growing with D) and *breaks* at $D \ge 0.02$ (convergence fraction 0, mean error 0.53 ticks); the second-order form Equation (8.5) (integral branch, $\mu_I = 0.002$) *recovers* 100% convergence up to $D = 0.05$ (mean error ≤ 0.05 ticks), confirming the integral branch is what a linear ramp requires [WITNESSED].

What remains [CONJECTURED] is exactly, and only, the global claim: uniform almost-sure convergence from an *arbitrary* initial offset. It is unchanged by these experiments — but now *bounded* by the measured finite basin ($|\hat{\varphi}_0| \lesssim 4$) and drift breakpoint ($D \approx 0.02$ first-order, recovered to $D = 0.05$ by Equation (8.5)). Figure 9 below plots one *representative* trajectory of the real band-counting detector from $\hat{\varphi}_0 = 2.5$, illustrating the regime migration of Corollary 8.10; the aggregate convergence evidence is the 500-run experiment reported in this paragraph, of which the plotted run is one instance.

Figure 9 shows the trajectory $\hat{\varphi}_n$ against the three regime bands of the classifier. The phase estimate leaves the miscalibration band within the first few iterations, crosses the bounded-temporal band, and settles inside the (1,0,0) band by $n = 11$; the residual fluctuations are consistent with the noise floor $\sigma = 0.4$ and the decaying step size.

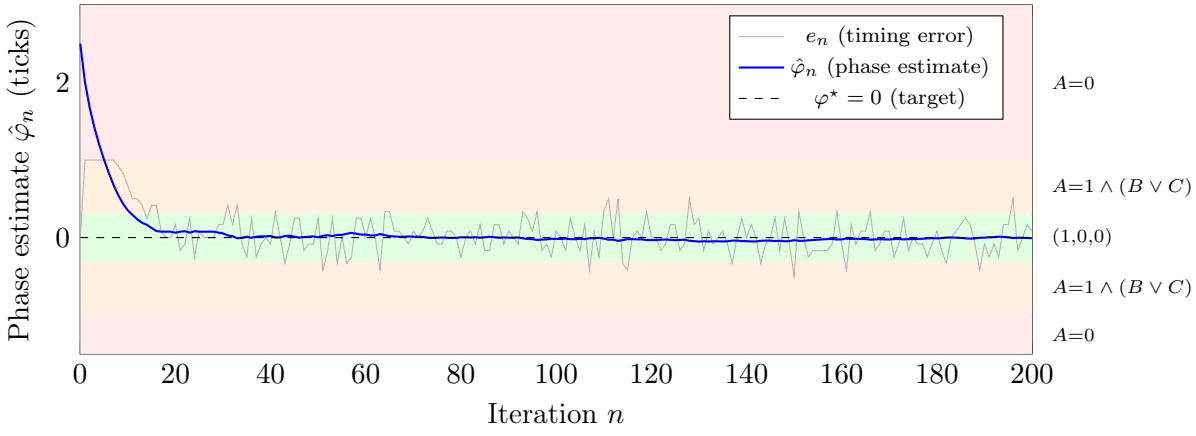


FIGURE 9. *Self-tuning re-reads the same implication up the observation-class ladder.* The vertical axis is the monitor's phase error $\hat{\varphi}_n$; each horizontal band translates that error into the window class the monitor would report *on one and the same physical implication.* A large error throws the trigger and its response into the wrong bands, so the implication is read as a miscalibration class ($|\hat{\varphi}_n| \geq 1$: A=0, red); a moderate error reads bounded-temporal ($0.3 \leq |\hat{\varphi}_n| < 1$: A=1 ∧ (B ∨ C), orange); a near-zero error reads the clean static-like class ($|\hat{\varphi}_n| < 0.3$: (1,0,0), green). As the self-tuning loop drives $\hat{\varphi}_n \to \varphi^\star = 0$ (blue), the observed class is *not* fixed: it climbs from miscalibrated toward (1,0,0) as the monitor calibrates. The trajectory settles in (1,0,0) by $n = 11$ and stays there, the timing error e_n (grey) then fluctuating around zero — the geometric form of Corollary 8.10 (Remark 8.9). This is a *single representative run* of the real band-counting detector (timing error from the n_0/n_1 counts of Definition 8.4) from $\hat{\varphi}_0 = 2.5$ under a fixed seed (`convergence_experiment.py`), not the abstract linear-field recursion; the aggregate witness is the 500-run de-circularised experiment above (100% of 500 runs, mean final error ≈ 0.012 ticks), of which this shows one run.

Table 2 reports the regime label at selected iterations of the same representative run. The monitor migrates from the miscalibration regime (A=0) at $n = 0$ through the bounded-temporal regime (A=1 ∧ (B ∨ C)) and settles in the static-like regime (1,0,0) by $n = 11$, realising Corollary 8.10 (Remark 8.9) for the *real* band-counting detector. Aggregate finite-prefix convergence over $n \leq N$ is witnessed by the 500-run de-circularised experiment of the preceding paragraph; the tail $n > N$ remains uncovered.

The same migration is confirmed at the population level by the four-epoch class distribution of Section 8.7.1 (Table 4).

8.2. **Powers as transductions on band-words.** The relaxation lattice above catalogues *which* axioms a power may break; we now record *what a power does to an observed word.* A power is naturally read as an operation on the band-language L^N_{band} of Section 6.1: it maps the observed band-word to an *enforced* band-word that the downstream verdict logic then classifies. Two families recur.

Definition 8.11 (Band-word powers, CONJECTURED framework)**.** Let $w \in L_{\text{band}}^N$ be an observed band-word.

INCLUDE(e, band $\to$ band$'$)**.:** Relabel an out-of-band event e as belonging to a target band under a *declared premise*. The paradigm case is a $Q^{\uparrow}$ landing in the `post` band — *late evidence* of a consequent validation — counted as an in-window validation. This is *attribution*, not manufacture: the event was genuinely observed (the inviolable floor Σ_2 of Remark 5.7 is untouched); only its band membership is re-assigned, and only under an explicitly stated observation premise. INCLUDE is the band-word face of the moving-window and phase powers (Γ_2, Γ_3) already carried by the self-tuning loop of Section 8.1: widening k or advancing $\hat{\phi}$ is precisely what draws a `post`-band $Q^{\uparrow}$ back inside the window.

EXCLUDE(e)**.:** Suppress an in-band glitch, making it *impermanent*: a spurious $Q^{\downarrow}$ inside the window is filtered out (debounce $\tau_{\min}$, Σ_1) or absorbed by a hysteresis dead-band (Θ_2). Again Σ_2 is respected — an event that corresponds to no real behaviour is what is removed, not a real one minted away.

The available powers assemble into a *transduction*

$$T \,:\, L_{\text{band}}^N \longrightarrow L_{\text{band}}^N, \qquad w \longmapsto T(w) \text{ (observed word} \mapsto \text{enforced word)},$$

bounded by the monitor's realisable operations and by the declared assumptions on the monitored system.

Remark 8.12 (Powers are the self-tuning catalogue, lifted to words). INCLUDE and EXCLUDE are not new operations: they are the relaxation-lattice powers of Section 5.4 read as edits on the observed band-word rather than as parameter moves. In particular INCLUDE realises the same late-evidence accounting that the declared theory ladder of the core paper [9] performs (Remark 5.11), and the self-tuning "power" of Section 8.1 is the online, closed-loop application of T. The framework of this subsection is therefore a reorganisation of established machinery; we tag it CONJECTURED because the transduction T has not been given a closed operational semantics here, only a schema.

Remark 8.13 (Open problem: rationality of the power transduction). Is T a *rational* transducer — one recognised by a finite-state transducer, hence regular in both its input and output projections? Because each band content is bounded by N (Proposition 6.1), a positive answer would make the enforcement objectives of Section 8.3 ω-regular and the reachability question decidable. The candidate construction is the transducer closure of the prefix trie of Proposition 6.7. This is left [OPEN].

A quantitative treatment of this lattice — relaxation costs and least-effort tuning paths — is left to future work in the monitor-parametrisation line.

8.3. **Enforcement as temporal reachability over the window-stream.** The self-tuning loop just analysed acts within a single trigger window. We now read its *repeated* action across an infinite stream of windows, and cast the objective of the powers of Section 8.2 as a temporal-reachability problem. Throughout, the epistemic discipline of the core paper [9] is enforced: this crosses from a finite window to an infinite stream, so no claim here is upgraded to PROVEN beyond the finite-prefix reach of Theorem 8.8 and Remark 8.9.

The window-stream. An infinite trace is a *stream of band-words*

$$W \;=\; w_0\, w_1\, w_2\, \cdots, \qquad w_i \in L^N_{\text{band}},$$

one observed signature per trigger window (Section 6.1). The powers of Definition 8.11 act windowwise through the transduction T, producing the *enforced stream* $T(W) = T(w_0)\,T(w_1)\,T(w_2)\cdots$. Fix a *target signature* $\sigma^* \in L^N_{\text{band}}$ — typically the antecedent-validated reference class R — and a start signature σ_0. Writing σ^* also for the per-window predicate "the enforced window equals σ^*", the enforcement objectives are the two lifts of Proposition 6.9 applied to the enforced stream.

Definition 8.14 (Enforcement objectives, CONJECTURED)**.** Over the enforced stream $T(W)$:

(a) **Recurrence (liveness).** $\mathbf{G}\,\mathbf{F}\;\sigma^*$: the enforced stream exhibits σ^* infinitely often — enforcement recurrently succeeds. By Proposition 6.9(c) this is an $\mathsf{L} \cap \mathsf{K}$ objective.

(b) **Persistence.** $\mathbf{F}\,\mathbf{G}\;\sigma^*$: eventually every enforced window is σ^* — enforcement *stabilises*.

The *enforcement goal* is that $\mathbf{F}\,\mathbf{G}\;\sigma^*$ or $\mathbf{G}\,\mathbf{F}\;\sigma^*$ be reachable from σ_0 under the transduction T, written $\sigma_0 \rightsquigarrow_T \sigma^*$.

Remark 8.15 (Persistence *is* the self-tuning convergence). The persistence objective $\mathbf{F}\,\mathbf{G}\;\sigma^*$ is not a new claim: it is exactly the self-tuning convergence of Theorem 8.8. As the phase estimate $\hat{\varphi}_n$ enters and stays in the basin B_ρ, the observation regime migrates to the static-like class (1,0,0) (Corollary 8.10) and every subsequent enforced window equals the reference signature — eventually-always σ^*. Consequently $\mathbf{F}\,\mathbf{G}\;\sigma^*$ *inherits the honest status* of Remark 8.9 verbatim: local almost-sure convergence inside B_ρ is PROVEN; global convergence from an arbitrary σ_0 is CONJECTURED; and the finite-prefix simulation of Section 8.1.6 WITNESSES it only for $n \le N = 200$, with the tail $n > N$ explicitly uncovered, beyond what any finite observation can discharge. We introduce *no* new PROVEN persistence claim.

Remark 8.16 (Realising the deferred third regularity source of the refinement framework, CONJECTURED). The refinement-confirmation framework of [7] (its contribution O5) lifts refutation-only verdict traces to finite-time confirmation of properties that, on the infinite trace, lie in non-monitorable cells such as $\mathsf{L} \cap \mathsf{K}$. It develops two sources of regularity — the intrinsic regularity of the verdict algebra and the extrinsic regularity of a process behind the monitor — and names a *third*, "regularity produced by the refinement system itself when the monitor is parametric and its parameters can be adapted online against a target Layer-2 property", which is explicitly placed outside that paper's scope and left to future dedicated treatment. The Stage-C self-tuning monitor of the present paper is a concrete realisation of exactly that deferred source: it is parametric in the observation-kernel vector $\boldsymbol{\theta}$ (Section 8.1) and adapts those parameters online against the target signature σ^* through the closed timing-recovery loop. On this reading the persistence objective $\mathbf{F}\,\mathbf{G}\;\sigma^*$ of Definition 8.14 is discharged by the *convergence* of the parametric loop (Theorem 8.8 and Corollary 8.10) rather than by runtime monitoring of an infinite trace — precisely the move that [7] defers. The realisation is tagged CONJECTURED: the cited framework is itself conjectured at the framework level (its scope remark for O5), and the realisation inherits the honest split of Remark 8.15 unchanged — local almost-sure convergence inside B_ρ stays PROVEN and global convergence from an arbitrary σ_0 stays CONJECTURED. No new PROVEN claim is introduced.

Remark 8.17 (Recurrence and reachability are conjectures). The recurrence objective $\mathbf{G}\,\mathbf{F}\ \sigma^*$ and the reachability relation $\sigma_0 \rightsquigarrow_T \sigma^*$ are genuinely infinite-trace claims with *no* finite witness: no finite prefix of $T(W)$ establishes "σ^* infinitely often", and reachability quantifies over the whole future of the stream. Absent the positive resolution of Remark 8.13 — which would make these ω-regular and decidable at the level of the *realised* transduction T — both are tagged CONJECTURED. Whether a given $\mathbf{G}\,\mathbf{F}\ \sigma^*$ or $\mathbf{F}\,\mathbf{G}\ \sigma^*$ is enforceable, given a power set and the declared system assumptions, is a *controllability* question (the monitor as a controller against its environment) that ties this framework to runtime enforcement and shielding; we position the *adversarial* form as future work, not a result of this paper. At the finite *class quotient*, and on a *cooperative* (edge-affording) signal, a modest *controllability* fact does hold, conditionally: the restricted class-quotient controllability proposition (Proposition 8.21) shows both objectives controllable with memoryless strategies once the estimate stays in the convergence basin (H1a), the required migration edges are afforded (H1b), and the transitions are causally coherent (H2). What this remark leaves CONJECTURED is exactly the *adversarial enforcement* problem — steering against an environment that may withhold edges (H1b dropped) — together with the two *unconditional* residuals dropped by the other hypotheses: (R1) basin-free global steering over an arbitrary initial offset, which the measured mean-field collapse (Remark 8.9) makes false for a fixed-span window; and (R2) CC-free soundness, which is the causal-necessity theorem owned by the core paper [9], a cross-paper dependency not re-proven here.

Remark 8.18 (Descriptive-set-theoretic level of the two objectives, WITNESSED). The two objectives of Definition 8.14 are not merely analogous to but are *instances* of the universal-attribution theorem of [7]: for any verdict monitor with verdict alphabet of size ≥ 2, reading the per-window predicate σ^* as a verdict makes the persistence pullback $\mathbf{F}\,\mathbf{G}\ \sigma^*$ land at level $\mathbf{\Sigma}^0_2$ and the response pullback $\mathbf{G}\,\mathbf{F}\ \sigma^*$ at level $\mathbf{\Pi}^0_2$; both are pre-images of $\mathsf{L}\cap\mathsf{K}$ properties of the verdict stream, and lie in $\mathsf{L}\cap\mathsf{K}$ themselves when the monitor is non-degenerate at σ^*. The Borel levels therefore come for free: the persistence ($\mathbf{F}\,\mathbf{G}$) objective is $\mathbf{\Sigma}^0_2$ and the recurrence ($\mathbf{G}\,\mathbf{F}$) objective is $\mathbf{\Pi}^0_2$. Once self-tuning converges the observation regime migrates to the static-like class (1,0,0) (Corollary 8.10). The absorbing-versus-non-absorbing dichotomy of [7] concerns the verdict monitor, not the class trajectory: a pullback lies in $\mathsf{L}\cap\mathsf{K}$ exactly when the monitor is non-absorbing at that verdict. It does not make the persistence objective finitely confirmable: that objective stays in $\mathsf{L}\cap\mathsf{K}$, neither finitely confirmable nor finitely refutable, and is discharged by the convergence proof (Remark 8.15), not by monitoring. This grounding is WITNESSED (an application of two proven external results); it does *not* upgrade the reachability claim of Remark 8.17, which stays CONJECTURED.

8.3.1. *A restricted, proven-conditional controllability result.* The objectives of Definition 8.14 are conjectural at the level of the *realised* transduction T (Remarks 8.13 and 8.17). On the finite class quotient, and on a *cooperative* (edge-affording) signal, a modest *controllability* fact holds, and it is worth isolating exactly how much this buys — and at what cost — because the gap between it and the adversarial enforcement question is the honest boundary of the enforcement layer.

Definition 8.19 (Class-quotient arena)**.** The *class-quotient arena* is the finite directed graph

$$\mathcal{A} \;=\; \big(Q,\ \mathrm{Act},\ \delta\big), \qquad Q = \{(1,0,0),\dots,(0,1,1)\}, \quad \mathrm{Act} = \{T_1,\dots,T_9\}\cup\{\mathrm{hold}\},$$

whose states are the eight observation classes, whose actions are the nine tuning operations of Table 10 together with a null action hold (apply no tuning), and whose transition relation $\delta \subseteq Q \times \mathrm{Act} \times Q$ contains (C_i, T_j, C_k) iff applying T_j to an C_i-classified observation yields an C_k-classified one — the distance-one migration edges underlying the reachability matrix of Table 11 — and the self-loop $(C_i, \mathrm{hold}, C_i)$ for the stationary re-reading of a class under no tuning. By Proposition A.1 $\mathcal{A}$ is *finite*, *strongly connected* (every ordered pair of classes is joined by a tuning path — at most two edges, three for (1,0,0)→(0,1,1) and (0,1,0)→(1,0,1) via (1,1,1)) and *trap-free* (every class has an outgoing migration).

Remark 8.20 (Rationality holds at the quotient, not for T as realised). Passing to $\mathcal{A}$ reframes Remark 8.13. The induced class-level transduction — "observed class ↦ enforced class under an action word" — is a relation over the finite set Q, and is therefore *rational by construction*: it is recognised by the finite automaton $\mathcal{A}$ itself. This does *not* resolve Remark 8.13, which asks whether T *as realised* is rational. The realised T carries the continuous adaptive memory $\hat{\varphi} \in \mathbb{R}$ of the self-tuning loop (Section 8.1) and is therefore *not* a finite-state object; only its projection onto the eight-class quotient is. The rationality we use below is the quotient's, and is bought precisely by discarding $\hat{\varphi}$ — the excluded interval we reintegrate in Remark 8.22.

Proposition 8.21 (Restricted class-quotient controllability, PROVEN-CONDITIONAL)**.** *Assume*

(H1a) (basin membership.) *The self-tuning estimate stays in the convergence basin $\hat{\varphi} \in B_\rho$ of Theorem 8.8;*

(H1b) (edge affordance.) *every migration edge of $\mathcal{A}$ used by the strategy is signal-afforded on the observed window (in particular the* hold *self-loop at* (1,0,0) *is afforded). This is an explicit assumption on the environment/signal — a* cooperative-environment *hypothesis — and is* not *a consequence of* H1a*: a transient silent window can leave $\hat{\varphi}$ in B_ρ while a needed edge is momentarily un-afforded;*

and, for soundness,

(H2) (causal coherence.) *every enforced transition satisfies the causal-coherence conditions CC1–CC4 and the causal-necessity theorem of the core paper* [9] *(soundness is* imported*, not re-proven here).*

Then, on the arena $\mathcal{A}$, the two objectives of Definition 8.14 are controllable *by the monitor, each via a* memoryless *(positional) strategy:*

(i) (Recurrence, a Büchi objective.) *For every target class $\sigma^* \in Q$ and every start class, the monitor has a positional strategy that visits σ^* infinitely often ($\mathbf{G}\,\mathbf{F}\,\sigma^*$).*

(ii) (Persistence to (1,0,0), a co-Büchi objective.) *The monitor has a positional strategy under which eventually every window is read in the absorbing static-like class* (1,0,0) *($\mathbf{F}\,\mathbf{G}\,$(1,0,0)). Here* hold *is the null action, so (ii) is the self-tuning convergence of Theorem 8.8 re-expressed as a co-Büchi persistence objective (Remark 8.15); it enforces nothing beyond convergence and is recorded as an honest observation, not a new contribution.*

Proof. Under H1b every action the strategy takes is afforded, so the monitor is the sole mover on $\mathcal{A}$ — a finite *one-player* (controllable) arena. Under H2 every such edge is sound, so a play is a *sound* adaptation. Both claims are then a three-line graph argument on the strongly connected finite arena $\mathcal{A}$ (Proposition A.1).

(i) Strong connectivity makes σ^* reachable from every class and puts it on a cycle through itself; the positional strategy "at each class take an edge on a fixed shortest tuning path toward σ^*" visits σ^* infinitely often (positional recurrence on a strongly connected finite graph).

(ii) (1,0,0) is reachable from every class (strong connectivity) and, under H1b, carries the afforded hold self-loop (Corollary 8.10); the positional strategy "head to (1,0,0) along a fixed shortest path, then hold" (reach-then-hold) keeps every window in (1,0,0) from some point on.

On a fixed finite arena decidability and the existence of memoryless strategies are automatic from strong connectivity, and are not the content here (standard for finite ω-regular arenas; see [15]). The finite arena and its strong connectivity and trap-freeness are supplied by Proposition A.1. □

Remark 8.22 (The excluded intervals: basin, collapse zone, and within-basin affordance failure). Proposition 8.21 lives on the finite quotient $\mathcal{A}$, and its guarantee is bounded by *three* excluded intervals, which we make explicit — without them the proposition would be epistemic inflation.

(1) The continuous phase memory. The quotient $\mathcal{A}$ discards the continuous phase memory $\hat{\varphi} \in \mathbb{R}$ carried by the realised monitor (Remark 8.20); the controllability claim is over the eight-class projection, not over the realised transduction T.

(2) The out-of-basin collapse zone. Outside B_ρ — concretely, for a measured initial offset $|\hat{\varphi}_0| \geq 5$ ticks — the band-counting mean-field collapses to $h(\theta) \approx 0$ (Remark 8.9, [WITNESSED]): the migrate-class action loses its drift and the hold self-loop at (1,0,0) drifts off. Hypothesis H1a excludes this zone (the finite basin $|\hat{\varphi}_0| \lesssim 4$).

(3) Within-basin affordance failure. Basin membership does *not* entail edge affordance: even with $\hat{\varphi} \in B_\rho$, a transient silent or un-afforded window can momentarily deny a migration edge or the hold self-loop the strategy needs. This within-basin affordance failure is exactly what hypothesis H1b assumes away; it is *not* the same as being outside the basin, and the earlier identification of "affordance failure" with "outside the basin" would be false. The proposition makes *no* claim on adversarial or transient within-basin un-affordance — an environment that may *withhold* edges rather than cooperate — which is the deferred adversarial-enforcement problem (Remarks 8.17 and A.2).

The honest scope is therefore: class-level $\mathbf{F}\,\mathbf{G}/\mathbf{G}\,\mathbf{F}$ *controllability* on the proven arena, *within the basin* and *on a cooperative (edge-affording) signal*, under CC1–CC4. The unconditional readings dropped by H1a, H1b and H2 — basin-free steering, adversarial enforcement, and CC-free soundness — remain open.

8.3.2. *Inclusion is liveness; exclusion is coliveness.* The two power families of Definition 8.11 carry, respectively, the two halves of the property algebra of [7], and this is the property-theoretic payoff of the powers.

- **Inclusion is liveness.** Including a useful out-of-band event — drawing a `post`-band $Q^{\uparrow}$ back into the window — is the recurrence property $\mathbf{G}\,\mathbf{F}$ include(e): progress is maintained until e is counted in the right band, infinitely often. It is a liveness/recurrence goal in the sense of [7], enforced by the INCLUDE powers.
- **Exclusion is coliveness.** Excluding an undesired in-band event — cancelling a glitch $Q^{\downarrow}$ — is the persistence property $\mathbf{F}\,\mathbf{G}\,\neg$ permanent(e): the glitch is *never permanent*, i.e. eventually-always cancelled. This is a never-finitely-confirmable (K-family, coliveness)

goal of [7] — one can never finitely certify that no future glitch will persist — enforced by the EXCLUDE powers.

Together the two drive $\sigma_0 \rightsquigarrow_T \sigma^*$: the liveness half reaches the target signature and the coliveness half cancels the undesired departures from it. Both lifts land in the *same* cell $\mathsf{L} \cap \mathsf{K}$ (Proposition 6.9(c); the witness-membership lemma of [7] places both $\mathbf{G}\,\mathbf{F}$ and $\mathbf{F}\,\mathbf{G}$ there): the two powers do not separate liveness from coliveness but target *opposite facets* of that one cell — INCLUDE the recurrence facet, EXCLUDE the persistence facet. The facets are not an ad-hoc reading: the $\mathsf{L} \cap \mathsf{K}$-bipartition remark of [7] shows that this cell splits, uniquely within the four-family algebra, into a Persistence polarity ($\mathbf{F}\,\mathbf{G}$, $\mathbf{\Sigma}_2^0$) and a Recurrence polarity ($\mathbf{G}\,\mathbf{F}$, $\mathbf{\Pi}_2^0$), the two being De Morgan-dual ($\neg\,\mathbf{F}\,\mathbf{G}\;p = \mathbf{G}\,\mathbf{F}\;\neg p$). The *classification* of the two powers onto the two facets — INCLUDE onto the Recurrence/$\mathbf{\Pi}_2^0$ facet, EXCLUDE onto the Persistence/$\mathbf{\Sigma}_2^0$ facet — follows from the two formulas above; the bipartition supplies the two facets, not the assignment. What remains HEURISTIC/CONJECTURED is the *enforcement* reading: the claim that the powers actually *achieve* the recurrence and persistence objectives inherits the tags of Definition 8.14 and Remarks 8.15 and 8.17 unchanged.

Remark 8.23 (The theory ladder as an INCLUDE-power liveness enforcement). The declared *theory ladder* of the core paper [9] — each rung of which *accounts for* a post-window consequent onset under a declared premise — is, in the vocabulary of this paper, exactly the `post`-band INCLUDE power of Definition 8.11 operating on the Out-of-band axis: a $Q^{\uparrow}$ that lands after the window is drawn back in as late evidence, an *attribution* of an event genuinely observed, never a manufacture (the floor Σ_2 of Remark 5.7 stands). Climbing a rung is thus an instance of the inclusion-is-liveness reading of Section 8.3.2: the recurrence goal $\mathbf{G}\,\mathbf{F}$ include($Q^{\uparrow}$) enforced under a stated observation premise. The identification of the ladder with the INCLUDE-power axis is HEURISTIC; the claim that climbing it *enforces* the target signature is CONJECTURED, inheriting the status of Remark 8.17. The conceptual ladder is owned by the core paper [9]; the formal band-word reading is the contribution of this section.

8.4. **Relation to Canonical Timing Recovery.** The error signal of Equation (8.3) belongs to a well-studied family of timing-error detectors (TEDs) in digital communications and clock/data recovery (CDR) [25]. We make the correspondence explicit, showing that the monitor's eight-tick partition provides a native realisation of three classical TED architectures.

Alexander (bang-bang) detector. The Alexander phase detector [1, 25] takes three samples — one before the data transition, one at the transition, and one after — and outputs a ternary signal $\{-1, 0, +1\}$. In our framework, these three points map to `x`, the `x`/`x+1` half-tick probe sample, and `x+1`, respectively. The resulting sign of e_n replicates the Alexander output: $\mathrm{sgn}(e_n) \in \{-1, 0, +1\}$ depending on whether the opener is early, aligned, or late.

Early–late gate. The early–late gate [13] samples the received signal at $t \pm \delta$ around the nominal decision instant and uses the amplitude difference as the error. Our n_0/n_1 partition (Definition 8.3) is the logical (binary) reduction of this principle: instead of analogue amplitudes, we count confirming events in `pre`/`x` (early) versus `x+1`/`m2` (late), which is the natural TED for a binary observation channel.

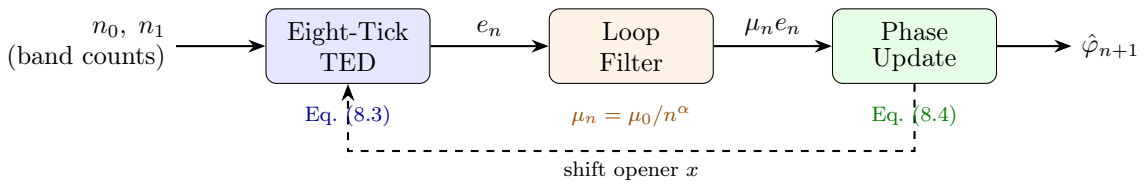


FIGURE 10. Self-tuning loop architecture. The eight-tick timing-error detector (TED) computes e_n from the early/late counts n_0, n_1 around the opener tick x; the loop filter applies the Robbins–Monro gain μ_n; and the phase update shifts the opener for the next observation window. Dashed line: feedback path.

Gardner TED.. The Gardner detector [13] requires exactly *two samples per symbol* — one at the decision instant and one at the midpoint between consecutive symbols — and computes the error as

$$e_n^{\mathrm{G}} \;=\; \big[x(nT) - x\big((n{-}1)T\big)\big]\, x\big(\big(n-\tfrac{1}{2}\big)T\big)\,, \tag{8.8}$$

where $x(\cdot)$ is the received signal and T the symbol period. The half-tick probe of $w^\star$ ($\psi = \frac{1}{2}$) provides precisely these two samples per tick: one at the opener x (`x`) and one at $x + \frac{1}{2}$, making the Gardner structure directly realisable. In our Boolean setting, the product in Equation (8.8) reduces to a sign comparison: if the half-tick sample confirms Q while the integer-tick samples at `x` and `x+1` disagree, the timing is off by approximately half a tick.

Metastability and hysteresis. In CDR circuits, timing uncertainty near the decision boundary manifests as *metastability* [14]; the threshold parameter θ in our TED (Definition 8.2) plays the role of a *hysteresis band* [30, 24] that suppresses spurious corrections when $|e_n|$ is small. In the eight-tick model, metastability-prone events are those where a confirming sample falls exactly on the `x`/`x+1` boundary ($t = x + \frac{1}{2}$); the half-tick probe resolves this to the extent possible at the monitor's timing granularity.

8.5. **Signal-Theoretic Parameter Mapping.** Table 3 summarises the systematic correspondence between monitor-model concepts and their CDR/signal-processing counterparts. Every row links a formal quantity in our framework to the standard engineering notion it instantiates.

8.6. **Self-Tuning Loop Architecture.** Figure 10 depicts the complete feedback loop. The architecture mirrors a standard digital PLL/CDR: the eight-tick TED produces the error signal e_n from the early/late counts around the opener x; a loop filter (Robbins–Monro gain schedule) smooths the error; and the phase update feeds back to shift the opener position for the next trigger event.

TABLE 3. Parameter mapping between the monitor model and classical CDR / signal synchronisation.

Monitor model	**CDR / sync. analogue**
Opener tick x (`x`)	Sampling instant (decision point)
`x+1` ($t = x{+}1$)	Post-cursor (first ISI tap)
$\hat{\varphi}$ (phase estimate)	NCO phase accumulator
n_0 (early count)	Early-gate integrator output
n_1 (late count)	Late-gate integrator output
e_n (timing error)	TED output (S-curve sample)
M^-, M^+ (margins)	Aperture window (sample-and-hold)
θ (decision threshold)	Hysteresis band [30, 24]
μ_n (step size)	Loop bandwidth / gain
K_d (TED gain, Equation (8.9))	S-curve slope at lock point
$\psi = \frac{1}{2}$ (half-tick probe)	2× oversampling [13]
None value (unresolvable)	Metastability [14]

8.7. **Small-Signal Stability.** To characterise local behaviour near the lock point, we linearise e_n around zero phase error. Let $\varphi = \hat{\varphi} - \varphi^\star$ be the residual offset, and assume the confirming-sample probability is Lipschitz in φ. Then for small $|\varphi|$:

$$\mathbb{E}[e_n] \;\approx\; K_d\,\varphi, \tag{8.9}$$

where $K_d > 0$ is the *TED gain* (slope of the S-curve at $\varphi = 0$). Substituting into Equation (8.4):

$$\varphi_{n+1} \;\approx\; (1 - \mu_n K_d)\,\varphi_n + \mu_n \xi_n, \tag{8.10}$$

where ξ_n is zero-mean noise. For $\mu_n K_d \in (0, 2)$ the recursion is contractive and the residual jitter variance satisfies

$$\operatorname{Var}(\varphi_\infty) \;\le\; \frac{\mu_\infty\,\sigma_\xi^2}{2K_d - \mu_\infty K_d^2}, \tag{8.11}$$

confirming that smaller step sizes yield tighter lock at the expense of slower tracking. This trade-off is the Boolean-observation analogue of the *jitter–bandwidth* trade-off familiar from phase-locked-loop design.

TABLE 4. Class distribution across adaptation epochs (percentages). The monitor migrates from miscalibration classes (A=0) to the optimal static-like class (1,0,0) as the phase estimator converges.

Epoch (n)	**M1**	**M2–M4**	**M5–M8**
$1--50$	86%	12%	2%
$51--100$	100%	0%	0%
$101--150$	100%	0%	0%
$151--200$	100%	0%	0%

8.7.1. *Class Distribution Migration.* Partitioning the $N = 200$ iterations into four epochs of 50, Table 4 records the class distribution migrating from the miscalibration classes (A=0) to the optimal static-like class (1,0,0) as the phase estimator converges.

8.8. **Kernel-Level Tuning Examples.**
Illustrative micro-example. To see how the chain operates end-to-end, consider a scenario with $k = 3$ in which the ground truth is a clean implication:

$$\text{Ground truth: } (\varnothing,\ P{=}\top,\ \varnothing,\ Q{=}\top,\ \varnothing,\ \varnothing).$$
$$\text{Measured: } (\varnothing,\ P{=}\top,\ P{=}\bot,\ Q{=}\top,\ \varnothing,\ \varnothing).$$

The $P{=}\bot$ at `x+1` is a single-tick channel glitch.

- *Configuration* $\boldsymbol{\theta}^{(1)}$ *(no pre-filter):* $\mathcal{F}_{\theta_1}$ passes the glitch. $\mathcal{D}_{\theta_3}$ records $w_p{=}\bot$ at `x+1`, setting C=1 (retraction). The observed trace maps to (1,0,1) — a noise-induced class shift away from the true implication.
- *Configuration* $\boldsymbol{\theta}^{(2)}$ *(*$\tau_{\min}$*=2 ticks, hysteresis):* $\mathcal{F}_{\theta_1}$ suppresses the single-tick $P{=}\bot$. The monitor sees $w_p{=}\top$ persisting, so C=0. The observed trace maps to (1,0,0) (ideal confirmation), recovering the ground-truth class.

The class difference ((1,0,1) vs. (1,0,0)) arises entirely from the pre-filter setting; no ground-truth signal is modified. Self-tuning (Section 8.1) adapts $\boldsymbol{\theta}$ so that the monitor's observations converge to the static-like regime (Corollary 8.10).

Region-dependent detection sensitivity. Measurement reliability is not uniform across the observation window. Near the opener tick (`x`, `x+1`), sampling jitter and clock skew are at their peak because the monitor's phase estimate $\hat{\varphi}$ has not yet converged for the current trigger; in `post`, signals arrive after the bounded-eventuality deadline and carry no semantic weight for Φ_k. A band-dependent detection threshold accounts for this varying signal-to-noise ratio:

$$\theta(t) = \theta_{\text{base}} + \delta_{b(t)}, \quad b(t) \in \{\texttt{pre}, \ldots, \texttt{post}\},$$

where δ_b is a per-band gain offset calibrated to the expected noise power in band b. In practice, the monitor may (i) raise the threshold in `x+1` to suppress $Q{=}\bot$ observations that are likely sampling artefacts, or (ii) discard $Q{=}\top$ detections in `post` that cannot contribute to bounded-eventuality satisfaction. This is the standard technique of *gated detection* in digital communications, where a correlator is enabled only during the expected symbol interval. Crucially, the kernel performs *noise suppression*, not evidence censorship: it does

not modify ground-truth signals, and the decision to attenuate a band is driven by the physical SNR of that band, not by the semantic content of the observation. No stage of $\mathcal{K}_{\boldsymbol{\theta}}$ introduces evidence; each stage only suppresses or rejects measurements based on classical detection-theoretic criteria (threshold comparison, gated correlation, phase alignment). The band offsets δ_b become additional elements of the tunable state $\boldsymbol{\theta}$; Self-tuning can adapt them alongside the phase estimate $\hat{\varphi}$. Because the offsets affect only *which* observations pass the threshold—not the crossing/touch classification—the eight window classes, and the eight-class completeness check that backs them, remain invariant (the feature-invariance remark [9]).

Example 8.24 (Oversampling resolves same-bin events). With $\Delta t = 10\,\text{ms}$, $P{\downarrow}$ at 35.2 ms and $Q{\uparrow}$ at 38.7 ms share tick 3 (the same-bin modality analysis [9]). At $\Delta t' = 1\,\text{ms}$ ($r = 10$): $P{\downarrow}$ maps to tick 35, $Q{\uparrow}$ to tick 38. The ordering $P{\downarrow} < Q{\uparrow}$ is now observable; the lag predicate $\Psi_{\text{lag},3}$ is evaluable with $\Delta = 3\,\text{ms}$, upgrading the monitor to interval-like regime (the observation-regime definition [9]). The specification Φ_k and the eight-class quotient are unchanged; only the evaluable predicate set widens.

8.9. **Detection Error and the Minimum-Risk Threshold.** The band-dependent threshold of the preceding paragraph fixes *which* threshold the decision device $\mathcal{D}_{\theta_3}$ applies; it does not yet say *how* that threshold should be calibrated against the two errors the device can make. We close that gap with the classical detection-error account of a single-threshold decision device, specialised to the observation channel $\mathcal{C}(L, S, D, J)$ of Section 5. These are properties of the decision *device*, not of the trace's property class.

Detection-error rates. Fix a window class and let $\mathcal{D}_{\theta_3}$ compare a per-band statistic against the threshold θ. Under the channel model $\mathcal{C}(L, S, D, J)$, an observed sample is the ground-truth level perturbed by additive, zero-mean measurement noise — the same noise whose per-band power the offsets δ_b track. Write

- $\alpha(\theta)$ — the *false-alarm* (false-positive) rate: the probability that $\mathcal{D}_{\theta_3}$ emits a confirmation or refutation token when the ground truth carries none;
- $\beta(\theta)$ — the *miss* (false-negative) rate: the probability that a genuine token is suppressed.

Raising θ makes the device more conservative, so under the declared additive-noise channel $\alpha(\theta)$ is non-increasing and $\beta(\theta)$ is non-decreasing in θ: the two errors trade off monotonically. The exact curves depend on the assumed noise law, so α, β are stated as functions of θ under that model and are WITNESSED only relative to it (model-dependent).

Minimum-risk threshold. Assign a cost c_{FP} to a false alarm and c_{FN} to a miss. The Bayes *risk* of the device at threshold θ is

$$R(\theta) \;=\; c_{\text{FP}}\,\alpha(\theta) \;+\; c_{\text{FN}}\,\beta(\theta), \tag{8.12}$$

and the *minimum-risk threshold* is its minimiser

$$\theta^{\star} \;=\; \operatorname*{argmin}_{\theta}\; R(\theta). \tag{8.13}$$

Equation (8.13) is the standard Bayes / minimum-risk decision rule of detection theory [18, 31]: at $\theta^{\star}$ the marginal reduction in miss cost balances the marginal increase in false-alarm cost, so no local shift of the threshold lowers R. This is a textbook analytic optimum *under*

the assumed noise model, not a new theorem of this paper; θ_{base} and the band offsets δ_b are the tunable handles through which Self-tuning approaches $\theta^\star$.

Realisation. The decision stage realises (8.13) with two classical components placed downstream of the PLL/CDR phase corrector of Section 8.1.3: a *matched filter* — the optimal linear detector for a known pulse shape in additive noise [31] — feeding a *gated correlator* enabled only during the expected symbol interval (the gated-detection technique of the preceding paragraph). The correlator output is thresholded at $\theta^\star$ by $\mathcal{D}_{\theta_3}$. These are named as the standard realisation of the decision device, not re-derived here.

Relation to monitor-type optimality. The threshold optimum $\theta^\star$ is a *different and complementary* optimisation from the monitor-type optimum of Theorem 7.4. The latter minimises signature *ambiguity* $\mathcal{A}$ (and opener sensitivity $\mathcal{S}$) at the level of the monitor *type* (ρ, ω, π): it selects *which* decision device to build. The minimum-risk $\theta^\star$ calibrates the decision *threshold within* a type already chosen. The two are nested and do not compete — type selection (Theorem 7.4) fixes the device; minimum-risk calibration (8.13) tunes its operating point — and they optimise distinct objectives: $\theta^\star$ minimises detection *risk*, whereas Definition 7.3 minimises classification *ambiguity*.

9. Physical Grounding

The causal-coherence conditions CC1–CC4 that underwrite this section are defined in the core paper [9] (Definition of causal coherence); we cite them and build on them here, constraining the selection criterion of Section 7 and the tuning loop of Section 8 with the physics of the observed process. This section supplies the *rely* half of Section 5: CC1–CC4 are the physical premises under which the guarantees S1–S8 are meaningful.

9.1. **Extended Optimality with Causal Coherence.**

The causal-coherence conditions motivate extending the monitor optimality criterion (Definition 7.3) with a *causal incoherence penalty*.

Definition 9.1 (Causal Incoherence Penalty)**.** Given a timing model $\mathcal{T}$ and a process Π with characteristic detection latency τ_{detect} and trigger-to-opener offset τ_{trigger}, the causal incoherence of monitor type w is:

$$\mathcal{C}(w) = \big|\, k \cdot \Delta t - \tau_{\text{detect}} \,\big| + \big|\, \hat{\varphi} \cdot \Delta t - \tau_{\text{trigger}} \,\big| + \max\big(0,\ \tau_{\min} - (M^- + M^+) \cdot \Delta t\big). \quad (9.1)$$

A monitor with $\mathcal{C}(w) = 0$ is *perfectly calibrated* to the physical process.

Definition 9.2 (Extended Monitor Optimality)**.** The *causally grounded* optimal monitor minimises the extended cost

$$w^\star_\Pi \;=\; \operatorname*{argmin}_w \big[\, \mathcal{A}(w) + \lambda\, \mathcal{S}(w) + \mu\, \mathcal{C}(w) \,\big], \qquad \lambda, \mu > 0, \quad (9.2)$$

subject to the same tie-breaking rule as Definition 7.3: among minimisers, the type with the finest timing-error detection resolution is preferred — *not* maximal $\mathcal{R}(w)$, which the half-tick probe lowers (Remark 7.7).

The added term $\mu\,\mathcal{C}(w)$ ensures that the optimal monitor is not only unambiguous ($\mathcal{A} = 1$) and stable ($\mathcal{S} = 0$), but also *physically consistent* with the observed process. The correspondence to the eight-tick partition is immediate:

- `x` ($t = x$) represents the *physical trigger detection instant*—the opener must coincide (within M^-) with the actual occurrence of P.
- `x+1` ($t = x{+}1$) covers the *minimal reaction latency region*: the earliest tick at which a causal consequence of P could become observable.
- `m2`/`cl` bands span the *physical causal window* $[\tau_{\min}, \tau_{\max}]$ within which Q is expected under the governing dynamics.

9.2. **Physically-Guided Self-Tuning.** The self-tuning loop of Section 8.1.3 drives $\hat{\varphi}_n \to \varphi^\star$ via stochastic approximation. Causal coherence imposes a *hard constraint*: the converged parameters must remain within the physically admissible region.

Corollary 9.3 (Physical Constraint on Self-Tuning)**.** *If the adaptation rule $\hat{\varphi}_{n+1} = \hat{\varphi}_n - \mu_n e_n$ (Equation (8.4)) converges to a fixed point $\varphi^\star$ such that*

$$k \cdot \Delta t \;<\; \tau_{\min}, \tag{9.3}$$

then the monitor is epistemically invalid*: its diagnostic output cannot distinguish a true causal chain $P \to Q$ from the absence of Q.*

Proof. By the causal-necessity theorem [9], violation of condition CC1 implies systematic false negatives. Convergence of the phase estimator does not remedy this, because the error signal e_n reflects timing *alignment* but not timing *sufficiency*: even a perfectly aligned monitor that closes the window before the causal response arrives will record $Q = \bot$ at every trigger. □

Remark 9.4 (The mis-verdict is the tuning error signal). The spurious refutation produced on a lone in-window $Q^\perp$ down-crossing — the (0,1,0)/(0,1,1) miscalibration regime of Section 8.1.6 — is exactly the control error the loop removes, and it splits across two actuators. A *phase* mis-alignment, in which the causal onset fell at or near the window edge, surfaces as the early–late asymmetry $n_0 \neq n_1$ of the timing-error detector (a false negative around the opener x) and is corrected by re-phasing, driving the readability orbit $(A{=}0) \to (A{=}1 \wedge (B \vee C)) \to (1{,}0{,}0)$ of Corollary 8.10. A *horizon* insufficiency, in which the window is genuinely too short (a CC1 violation), is by Corollary 9.3 *not* correctable by e_n — timing alignment is not timing sufficiency — and is instead routed to the adaptive-k calibration clamp (Γ_3/S6) that widens k. In both branches the mis-verdict is read as the tuning error signal.

In practice, this means:

(1) The window bound k must be *initialised* using a theoretical or empirical estimate of τ_{detect} (Equation (11.2)). The self-tuning loop may subsequently refine the phase $\hat{\varphi}$ and margins M^-, M^+, but must not shrink k below $\lceil \tau_{\min}/\Delta t \rceil$.
(2) The margin update rules (Definition 8.7) should be clipped: $M^-_{n+1} \geq M^-_{\min}$ and $M^+_{n+1} \geq M^+_{\min}$, where the lower bounds are derived from the physical latency budget.
(3) The step size μ_n must decay fast enough to prevent the phase estimate from drifting outside the admissible region during transients.

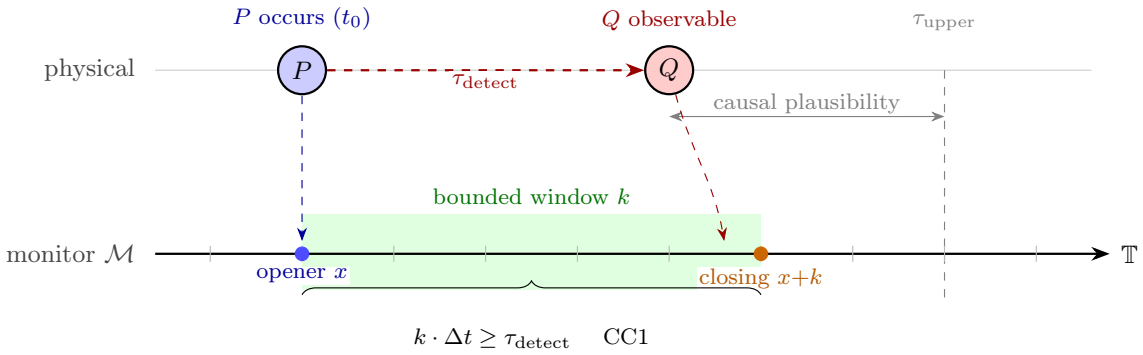


FIGURE 11. Causal latency alignment between physical reaction dynamics (top) and the monitor observation model (bottom), in the event–window idiom of the event–trigger–window diagram [9]. The physical process emits P at t_0; Q becomes observable only after the causal detection latency τ_{detect} and before the upper plausibility bound τ_{upper}. On the monitor lane the opener tick x aligns with P and the bounded window $[x, x+k]$ (green) must be long enough to contain the Q-observation: $k \cdot \Delta t \geq \tau_{\text{detect}}$, condition CC1 of the causal-coherence conditions [9].

9.3. **Causal Latency Alignment Diagram.** Figure 11 visualises the alignment requirement. The physical timeline (top) shows P occurring at t_0 and Q becoming observable after the causal detection latency τ_{detect}. The monitor lane (bottom) shows the observation ticks and bounded window: the opener tick x must coincide with P, and the closing tick $x+k$ must fall within the causally plausible region $[\tau_{\text{detect}}, \tau_{\text{upper}}]$.

9.4. **Implementation Discipline.** All counts, percentages, and crosswalk entries in this section are produced by the causal-analysis script of the reproducibility package, which:

(1) imports the same eight-tick enumeration engine used by the enumeration script of the reproducibility package;
(2) evaluates five signature-level predicates (Definition 10.1) over all 50000 signatures;
(3) generates the crosswalk table (Table 6) and the window-sweep data (Table 7, Fig. 12) as LaTeX fragments;
(4) passes 10 unit tests verifying predicate non-emptiness, structural invariants, and combined coverage.

No signature was inspected manually. The script belongs to the authors' reproducibility material shared with the core paper [9]; it is not part of either paper's deposited record.

10. Pathologies and Empirical Diagnostics

10.1. **Trace-Level Taxonomy of Causal Pathologies.** The causal-coherence conditions CC1–CC4 [9] are abstract constraints. We now identify the *concrete* failure modes that arise when one or more conditions are violated, and map every failure mode onto the eight-tick signature space [9].

TABLE 5. Signature-level coverage of the five causal pathologies over all 50000 signatures ($k = 3$). "Cov." is the number of matching signatures; "CC" is the violated condition.

Failure Mode	**CC Viol.**	**Cov.**	**Dominant Outcome**
F1 Premature Falsification	CC1 (short)	25600	Violated / Vacuous
F2 Causal Inflation	CC1 (long)	24400	OK (spurious)
F3 Parallel Q	CC1 + CC4	10000	OK (spurious)
F4 Probabilistic Causation	CC1–CC3	9600	OK (mixed)
F5 Noise-Induced FP	CC3	32000	OK (transient)
Combined coverage			*complete by construction over the eight classes*

The diagnostics of this section quantify the five causal failure modes F1–F5 of [9]; we recall only their labels and defining conditions, deferring the causal-coherence theory to the core paper.

Definition 10.1 (Causal pathologies, F1–F5 of [9])**.** For a monitor with window length k and threshold θ, the five trace-level failure modes, by label and defining condition:

F1. **Premature Falsification** — $k \cdot \Delta t < \tau_{\min}$: the window closes before Q can arrive ($A = 0$, marked Violated or Vacuous); a CC1 violation (window too short).

F2. **Causal Inflation** — $k \cdot \Delta t \gg \tau_{\max}$: an *independent* Q falls inside the window ($A = 1$, "OK" spurious); a CC1 violation (window too long).

F3. **Parallel (Independent)** Q — Q is already True at the opener tick, impossible under a causal latency $\tau > 0$; violates both CC1 and CC4.

F4. **Probabilistic Causation** — the link $P \to Q$ has probability $p < 1$, so q_T and q_F both occur in the window ($A = 1$, $B = 1$); stresses CC1–CC3 jointly.

F5. **Noise-Induced False Positive** — a transient fluctuation pushes Q above θ in one band amid q_F neighbours; violates CC3 (threshold detectability).

Each failure mode is captured by a *signature-level predicate* that inspects only the eight-tick signature σ and the associated pattern π; no manual inspection of individual traces is required. Table 5 summarises the five predicates and their coverage.

The five predicates were *constructed* to be jointly exhaustive over the eight-tick signature space, so their combined 100% coverage is a *completeness* property of the taxonomy — a statement about how the predicate family was built — rather than an empirical result, and it is *not* evidence that the CC conditions discriminate coherent from incoherent operation. Note that the classes *overlap*: the per-mode counts $25600 + 24400 + 10000 + 9600 + 32000 = 101600$ exceed the 50000 signatures (a $2.03\times$ overlap), so the predicates form a *covering*, not a partition, and a given signature typically matches several modes at once. A flag that fires on 100% of inputs separates nothing by itself; discrimination between coherent and incoherent operation is carried not by the coverage figure but by the specific parameter regimes inside F1–F5 (the $\tau_{\min}$, $\tau_{\max}$, and θ thresholds), which the window-length sweep (Figure 12) and the crosswalk of Section 10.2 actually exhibit.

TABLE 6. Crosswalk between causal pathologies, CC violations, window classes, and canonical scenarios. All counts are computed automatically by the causal-analysis script of the reproducibility package over the 50000 eight-tick signatures at $k = 3$.

Failure	**CC Viol.**	**#Sigs**	**Dom. M_i**	**Outcome**
Premature Falsification	CC1 ($k{\cdot}\Delta t < \tau_{\min}$)	192	M8, M7, M6	Violated:47%, Vacuous:52%
Causal Inflation	CC1 ($k{\cdot}\Delta t \gg \tau_{\max}$)	192	M4, M2, M3	OK:100%
Parallel Q	CC1 + CC4 (independent Q)	80	M4, M3, M2	OK:100%
Probabilistic Causation	CC1–CC3 (stochastic $P{\to}Q$)	180	M4, M2	OK:100%
Noise-Induced FP	CC3 (θ miscalibration)	160	M4, M2	OK:100%

10.2. **Crosswalk: Pathologies, Window Classes, and Scenarios.** Table 6 maps each failure mode onto the eight window classes, the outcome distribution, and the affected scenario IDs.
Several structural observations emerge:

- **Premature Falsification** concentrates in the $A{=}0 \wedge (B \vee C)$ classes (three of the four $Q^{\top}$-absent classes, $A = 0$), confirming that a short window systematically hides the causal link.
- **Causal Inflation** and **Parallel** Q concentrate in the $A{=}1{\wedge}(B{\vee}C)$ classes (the "bounded" classes where $A = 1$), showing that an excessively long window or an independent Q inflates the OK count.
- **Probabilistic Causation** spans (1,1,0) and (1,1,1) (both $A = 1$ classes with $B = 1$), forming the largest single pathology class (180 signatures) due to the mixed evidence footprint.
- **Noise-Induced FP** appears exclusively in OK-outcome classes, confirming that transient threshold crossings masquerade as genuine Q detections.

10.3. **Window-Length Sensitivity Sweep.** To quantify the two opposing CC1 regimes (window too short vs. too long), we sweep the window parameter k across six values and recompute the full signature catalogue and outcome distribution for each. The results appear in Fig. 12; the underlying statistics table is Table 7.

Two regimes are clearly visible:

(1) **Short window ($k = 2$):** the Violated rate rises sharply. This is the Premature Falsification regime: many genuine $P \Rightarrow Q$ links are falsely rejected because Q cannot arrive within two ticks. (The window-length sweep is the physical, k-parameterised outcome study of Figure 12, a lens distinct from the fixed classification catalogue; the

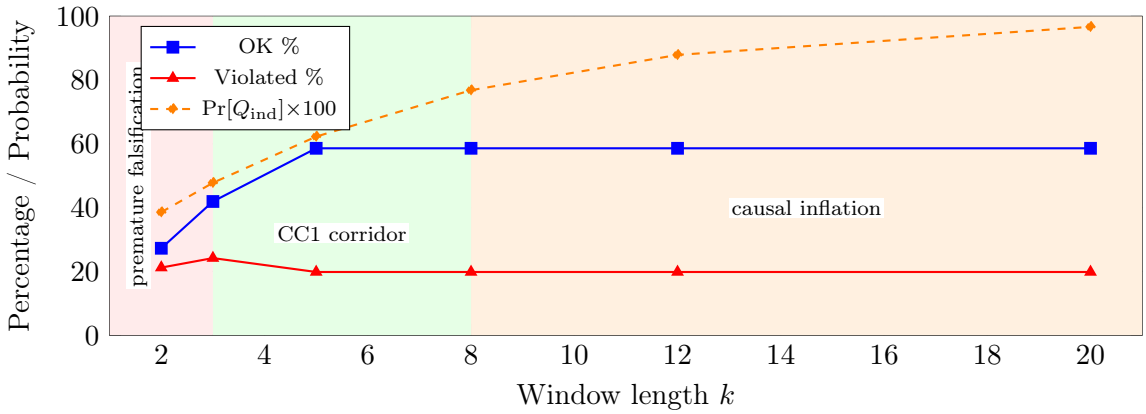


FIGURE 12. Window-length sensitivity sweep, read as the CC1 corridor. Shaded zones mark the premature-falsification regime ($k < 3$: Violated rate rises, OK rate collapses), the admissible corridor ($\tau_{\min} \le k \cdot \Delta t \le \tau_{\max}$), and the causal-inflation regime ($k \ge 8$), where the probability of independent-Q contamination (orange, dashed) exceeds 73% and reaches $\approx 97\%$ at $k = 20$. The Vacuous share ($= 100\% - \text{OK} - \text{Violated}$) is omitted for clarity; the full distribution is tabulated in Table 7.

TABLE 7. Window sweep: signature-space statistics as a function of window length k. "Infl. Idx" is the fraction of OK signatures that match the Causal Inflation predicate; $\Pr[Q_{\text{ind}}]$ is the probability that an independent Q with base rate $\rho_Q = 0.15$ per tick appears at least once in the window.

k	#Sigs	OK %	Viol. %	Vac. %	Infl. Idx	$\Pr[Q_{\text{ind}}]$
2	66	27.3	21.2	51.5	0.444	0.386
3	248	41.9	24.2	33.9	0.615	0.478
5	464	58.6	19.8	21.6	0.706	0.623
8	464	58.6	19.8	21.6	0.706	0.768
12	464	58.6	19.8	21.6	0.706	0.879
20	464	58.6	19.8	21.6	0.706	0.967

eight-tick catalogue of Section 6 is its k=3 instance, with 50000 signatures — monitor-outcome split 48.8% OK / 29.6% Violated / 21.6% Vacuous, machine-computed by `eightick_pathology.py`.)

(2) **Long window ($k \ge 8$):** the combinatorial catalogue saturates (the middle-band summary absorbs additional ticks), but the *physical* risk of Causal Inflation grows: $\Pr[Q_{\text{ind}}]$ exceeds 73% at $k = 8$ and reaches 97% at $k = 20$, meaning nearly every window is contaminated by an independent Q occurrence for any base rate $\rho_Q \ge 0.15$.

This quantitative sweep provides the empirical grounding for the heuristic rule $\tau_{\min} \le k \cdot \Delta t \le \tau_{\max}$ stated in CC1 [9].

11. Case Studies

11.1. **Example: Chemical Reaction Monitoring.** We illustrate the causal-coherence conditions CC1–CC4 [9] with a fully worked physical example.

Scope disclaimer. This example assumes a *single-source* P and Q timeline with deterministic reaction kinetics and no exogenous triggers; this is a strict subcase of the general observation model (the observation-model section [9]). The full eight-class catalogue is not reduced by these assumptions; they only restrict the set of reachable classes in this particular deployment.

Setup. Consider a batch reactor in which reactants A and B are mixed at time t_0 (event P: "mixing initiated") and a product C is subsequently formed (event Q: "product detected"). The reaction follows first-order kinetics:

$$\frac{\mathrm{d}[C]}{\mathrm{d}t} \;=\; k_r\,[A]\,[B], \tag{11.1}$$

where k_r is the rate constant. The characteristic reaction time is $\tau_{\mathrm{rxn}} \approx 1/k_r$, and the minimum time for the detector to register Q is

$$\tau_{\mathrm{detect}} \;=\; \tau_{\mathrm{rxn}} + \tau_{\mathrm{meas}}, \tag{11.2}$$

where τ_{meas} is the instrument response time.

Parameter mapping. Table 8 maps the monitor's tunable state to physically measurable quantities.

Causal coherence constraints. Applying the causal-coherence conditions CC1–CC4 [9]:

- CC1: $k \cdot \Delta t \geq \tau_{\mathrm{detect}}$ (the window must be long enough for the product to form *and* be detected). Simultaneously, $k \cdot \Delta t \leq \tau_{\mathrm{upper}}$, where τ_{upper} is a chemistry-derived bound beyond which an observed $[C] > 0$ is more likely due to contamination, side reactions, or baseline drift than to the $A{+}B$ mechanism.
- CC2–CC3: the detector response time τ_{meas} bounds M^+ and θ.

Failure modes.

$k \cdot \Delta t < \tau_{\mathrm{detect}}$**::** The product C has not yet formed (or is below θ) when the window closes. The monitor reports $P \Rightarrow Q$ as *false*—a systematic false negative.

$k \cdot \Delta t \gg \tau_{\mathrm{detect}}$**::** A spurious concentration spike from an unrelated process may fall inside the wide window. The monitor mistakenly confirms $P \Rightarrow Q$ from the correlation $P \wedge Q$, without causal necessity. This directly supports the paper's central claim that *the monitor must not infer* $P \Rightarrow Q$ *from* $P \wedge Q$.

TABLE 8. Parameter mapping for chemical-reaction monitoring.

Monitor parameter	**Physical meaning**
$\hat{\varphi}$	Detection phase offset: delay between mixing pulse and first detector sample.
M^-	Pre-trigger measurement slack accounting for mixing non-ideality.
M^+	Post-trigger observation slack for slow or noisy detectors.
$k \cdot \Delta t$	Bounded observation horizon. Must satisfy $\tau_{\text{detect}} \leq k \cdot \Delta t \leq \tau_{\text{upper}}$.
θ	Concentration detection threshold of the sensor (e.g., minimum $[C]$ for a colorimetric or spectrometric readout).
Δt	Tick duration (sampling interval of the monitoring instrument).

11.1.1. *Measurement chain.* The physical pipeline is:

$$\text{chemical event} \xrightarrow{\tau_{\text{rxn}}} \text{detector (PMT/photodiode)} \xrightarrow{\tau_{\text{det}}} \text{analog voltage} \xrightarrow{\text{amp/filter}} \text{ADC} \xrightarrow{S_{\Delta t}} \tau_{\text{meas}} \xrightarrow{\mathcal{K}_{\boldsymbol{\theta}}} \tau_{\text{obs}}$$

The monitor model maps onto this chain as follows: the sampling stage $S_{\Delta t}$ corresponds to the ADC sample clock; the pre-filter $\mathcal{F}_{\theta_1}$ to the amplifier/filter stage and any software debounce; the phase-alignment $\mathcal{A}_{\theta_2}$ and decision $\mathcal{D}_{\theta_3}$ to the threshold comparator; and the gating $\mathcal{M}_{\theta_4}$ to the instrument's trigger-and-acquisition window.

11.1.2. *Numeric scenario.* Consider monitoring the implication "reagent addition (P) implies fluorescence rise (Q) within k ticks" on a stopped-flow instrument. This is the single quantitative walkthrough of the case study.

Ideal trace. The ideal trace has $t_{P\uparrow} = 10.2\,\text{ms}$, $t_{P\downarrow} = 17.8\,\text{ms}$ (injection pulse $|I_P| = 7.6\,\text{ms}$), $t_{Q\uparrow} = 15.1\,\text{ms}$, $t_{Q\downarrow} = 32.0\,\text{ms}$ (fluorescence rise after 4.9 ms lag). Hence $\tau_{\text{ideal}} \models \Psi_{\text{lag},\Delta}$ with $\Delta = t_{Q\uparrow} - t_{P\downarrow} \approx 0$ (overlap regime).

Scenario A: coarse sampling ($\Delta t = 10\,\text{ms}$). At $\Delta t = 10\,\text{ms}$, the sampling bins are $B_0 = [0, 10)$, $B_1 = [10, 20)$, $B_2 = [20, 30)$, ... Both $t_{P\uparrow}$ and $t_{Q\uparrow}$ fall in bin B_1: this is same-bin ambiguity (the same-bin modality analysis [9]). The monitor cannot determine whether $P\uparrow$ preceded $Q\uparrow$. With $k = 3$ ticks (30 ms record), the monitor observes w_p and w_q both active at tick 1, producing a (1,1,1)-class pattern (both P and Q in $\mathtt{x}$). Edge-preservation

fails because $\delta = \min(|I_P|, t_{Q\uparrow} - t_{P\uparrow}) = \min(7.6, 4.9) = 4.9\,\text{ms} < 2\Delta t = 20\,\text{ms}$, violating condition (ii) of the observation-soundness proposition [9].

Scenario B: fine sampling ($\Delta t = 1\,\text{ms}$). At $\Delta t = 1\,\text{ms}$, $t_{P\uparrow}$ falls in B_{10} and $t_{Q\uparrow}$ in B_{15}: five bins apart, so edge ordering is resolved (the bin-ordering lemma [9]). The standard debounce (Algorithm A, $\tau_{\min} = 2$ ticks) suppresses any sub-millisecond glitch. With $k = 20$ ticks (20 ms record), the monitor observes $P^\top$ at tick 10, $Q^\top$ at tick 15: a (1,1,0)-class pattern (overlap, Q in `x+1`). The trace $\tau_{\text{obs}} \in \text{Conf}_{\text{iv}}^{\text{ov}}(k)$.

Scenario C: glitch suppression. Suppose a turbulence-induced spike at $t = 12.3\,\text{ms}$ causes a transient Q-above-threshold lasting 0.4 ms (sub-bin duration at $\Delta t = 1\,\text{ms}$). Without debounce, the monitor reports a spurious $Q^\top$ at tick 12. Algorithm A with $\tau_{\min} = 2$ ticks suppresses this event because the pulse width ($0.4\,\text{ms} < 2\,\Delta t$) does not meet the persistence criterion. The genuine $Q^\top$ at tick 15 passes because the fluorescence rise persists for ≥ 2 ticks.

11.1.3. *Regime migration and recovery.* The three scenarios above illustrate the regime migration of Corollary 8.10:

$$\text{A: } \mathcal{K}_{\boldsymbol{\theta}}(\tau_{\text{meas}}) \notin \text{Conf}_{\text{iv}}^{\text{ov}}(k) \ \text{(unresolved)},$$
$$\text{B: } \mathcal{K}_{\boldsymbol{\theta}'}(\tau_{\text{meas}}) \in \text{Conf}_{\text{iv}}^{\text{ov}}(k), \quad \Delta t' = \tfrac{\Delta t}{10}.$$

The logical specification Φ_k is unchanged; only the observational projection parameters changed. This is a concrete instance of the invariance stated in the tuning-invariance remark [9]: tuning alters observational classification, not logical semantics.

Reproducibility note. All three scenarios are constructed from explicit parameter values that can be verified against the bin-ordering lemma and the observation-soundness proposition [9]; no experimental data are used. The parameter ranges in the case-study parameter table [9] are drawn from published stopped-flow specifications [27] and are intended as a plausibility demonstration, not an empirical benchmark.

11.2. **Scenario 2: Finite Refutation (Cosafety).**
Setup. Same monitor configuration ($L = 1$, $S = 0$, $D = 0$, $J = 0$, $k = 3$), but $\mathcal{Q}$ does *not* emit any event: e_Q never occurs.

Observed trace. $\mathcal{P}$ emits at $t_P = 3$. With $\epsilon_P = 0$, the monitor sees w_p at $t_M^P = 4$. No w_q is ever observed. Opener tick $x = 4$; window $[4, 7]$.

tick	0	1	2	3	4	5	6	7	8	9	10…
w_p					$\top$						
w_q											

Eight-tick projection. Relative to $x = 4$:
- *x* (t=4): $w_p = \top$;
- *x+1* (t=5): $\varnothing$;
- *m2* (t=6): $\varnothing$;
- *cl* (t=7): $\varnothing$;
- *post* ($t > 7$): $\varnothing$.

Signature: $\sigma = (P^\top, \varnothing, \varnothing, \varnothing, (0, 0))$.

Classification. Features: $A = 0$, $B = 0$, $C = 0$. Window class: (0,0,0) (A=0, B=0, C=0).

Property fragment. The trace prefix $\sigma[0..7]$ is a *finite witness* to the violation of Φ_3: w_p holds at tick 4 but w_q does not hold at any tick in $\{4, \ldots, 7\}$. This is a bad prefix certifying that $\sigma \not\models \Phi_3$.

The dual perspective is that $\sigma[0..7]$ *satisfies* the cosafety property (cell $\mathsf{C} \cap \mathsf{L}$) $\mathcal{L}_4 = \mathbf{F}(w_p \wedge \neg \mathbf{F}_{\leq k} w_q)$ from the fragment-classification table [9]: a single finite prefix suffices to confirm a violation of the implication. The single-confirmation **F**-shape is a direct skeleton read-off — finitely confirmable (cosafety C) yet with no finite bad prefix (liveness L), matching $\Phi_{Fp} \in \mathsf{C} \cap \mathsf{L}$ of the canon [7].

Interpretation. The (0,0,0) pattern captures the *silent window*: the trigger fired but no response arrived. Under the calibration theorem (the calibration theorem [9]), this means either (a) the causal relation $P \Rightarrow Q$ does not hold, or (b) the observation parameters underestimate the true delay. A monitor for Φ_3 would emit a *negative verdict* at tick 7 — a finite-time decision, confirming the cosafety character of the refutation. The framework thus separates the *structural cause* of the verdict ((0,0,0): no Q at all) from the *parametric cause* (k possibly too small), enabling targeted diagnostics.

11.3. **Scenario 3: Unbounded Regime (Liveness).**

Setup. $\mathcal{M}$ has parameters $L = 1$, $S = 0$, $D_{\max} \to \infty$ (e.g., the drift rate is unknown and potentially unbounded), $J = 1$. Because $D_{\max}$ is unbounded, $\Delta = \infty$ and no finite k satisfies $k \geq 2\Delta$.

Ground process. $\mathcal{P}$ emits at $t_P = 3$; $\mathcal{Q}$ emits at $t_Q = 5$. The causal relation holds, but the observation gap $|t_M^Q - t_M^P|$ is unbounded because D is unbounded.

Trace structure. Under a particular drift realisation $D = 50$: $t_M^P = 3{+}1{+}50{\cdot}3{+}\epsilon_P = 154{+}\epsilon_P$; $t_M^Q = 5 + 1 + 50 \cdot 5 + \epsilon_Q = 256 + \epsilon_Q$. The gap is ≥ 100 ticks. For any chosen k, a larger D would push t_M^Q beyond the window.

Classification. No bounded-eventuality formula Φ_k can serve as a safety monitor. The implication must be expressed as $\Phi_\infty = \mathbf{G}(w_p \to \mathbf{F}\, w_q)$, which lies in the cell $\mathsf{L} \cap \mathsf{K}$ (the unbounded-response formula $\mathcal{L}_3$ from the fragment-classification table [9]): no finite good prefix confirms it (coliveness K) and no finite bad prefix refutes it (liveness L), so it is non-monitorable. This is the paper's one case where the class is *not* read off the leading $\mathbf{G}$: the outer $\mathbf{G}$ is safety-shaped, but the inner unbounded $\mathbf{F}$ flips the whole formula to $\mathsf{L} \cap \mathsf{K}$, matching $\Phi_{GFp} \in \mathsf{L} \cap \mathsf{K}$ of the canon [7].

At any *fixed* point in time, the monitor has observed a prefix that is either consistent with eventual w_q (no verdict) or has already seen w_q (positive evidence, but not a safety verdict). The property admits no finite refutation: every finite prefix that lacks w_q could still be extended to a satisfying trace where w_q appears later.

Interpretation. This scenario demonstrates the *liveness boundary* of the framework. When $\Delta = \infty$, the eight-class classification remains well-defined for any concretely observed finite prefix, but the *verdict* is necessarily inconclusive: the monitor can accumulate evidence but can never certify Φ_∞ or $\neg\Phi_\infty$ in finite time.

This is precisely the content of the causal-observability lemma [9]: unbounded observation parameters are incompatible with safety monitoring. The framework's response is not to *fail* but to *reclassify*: the property moves from $\mathcal{L}_2$ to $\mathcal{L}_3$ (the unbounded-regime definition [9]), and the monitor's role changes from *certifier* to *evidence accumulator*.

12. Reproducibility

All quantitative artifacts in this paper are regenerated by the authors' reproducibility scripts, shared with the core paper [9]. The 50000-signature catalogue at $k = 3$ is enumerated by this paper's `eightick_derive.py` (Remark 6.5); the ambiguity/robustness/sensitivity scores behind the selection result of Section 7, and the class-migration statistics of Section 8, are produced by the signature-enumeration and simulation scripts of that material; in addition, the per-type $(\mathcal{A}, \mathcal{S})$ table behind Theorem 7.4 is computed by the self-contained script `eightick_selection.py`, which rebuilds the $390625 \to 50000 \to 8$ catalogue and evaluates $\mathcal{A}(w)$ and $\mathcal{S}(w)$ for all eight monitor types; the causal-pathology crosswalk (Table 6) and the window-length sweep (Figure 12 and Table 7) are produced by its causal-analysis script, which evaluates the five signature-level predicates of Definition 10.1 over the full catalogue and verifies combined coverage by unit test. The convergence trajectory of Figure 9 and the checkpoints of Table 2 are a single representative run of the real band-counting detector, produced by the self-contained script `convergence_experiment.py` under a fixed seed (see below); the epoch-wise class distribution of Table 4 is produced by the notebook suite of the core package, whose reachability/phase-recovery notebook `nb04_schmitt_gallery_reachability` realises the regime migration of Corollary 8.10 under the parameters stated. The quantitative convergence claim of Section 8.1.6 — non-circular finite-prefix convergence of the real band-counting detector, the measured mean-field $h(\theta)$, the finite basin, and the first-/second-order drift breakpoints — is backed by the self-contained script `convergence_experiment.py` (with output `convergence_experiment_out.json`), which derives e_n directly from the eight-tick n_0/n_1 counts of Definitions 8.3 and 8.4, runs the update Equations (8.4) and (8.5), and anchors itself to the machine-checked catalogue by importing `eightick_derive.py`

and asserting the $390625 \to 50000 \to 8$ counts. No figure or table in this paper depends on data inspected by hand.

13. Limitations and Open Problems

We state the boundary of validity plainly. The paper's guarantees are uneven in strength, and the honest reading of what is and is not settled is part of the contribution.

Local, not global, convergence.: The self-tuning guarantee (S4, Theorem 8.8) is *local*: almost-sure convergence to the true phase holds within a basin of attraction under the sign/regression condition (C5). Global convergence of the band-counting timing-error detector is *conjectured*; the de-circularised experiment (Section 8.1.6) now witnesses convergence of the *real* detector on a finite prefix from within the basin, and *measures* that basin to be finite ($|\hat{\varphi}_0| \lesssim 4$ ticks) with a first-order drift breakpoint at $D \approx 0.02$ (recovered to $D = 0.05$ by the second-order loop) (Remark 8.9); the uncovered region — uniform behaviour outside the measured basin and beyond the simulated horizon — is not covered by proof.

Rationality of the power transduction is open.: The tuning powers assemble into a transduction T on band-words (Section 8.2), but whether T is a rational (finite-state) transduction is an open problem (Remark 8.13). Until it is resolved, the algebraic closure properties one would want of the power set — and the decidability of the enforcement question below — remain unavailable.

Only a conditional *controllability* fact is proven, on the cooperative class quotient; adversarial enforcement is open.: The ω-level enforcement objectives (Definition 8.14) — that the tuning powers can drive the enforced window-stream to $\mathbf{G}\,\mathbf{F}\ \sigma^*$ or $\mathbf{F}\,\mathbf{G}\ \sigma^*$ — yield, at the finite class-quotient arena and on a cooperative (edge-affording) signal, only a modest controllability fact: Proposition 8.21 shows both objectives controllable by memoryless strategies, PROVEN-CONDITIONAL on basin membership, an explicit edge-affordance assumption, and the causal-coherence hypotheses (decidability and memoryless strategies are automatic on the fixed finite arena, not a contribution). Three things stay open (Remark 8.17): *adversarial* enforcement — steering against an environment that may withhold edges (H1b dropped); (R1) the *unconditional*, basin-free reading, which the measured mean-field collapse (Remark 8.9) makes false for a fixed-span window; and (R2) CC-free soundness, owned by the core paper's causal-necessity theorem [9]. The claim over the *realised* transduction T (rather than its quotient) further awaits the open transduction-rationality problem (Remark 8.13).

The causal-incoherence penalty is heuristic.: The physical-grounding penalty $\mathcal{C}(w)$ (Contribution 4) is a design objective whose calibration is [HEURISTIC] and is *not* numerically evaluated in the case studies; only the admissibility *constraint* it accompanies is proven.

One heuristic structural analogy.: The identification of the `x`/`x+1` split with the entry/exit-edge refinement of the convention-algebra taxonomy (Remark 7.9) is a structural analogy tagged HEURISTIC, adopted to show the band separation is principled rather than ad hoc; it is not a literal instantiation and carries no proof obligation.

None of these boundaries undermine the paper's proven core — the witnessed selection result, the local convergence theorem, the proven admissibility constraint, and the machine-checked signature enumeration — but each marks a place where a stronger claim would require work not done here.

14. Future Work

The open items above define the natural next steps, in rough order of tractability.

(1) **Close the global-convergence gap.** Replace the finite-prefix witness of Remark 8.9 with a Lyapunov or supermartingale argument establishing convergence of the band-counting detector outside the local basin, or exhibit a counterexample delimiting the basin exactly.
(2) **Decide transduction rationality.** Settle Remark 8.13: either construct a finite-state transducer for the power transduction T, or prove no such transducer exists, and propagate the consequence to the decidability of the enforcement objectives.
(3) **From cooperative controllability to adversarial enforcement, and from the quotient to the realised transduction.** Proposition 8.21 gives only a conditional controllability fact on the cooperative class quotient. The next step is to drop the cooperative edge-affordance assumption (H1b) and treat the environment as *adversarial* — a genuine enforcement/shield problem — discharge the two residuals (basin-free steering (R1) and CC-free soundness (R2)), and lift the result from the quotient to the *realised* transduction T (contingent on Remark 8.13), ideally by connecting the tuning powers to a shield-synthesis construction [6] over the window-stream.
(4) **Calibrate the causal penalty.** Give $\mathcal{C}(w)$ a quantitative, instrument-grounded calibration and evaluate it on the case studies, promoting Contribution 4 from HEURISTIC toward WITNESSED.

These directions are stated as open problems, not as claimed results; the paper's contribution is the control layer and its proven and witnessed components, and the map above of what remains.

15. Conclusions

The result to carry away is a single one: this is the first end-to-end engineering of a monitor for the bounded-implication observation task — optimal type *selection* (Theorem 7.4) and a convergent *self-tuning* loop (Theorem 8.8) over a machine-checked generative signature space ($390625 \to 50000 \to 8$) — realising the active-tuning confirmation source that the classification companion [7] defers (its O5 direction).

We have argued, and largely demonstrated, that the observation class of a temporal-implication monitor is a *controlled quantity* — optimally selected and held against drift by a convergent self-tuning loop that steers the monitor across the observation-class partition without manufacturing truth. Where prior runtime verification *fixes* the observation class, this paper *steers* it. Selection minimises diagnostic ambiguity by choosing the monitor type that maps every reachable signature to a single window class with zero opener sensitivity, at the finest detection resolution among the minimisers (a witnessed result over the finite catalogue for ambiguity and sensitivity; the stability radius is analytic). Self-tuning holds the chosen type phase-aligned at runtime and provably migrates the observed class toward the static-like regime within a basin of attraction under a Robbins–Monro guarantee (global convergence of the band-counting detector conjectured, witnessed on a finite simulation prefix). Physical grounding keeps both honest: a causal-coherence penalty and an admissibility constraint forbid convergence to an operating point that would report confirmations the process dynamics cannot support, and a window-length sweep locates the admissible corridor between premature falsification and causal inflation. Together with the

eight-class worked gallery and the signature-level pathology diagnostics, these results turn the observation-relative monitorability of [9] from a descriptive thesis into an engineering discipline: a monitor can be *steered* across the observation-class partition, with provable guarantees and a hard physical boundary that keeps tuning from manufacturing truth.

Appendix A. Trace Classification, Calibration, and Reachability Details

This appendix collects the exhaustive trace-level enumeration under the M1-edge encoding, the tuning-induced calibration tree, and the reachability details supporting the body's selection and transition results; the worked per-class realisations themselves are Figure 1–Figure 8 in Section 4.

Full trace classification under $\mathbf{M1_{edge}}$. The eight figures Figures 1 to 8 together depict one canonical realisation per window class (one per (A,B,C) cell). Under $\mathrm{M1}_{\mathsf{edge}}$, however, each class admits *multiple* observationally distinguishable sub-classes according to the ordering and pre-window-state choices. This subsection enumerates them.

Combinatorial principles. Three structural rules govern the enumeration:

(R1) **Q-alternation.** Edges of $\tilde{q}$ across θ alternate by sign: $Q^{\top}$ and $Q^{\perp}$ events must interleave.

(R2) **Pre-state constraint.** Whether $\tilde{q}$ is above or below θ at the start of the window ($\tilde{q}_{\mathrm{pre}}$) determines the polarity of the *first* Q-event: if $\tilde{q}_{\mathrm{pre}} < \theta$, the first Q-edge is rising; if $\tilde{q}_{\mathrm{pre}} > \theta$, the first Q-edge is falling.

(R3) **Trigger convention.** $P^{\top}$ is recorded at $n = 0$ by definition of trigger; $\tilde{p}_{\mathrm{pre}} < \theta$ is forced.

Sub-class enumeration. Table 9 lists every observationally distinguishable token row in the window, partitioned by the primary feature triple (A, B, C).

Counts (Body only). By feature-triple class:

$$\#(1,0,0) = 1,\ \ \#(1,1,0) = 2,\ \ \#(1,0,1) = 2,\ \ \#(1,1,1) = 6,\ \ \#(0,0,0) = 1{+}1,$$
$$\#(0,1,0) = 1,\ \ \#(0,0,1) = 1{+}1,\ \ \#(0,1,1) = 2,$$

totalling 16 observationally distinct token rows and 18 physical realisations *within* the window. This count is specific to $\mathrm{M1}_{\mathsf{edge}}$; level-triggered M1 produces a strictly larger taxonomy (since every tick yields a binary token) while extremum-based M3 produces a strictly smaller one (constant signals emit no events) [8].

Composing the 18 Body realisations with the Post continuations yields the full finite-trace catalogue ($18 \cdot 3^h$ rows at horizon h); the positional enumeration and its verification are machine-checked in the reproducibility package of the core paper [9].

(A, B, C)	$\tilde{q}_{\text{pre}}$	**Event ordering in Body**	**Token row (n=1..3)**
$(1,0,0)$	below	$Q^{\top}$	$Q^{\top}\,\varnothing\,\varnothing$
$(1,1,0)$	below	$Q^{\top} \to Q^{\perp}$	$Q^{\top}\,Q^{\perp}\,\varnothing$
$(1,1,0)$	above	$Q^{\perp} \to Q^{\top}$	$Q^{\perp}\,Q^{\top}\,\varnothing$
$(1,0,1)$	below	$Q^{\top} \to P^{\perp}$	$Q^{\top}\,P^{\perp}\,\varnothing$
$(1,0,1)$	below	$P^{\perp} \to Q^{\top}$	$P^{\perp}\,Q^{\top}\,\varnothing$
$(1,1,1)$	below	$Q^{\top} \to P^{\perp} \to Q^{\perp}$	$Q^{\top}\,P^{\perp}\,Q^{\perp}$
$(1,1,1)$	below	$Q^{\top} \to Q^{\perp} \to P^{\perp}$	$Q^{\top}\,Q^{\perp}\,P^{\perp}$
$(1,1,1)$	below	$P^{\perp} \to Q^{\top} \to Q^{\perp}$	$P^{\perp}\,Q^{\top}\,Q^{\perp}$
$(1,1,1)$	above	$Q^{\perp} \to P^{\perp} \to Q^{\top}$	$Q^{\perp}\,P^{\perp}\,Q^{\top}$
$(1,1,1)$	above	$Q^{\perp} \to Q^{\top} \to P^{\perp}$	$Q^{\perp}\,Q^{\top}\,P^{\perp}$
$(1,1,1)$	above	$P^{\perp} \to Q^{\perp} \to Q^{\top}$	$P^{\perp}\,Q^{\perp}\,Q^{\top}$
$(0,0,0)$	above/below‡	–	$\varnothing\,\varnothing\,\varnothing$
$(0,1,0)$	above	$Q^{\perp}$	$Q^{\perp}\,\varnothing\,\varnothing$
$(0,0,1)$	above/below‡	$P^{\perp}$	$P^{\perp}\,\varnothing\,\varnothing$
$(0,1,1)$	above	$Q^{\perp} \to P^{\perp}$	$Q^{\perp}\,P^{\perp}\,\varnothing$
$(0,1,1)$	above	$P^{\perp} \to Q^{\perp}$	$P^{\perp}\,Q^{\perp}\,\varnothing$

‡ Constitutive ambiguity: two physical realisations ($\tilde{q}$ above or below θ) produce the *same* token row.

TABLE 9. Full trace classification under $\text{M1}_{\mathsf{edge}}$ at window length $k = 3$. Each row is an observationally distinguishable token sequence. Total: 16 distinguishable rows + 2 hidden ambiguity branches in (0,0,0) and (0,0,1) = **18 physical realisations**. The trigger token $P^{\top}$ at $n = 0$ is implicit in every row. Token-row positions shown for the in-band window ticks $n \in \{1, 2, 3\}$ (the ticks `x+1`, `m2`, `cl`); ticks in `Post` ($n > 3$) are uniformly $\varnothing$ in each realisation.

Tuning-induced trace migrations: the calibration tree. The eight figures of Figures 1 to 8 together with the enumeration of Table 9 cover *static* classifications: given a kernel state $\boldsymbol{\theta}$ and a signal τ, the window class is determined. The monitor, however, can *tune* $\boldsymbol{\theta}$ between observations. Each tuning operation T_j shifts $\boldsymbol{\theta} \to \boldsymbol{\theta}'$, and may change the window class of the *same* underlying signal. This subsection formalises the resulting tree of reachable classifications — a kind of chess board where each move is a tuning operation and each position is a window class — and develops the exhaustive witness behind this paper's *universal-readability* proposition (Proposition A.1), whose logic runs in three steps. *(a) Universality:* at the abstract level the tuning space is universal — every classification is reachable from every other in at most three elementary operations, at most two for all but two class pairs (Proposition A.1). *(b) Feasibility:* per signal, universality is sharpened by a realisability criterion — which target readings are attainable is decided by the geometry of the signal,

channel-wise, by the synthesis machinery of [8], with the two channels coupled through the opener-anchored window (the two-signal feasibility clause of Proposition A.1) and the touch probe of Proposition A.3 (illustrated in Figure 14) deciding the silent case. *(c) Epistemic guard:* arbitrary *readability* never becomes arbitrary *truth* — all nine operations are $\mathcal{L}_1^{\text{tun}}$ (monitor-internal), so the non-circularity guarantee of the core paper [9] holds at every node of the tree.

Refinement of the band partition. We have seen the 4-band partition (`Pre` | `X` | `Body` | `Post`) and its eight-tick refinement (`pre` | `x-1` | `x` | `x+1` | `m2` | `cl` | `cl+1` | `post`). For calibration analysis, a further refinement is useful:

$$\underbrace{\texttt{prePre}}_{t\ll 0} \mid \underbrace{\texttt{prePeri}}_{-m\le t<0} \mid \texttt{x} \mid \texttt{x}+1 \mid \texttt{m2} \mid \texttt{cl} \mid \underbrace{\texttt{postPeri}}_{k<t\le k+m'} \mid \underbrace{\texttt{postFar}}_{t>k+m'}$$

The 8-band partition isolates the *calibration-relevant near-states* (`prePeri`, `postPeri`) from the asymptotic regimes (`prePre`, `postFar`). Crucially, `postPeri` can *resolve the constitutive ambiguity* of (0,0,0) and (0,0,1): a Q-edge there reveals the prior Q-state that was invisible during the window.

Tuning operations. Table 10 lists the nine elementary tuning operations of the kernel. Each modifies a single parameter of $\boldsymbol{\theta}$ and has a predictable effect on the observed token row.

Op	**Parameter**	**Effect**
T_1	$\theta \to \theta \pm \Delta\theta$	Threshold shift
T_2	$\hat{\varphi} \to \hat{\varphi} + \Delta\varphi$	Phase alignment (PLL)
T_3	$k \to k \pm 1$	Window length adjustment
T_4	$\tau_{\min}$	Pre-filter / debounce strength
T_5	$M^{\pm}$	Margin extension (gating)
T_6	Δ_{hyst}	Hysteresis width ($\text{M1}_{\text{edge}} \leftrightarrow$ M2)
T_7	Δt	Sample-rate / oversampling
T_8	Robbins–Monro $\hat{\varphi}_n$	Self-tuning auto-convergence
T_9	causal coherence filter	Accept / reject under CC1–CC4

TABLE 10. Nine elementary tuning operations. T_1, T_3, T_6 each admit a directional variant ($\pm$); the remaining operations are parameterised differently but encode comparable degrees of freedom.

The calibration tree. Fix a physical signal τ and an initial kernel state M_0. The *calibration tree* $\mathcal{T}(\tau, M_0)$ is rooted at the pair $(M_0, \text{class}(\mathcal{K}_{M_0}(\tau)))$. Its children at depth one are

$$\big\{\big(T_j(M_0),\, \text{class}(\mathcal{K}_{T_j(M_0)}(\tau))\big) \,:\, j \in \{1,\dots,9\}\big\}.$$

Children at depth two apply a second operation to each depth-one state, and so on up to a chosen bound d. The leaves of $\mathcal{T}(\tau, M_0)$ are the window classes reachable from $\mathcal{K}_{M_0}(\tau)$ within d tuning steps.

This tree formalises the operational sense of Quaestio-escape (Table 11): from any $(M_0, (0,0,0))$ pair there exists a depth-one neighbour (M_1, C) with window class $C \neq (0,0,0)$, provided τ's amplitude is greater than zero.

Reachability matrix. Table 11 gives the min-distance matrix $d(C_i, C_j)$ over all pairs, computed by breadth-first search over the tuning operators at depth $d = 2$ from each starting class. Entry d_{ij} is the minimum number of tuning operations needed to migrate from class C_i to C_j; the symbol ∞ marks pairs unreachable within two moves (these become reachable at $d = 3$ or higher).

d_{ij}	(1,0,0)	(1,1,0)	(1,0,1)	(1,1,1)	(0,0,0)	(0,1,0)	(0,0,1)	(0,1,1)
(1,0,0)	0	1	1	2	1	2	2	∞
(1,1,0)	1	0	2	1	1	1	2	2
(1,0,1)	1	2	0	1	1	2	1	2
(1,1,1)	2	1	1	0	1	1	1	1
(0,0,0)	1	1	2	2	0	1	1	2
(0,1,0)	2	1	∞	2	1	0	2	1
(0,0,1)	2	2	1	2	1	2	0	1
(0,1,1)	2	2	2	1	1	1	1	0

TABLE 11. Minimum tuning distance $d(\cdot, \cdot)$ between window classes, at exploration depth $d = 2$. Most migrations are 1 or 2 moves; the only pairs requiring more are (1,0,0) $\to$ (0,1,1) and (0,1,0) $\to$ (1,0,1), both of which involve large changes in the feature triple. No class is a trap (every class has at least one outgoing migration), confirming that the calibration space is connected.

Operational consequences. The matrix says, concretely:

- Every Quaestio class ((1,1,0), (1,1,1), (0,0,0)) admits a one-step escape: (0,0,0) to (1,0,0) via threshold lowering, (1,1,0) to (1,0,0) via Q-debounce, (1,1,1) to (1,0,1) via window contraction.
- The genuine-verdict classes (1,0,0) and (0,1,0) are also *maximally distant* pairs: 2 moves apart. This reflects the underlying (A, B, C) feature distance.
- No class is a trap. The calibration space is therefore *strongly connected*: given enough tuning steps, any starting classification can be migrated to any target.

Proposition A.1 (Universal readability in at most three tuning moves)**.** *For every ordered pair of observation classes (C_i, C_j) the kernel can be retuned from a C_i-classified observation to C_j by composing at most two tuning relaxations — three for the two exceptional pairs* (1,0,0) $\to$ (0,1,1) *and* (0,1,0) $\to$ (1,0,1)*, which route through the intermediate class* (1,1,1)*. The depth-two reachability matrix of Table 11, computed by breadth-first search over the nine-operation tuning catalogue of Table 10, is the exhaustive witness; no class is a trap, so the calibration space is strongly connected (this reachability and strong-connectivity claim is* PROVEN*: an exhaustive breadth-first search over the finite nine-operation catalogue). The distance bound is an upper bound only: an entry is* realisable *on a given trace iff the signal supports the required in-window event pattern (realisability is* WITNESSED*, per trace and per signal), a per-signal feasibility that decomposes channel-wise — each of the P- and Q-channels prescribing a single-signal feasibility instance, coupled by the alignment knobs $(k, \hat{\varphi}, \Delta t)$ that place the Q-events relative to the P-channel opener — the two-signal lift of the single-signal synthesis of* [8].

Remark A.2 (Reachability is operational; soundness is semantic). Proposition A.1 is a *monitor-side*, operational fact: it says which observation class the kernel can be *tuned* to reach, and is silent on whether the reached reading may be attributed TRUE. A reachable transition is *sound* only when it respects the causal-coherence constraint owned by the core paper [9] — its causal-necessity theorem and its conditions CC1–CC4 — a *trace-side*, semantic condition on the underlying physical process. Reachability without that constraint is exactly the truth-manufacture that the non-circularity boundary $\mathcal{L}_1^{\text{tun}}$ of [8] forbids.

Limits of the abstraction. The reachability matrix abstracts away the physical signal τ: a migration (0,0,0) → (1,0,0) is *possible* in one move iff τ has a Q-peak just below θ; if τ is genuinely silent (Q constant well below θ), no tuning can escape the Quaestio. The matrix entries are therefore *upper bounds on signal-realisable migrations.* A signal-specific tree (with branches conditioned on τ's amplitude, frequency, and phase) is generated on demand by the same enumeration script.

This escapable-versus-trapped dichotomy is not merely a limit of the abstraction: it is *observable*, at the cost of one memoryless observation channel. The touch modality of the crossing/touch observation primitives [9] — so far unused by the crossing-only machinery of this paper — is exactly the detector of the sub-threshold peak on which the dichotomy turns.

Proposition A.3 (Touch-based disambiguation of the silent window)**.** *Let τ be a trace classified* (0,0,0) *(silent window: $A = B = C = 0$), so that the in-window Q-signal s_q never attains θ_q, and let $\delta > 0$ be a probe margin. Augment the monitor with a touch channel* $\text{Touch}^+(\theta_q - \delta)$ *— the memoryless extremum detector of the crossing/touch observation primitives* [9, 8]*, with capability cost* $(w, b, k) = (3, 0, 1)$ [8]*. Then:*

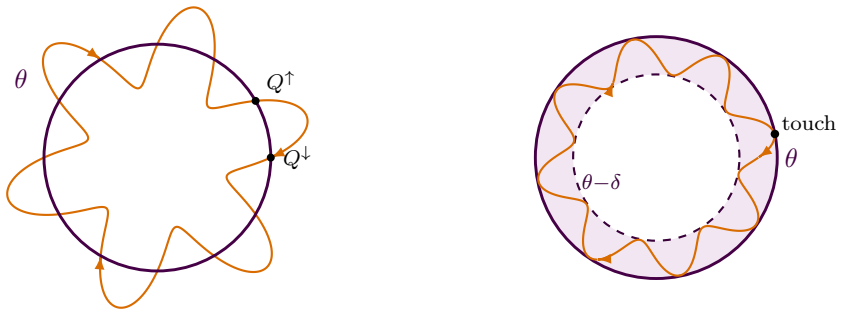
(i) (*δ*-escapable.) *If the touch channel fires at an in-window tick m — i.e. s_q attains a strict local maximum with $s_q(m) \in [\theta_q - \delta, \theta_q)$ — and s_q enters the window below some threshold $\theta' \le s_q(m)$, then the single $\mathcal{L}_1^{\text{tun}}$ tuning (T_1, threshold shift) that lowers θ_q to θ' produces an in-window up-crossing: the retuned trace has $A = 1$ and leaves* (0,0,0).

(ii) (*δ*-trapped.) *If the touch channel is silent — $s_q(t) < \theta_q - \delta$ at every in-window tick t — then every threshold shift of magnitude at most δ leaves the trace in* (0,0,0).

In particular, touch silence certifies *δ-trappedness, and a touch firing together with one additional sample comparison at the window opening (does s_q enter below θ'?) certifies δ-escapability: the dichotomy — invisible to the crossing-only alphabet Σ — is decided by one memoryless channel plus a single opening-tick comparison.*

Proof. (i) The firing at m witnesses $s_q(m-1) < s_q(m) \ge s_q(m+1)$ with $s_q(m) \ge \theta_q - \delta$. Since s_q enters the window below θ' and attains $s_q(m) \ge \theta'$ at m, there is a first in-window tick $t^* \le m$ with $s_q(t^*-1) < \theta' \le s_q(t^*)$: an up-crossing of θ', which the retuned crossing monitor latches as $Q^\top$, giving $A = 1$. (ii) For any shifted threshold $\theta'' \ge \theta_q - \delta$, no in-window sample reaches θ'', so neither a Q-validation nor a Q-invalidation can fire; the feature triple remains $(0, 0, 0)$. □

Remark A.4 (The observation-retuning simulator and its declared domain). Read as a transformation range, Proposition A.1 is the *observation-retuning* simulator: for any *active* (*δ*-escapable, Proposition A.3(i)) signal, the retuning maps the observed reading onto any target observation class in at most three moves. The *abstract* reachable set is the strongly-connected window class graph, which is PROVEN (exhaustive breadth-first search, Proposition A.1); its



(a) crossing: one circle θ

(b) touch: band $[\theta-\delta, \theta]$

FIGURE 13. The circular (period-wrapped) monitor. The length-T tick window is glued end-to-start onto the circle $\mathbb{R}/T\mathbb{Z}$ and the threshold becomes a concentric circle; the arrowheads on each signal mark the *traversal sense* (clockwise), which fixes the polarity of the radial events. (a) Crossing: the signal winds in and out of the single threshold circle θ; along the traversal sense an *outward* crossing is a validation $Q^{\uparrow}$ and an *inward* crossing an invalidation $Q^{\downarrow}$. (b) Touch: the probe band $[\theta-\delta, \theta]$ of Proposition A.3 is the annulus between the two concentric circles, and the signal *oscillates within the band*; each outward peak that reaches θ is a touch (a peak reaching the band is δ-escapable; a signal confined below $\theta-\delta$ would be δ-trapped, touch-silent).

per-signal realisation is WITNESSED (one active signal at a time, via the affordance hypothesis H1b). The domain is restricted, and stated as such: the retuning simulator ranges *from any active signal*, *not* from any trace — a δ-trapped silent signal (Proposition A.3(ii)) is outside the domain, and H1b is precisely the *declared domain* of the construction, not an apology for a gap. The inversion is the point: the δ-trapped boundary is not a hole, it *is* the event-soundness floor of Theorem 5.8 seen from the reachability side — the retuning cannot forge the one token it must not, so the very signals it cannot move are exactly those on which forging a crossing would violate Σ_2. (This "retuning simulator" is the observation-retuning *transform*; it is distinct from, and unrelated to, the numerical finite-prefix study of Section 8.1.6.)

The circular monitor: a geometric representation. The crossing/touch observations of Proposition A.3 admit a compact geometric rendering: glue the length-T tick window end-to-start onto the circle $\mathbb{R}/T\mathbb{Z}$ and draw the threshold as a concentric circle. Figure 13 shows the two modalities in this picture; the picture is a change of coordinates only, as Proposition A.5 records.

Proposition A.5 (The circular rendering is event-preserving, PROVEN, elementary)**.** *Take the* ω-periodic (lasso) *rendering: the length-T tick window is glued end-to-start onto $\mathbb{R}/T\mathbb{Z}$ and retraversed each period, so τ_0 continues τ_{T-1} and the seam edge $(T-1)\to 0$ is a genuine period-boundary transition. Mapping each threshold θ (respectively probe band $[\theta-\delta, \theta]$) to a concentric circle (respectively annulus) then induces a bijection between the radial outgoing/incoming transitions (respectively in-annulus strict extrema) of the wrapped signal and*

the crossing (respectively touch) token stream of the crossing/touch monitors of [8]*, run on the wrapped window with the seam edge evaluated as a period-boundary transition — on the* $T-1$ *interior edges the bijection is with the plain finite-window monitor, the seam carrying the one period-boundary token (none, on a non-periodic open window). The rendering changes coordinates, not events.*

Proof. A crossing (respectively touch) token fires exactly when a predicate on the pair (parameter index, value) holds. The wrap $m \mapsto m \bmod T$ is a homeomorphism of the (now circular) parameter axis that fixes the value axis pointwise; under the ω-periodic reading the seam $(T-1)\to 0$ is a real period-boundary tick evaluated like any other, so the predicate's truth is preserved at every tick, the seam included. The firing set — hence the emitted token stream — is therefore unchanged. This is a re-coordinatisation of the same events, not a deep result. □

Remark A.6 (Winding versus confinement, WITNESSED). Under the ω-periodic (lasso) rendering, in which the loop is retraversed once per period, the $\mathsf{L} \cap \mathsf{K}$ bipartition of the enforcement objectives (Definition 8.14) becomes topological. A recurrence objective $\mathsf{GF}\,\sigma^*$ holds iff the signal crosses the threshold circle on cofinally many laps — it winds in and out forever — while a persistence objective FG holds iff the signal is eventually confined to one side of the circle (inside or outside). Perpetual winding versus eventual confinement is the geometric face of the recurrence-versus-persistence split of Remarks 8.15 and 8.18. This is a geometric re-expression of the already-proven classification, not a new theorem.

Definition A.7 (Π-observation-security model, with a HEURISTIC cryptographic analogy)**.** Fix a *power budget* $\Pi \subseteq \{T_1, \ldots, T_8, \text{EXCLUDE}, \text{INCLUDE}\}$, measured in the capability calculus $\kappa = (w, b, k)$ of [8] through the lattice identification of Remark 5.6 (each power a bounded coordinate increment $(\Delta w, \Delta b, \Delta k)$ toward the join $\kappa = (3, 2, 1)$). The Π-*observation-security model* is the reachability-equivalence $\sim_\Pi$ that Π induces on observed readings: two readings are Π-equivalent iff each is reachable from the other by a composition of powers in Π (Proposition A.1 computes $\sim_\Pi$ for the full crossing budget). Its *un-forgeable core* is the Σ_2-token set $\mathsf{Sound}(\tau)$: by Theorem 5.8 this core is invariant under every *read-only* power, so no re-parameterisation budget can forge it. Every power in the budget above only re-*reads* τ; the un-forgeability is therefore a property of that *read-only* restriction, not an absolute. A strictly stronger power lies *outside* the lattice — a *signal-write* power W, the ability to modify or synthesise τ itself. This read-only/signal-write stratification is *not new here*: it is the Layer-1-versus-Layer-3 distinction of the strict tuning-layer hierarchy $\mathcal{L}_1 \subsetneq \mathcal{L}_2 \subsetneq \mathcal{L}_3$ of [8] (parameter retuning that leaves the signal unchanged $\subsetneq$ tick-level copy-paste, which fabricates no new sample value $\subsetneq$ full signal synthesis). Our contribution is only the *event-soundness reading* of that boundary: W is exactly the Layer-3 power the no-synthesis premise of Theorem 5.8 withholds, and Σ_2-closure is the statement that the read-only sublattice cannot forge a token. Granting W *unlocks* forgery: a monitor that may write τ can make a transit real and emit a token backed by the signal it now controls, sound relative to the written τ but fabricated relative to the world. The security lattice thus carries *two* boundaries, one per soundness level: the signal-write boundary W bounds *syntactic* Σ_2 soundness (crossing it forges a token), and CC3 bounds *meaningful* soundness (crossing it confirms below resolution). In κ-space, crossing CC3 means a budget whose threshold precision exceeds the instrumental resolution — it leaves Σ_2 intact yet confirms below resolution (Remark 5.10). Σ_2-closure is accordingly *read-only-conditional*, and this is the intended content: soundness is a function of the powers granted the simulator, precisely

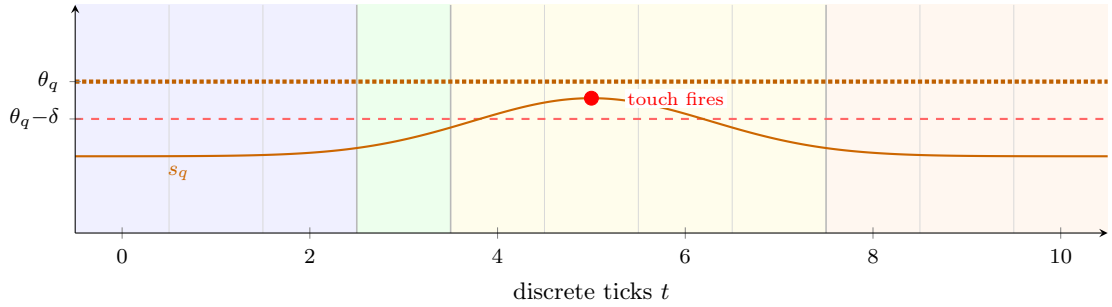


FIGURE 14. Touch-based disambiguation of the silent window (Proposition A.3). The Q-signal never crosses θ_q, so the crossing-only monitor observes nothing ((0,0,0), Quaestio). A memoryless touch channel at the probe level $\theta_q-\delta$ [8] fires at the sub-threshold peak, witnessing that the window is δ-escapable: lowering θ_q by at most δ (tuning T_1) converts the peak into an in-window validation. A silent touch channel certifies δ-trappedness instead.

as a simulation-based model prescribes. Different κ-budgets induce different $\sim_\Pi$, hence different security models.

Scope. This is a *bespoke, combinatorial/reachability-based observation-security* model, *not* standard cryptographic simulation-based security: there is no probability space, no advantage function, and no adversary-parameterised game; the "indistinguishability" $\sim_\Pi$ is *exact and combinatorial*. We adopt the phrase "security model" only as a HEURISTIC analogy with simulation-based security — an analogy of *form* (power budget $\leftrightarrow$ adversary resource; un-forgeable core $\leftrightarrow$ security invariant), not a transfer of cryptographic machinery. The identification of the power budget with the capability triple κ is itself WITNESSED (Remark 5.6, grounded in [8]).

Connection to causal coherence. Tuning operation T_9 (causal coherence filter, conditions CC1–CC4 [9]) acts as a *rejection filter* on the tree: it removes branches whose post-tuning kernel violates the theory-derived bounds on timing. Under T_9, the calibration tree becomes *pruned* to physically admissible moves only, restoring the link between calibration and theory coherence.

In kernel-parameter space each window class occupies an open region whose boundaries are one-tuning-step migrations, as the reachability matrix of Table 11 records.

Reading the eight panels jointly. Figures 1 to 8 together expose the operational meaning of the (A, B, C) feature triple. In each figure the *same* Schmitt rule with *the same* threshold θ is applied to *different* continuous signals; what changes is the shape of the trajectory relative to θ, not the decision rule. This makes precise the claim of the observation-not-omniscience remark [9]: the window class is a property of the *observed* trace, not of the ground-truth process—two processes whose signals coincide outside the dead-band will be classified identically, while two processes whose signals are physically distinct but happen to share the same trajectory through the dead-band region will be classified differently. The dead-band thickness $\Delta = \theta_{\text{high}} - \theta_{\text{low}}$ is therefore an *epistemic control*, not a measurement error: it parameterises how much observational structure the kernel exposes to the crossing/touch classification (the crossing/touch observation primitives [9]).

Calibration as dead-band narrowing. The calibration theorem $\Delta < \Delta_{\max}$ admits a direct geometric reading on Figure 3: if the dead-band is narrowed until the $\tilde{q}$ oscillator's positive peak crosses θ_{high}, the trace migrates from (0,0,0) to (1,0,0); if both peaks cross, the trace migrates to (1,1,0); if instead $\tilde{p}$ enters its own dead-band, the trace migrates to (1,0,1). The calibration parameter Δ is operationally the *vertical extent* of the gray strip.

The three encodings are parameter points of the single augmented monitor of the threshold-monitoring paper [8], where the tuning operators of this section acquire their parametric meaning.

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